

Serial Communication Interface (SCI)/ Local Interconnect Network (LIN) Module

This chapter describes the serial communication interface (SCI) / local interconnect network (LIN) module. The SCI/LIN is compliant to the LIN 2.1 protocol specified in the *LIN Specification Package*. This module can be configured to operate in either SCI (UART) or LIN mode.

NOTE: This chapter describes a superset implementation of the LIN/SCI module that includes features and functionality that require DMA. Since not all devices have DMA capability, consult your device-specific datasheet to determine applicability of these features and functions to your device being used.

Topic	Page
28.1 Introduction and Features	1513
28.2 SCI Communication Formats	1518
28.3 SCI Interrupts	1526
28.4 SCI DMA Interface	1529
28.5 SCI Configurations	1530
28.6 SCI Low-Power Mode	1532
28.7 LIN Communication Formats	1533
28.8 LIN Interrupts	1551
28.9 LIN DMA Interface	1551
28.10 LIN Configurations	1552
28.11 Low-Power Mode	1554
28.12 Emulation Mode	1556
28.13 SCI/LIN Control Registers	1557
28.14 GPIO Functionality	1606

28.1 Introduction and Features

The SCI/LIN module can be programmed to work either as an SCI or as a LIN. The core of the module is an SCI. The SCI's hardware features are augmented to achieve LIN compatibility.

The SCI module is a universal asynchronous receiver-transmitter that implements the standard nonreturn to zero format. The SCI can be used to communicate, for example, through an RS-232 port or over a K-line.

The LIN standard is based on the SCI (UART) serial data link format. The communication concept is single-master/multiple-slave with a message identification for multi-cast transmission between any network nodes.

Throughout the chapter Compatibility Mode refers to SCI Mode functionality of SCI/LIN Module. The initial part of the chapter explains about the SCI functionality and later part about the LIN functionality. Though the register are common for LIN and SCI, the register descriptions has notes to identify the register / bit usage in different modes.

28.1.1 SCI Features

The following are the features of the SCI module:

- Standard universal asynchronous receiver-transmitter (UART) communication
- Supports full- or half-duplex operation
- Standard nonreturn to zero (NRZ) format
- Double-buffered receive and transmit functions in compatibility mode
- Supports two individually enabled interrupt lines: level 0 and level 1
- Configurable frame format of 3 to 13 bits per character based on the following:
 - Data word length programmable from one to eight bits
 - Additional address bit in address-bit mode
 - Parity programmable for zero or one parity bit, odd or even parity
 - Stop programmable for one or two stop bits
- Asynchronous or isosynchronous communication modes
- Two multiprocessor communication formats allow communication between more than two devices
- Sleep mode is available to free CPU resources during multiprocessor communication and then wake up to receive an incoming message
- The 24-bit programmable baud rate supports 2^{24} different baud rates provide high accuracy baud rate selection
- At 100MHz Peripheral Clock, 3.125 Mbits/s is the Max Baud Rate achievable
- Capability to use Direct Memory Access (DMA) for transmit and receive data
- Five error flags and Seven status flags provide detailed information regarding SCI events
- Two external pins: LINRX and LINTX
- Multi-buffered receive and transmit units

NOTE: SCI/LIN module does not support UART hardware flow control. This feature can be implemented in software using a general purpose I/O pin.

28.1.2 LIN Features

The following are the features of the LIN module:

- Compatibility with LIN 1.3, 2.0 and 2.1 protocols
- Configurable Baud Rate up to 20 Kbits/s
- Two external pins: LINRX and LINTX.
- Multi-buffered receive and transmit units
- Identification masks for message filtering
- Automatic master header generation
 - Programmable synchronization break field
 - Synchronization field
 - Identifier field
- Slave automatic synchronization
 - Synchronization break detection
 - Optional baud rate update
 - Synchronization validation
- 2^{31} programmable transmission rates with 7 fractional bits
- Wakeup on LINRX dominant level from transceiver
- Automatic wakeup support
 - Wakeup signal generation
 - Expiration times on wakeup signals
- Automatic bus idle detection
- Error detection
 - Bit error
 - Bus error
 - No-response error
 - Checksum error
 - Synchronization field error
 - Parity error
- Capability to use Direct Memory Access (DMA) for transmit and receive data.
- 2 Interrupt lines with priority encoding for:
 - Receive
 - Transmit
 - ID, error, and status
- Support for LIN 2.0 checksum
- Enhanced synchronizer finite state machine (FSM) support for frame processing
- Enhanced handling of extended frames
- Enhanced baud rate generator
- Update wakeup/go to sleep

28.1.3 Block Diagram

The SCI/LIN module contains core SCI block with added sub-blocks to support LIN protocol.

Three Major components of the SCI Module are:

- Transmitter
- Baud Clock Generator
- Receiver

Transmitter (TX) contains two major registers to perform the double- buffering:

- The transmitter data buffer register (SCITD) contains data loaded by the CPU to be transferred to the shift register for transmission.
- The transmitter shift register (SCITXSHF) loads data from the data buffer (SCITD) and shifts data onto the LINTX pin, one bit at a time.

Baud Clock Generator

- A programmable baud generator produces either a baud clock scaled from VBUSP CLK.

Receiver (RX) contains two major registers to perform the double- buffering:

- The receiver shift register (SCIRXSHF) shifts data in from the LINRX pin one bit at a time and transfers completed data into the receive data buffer.
- The receiver data buffer register (SCIRD) contains received data transferred from the receiver shift register

The SCI receiver and transmitter are double-buffered, and each has its own separate enable and interrupt bits. The receiver and transmitter may each be operated independently or simultaneously in full duplex mode.

To ensure data integrity, the SCI checks the data it receives for breaks, parity, overrun, and framing errors. The bit rate (baud) is programmable to over 16 million different rates through a 24-bit baud-select register. [Figure 28-1](#) shows the detailed SCI block diagram.

The SCI/LIN module is based on the standalone SCI with the addition of an error detector (parity calculator, checksum calculator, and bit monitor), a mask filter, a synchronizer, and a multi-buffered receiver and transmitter. The SCI interface, the DMA control subblocks and the baud generator are modified as part of the hardware enhancements for LIN compatibility. [Figure 28-2](#) shows the SCI/LIN block diagram.

Figure 28-1. Detailed SCI Block Diagram

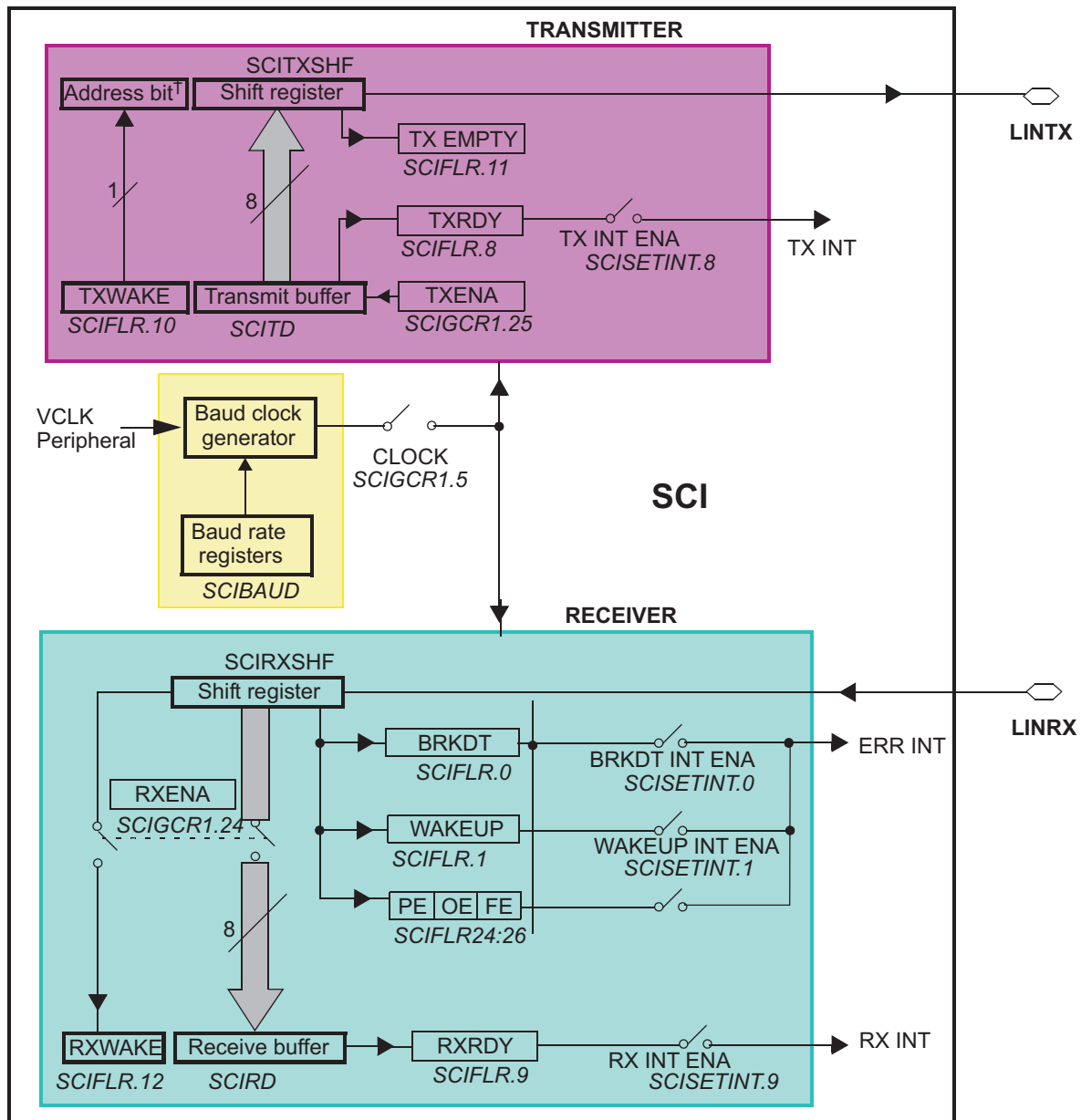
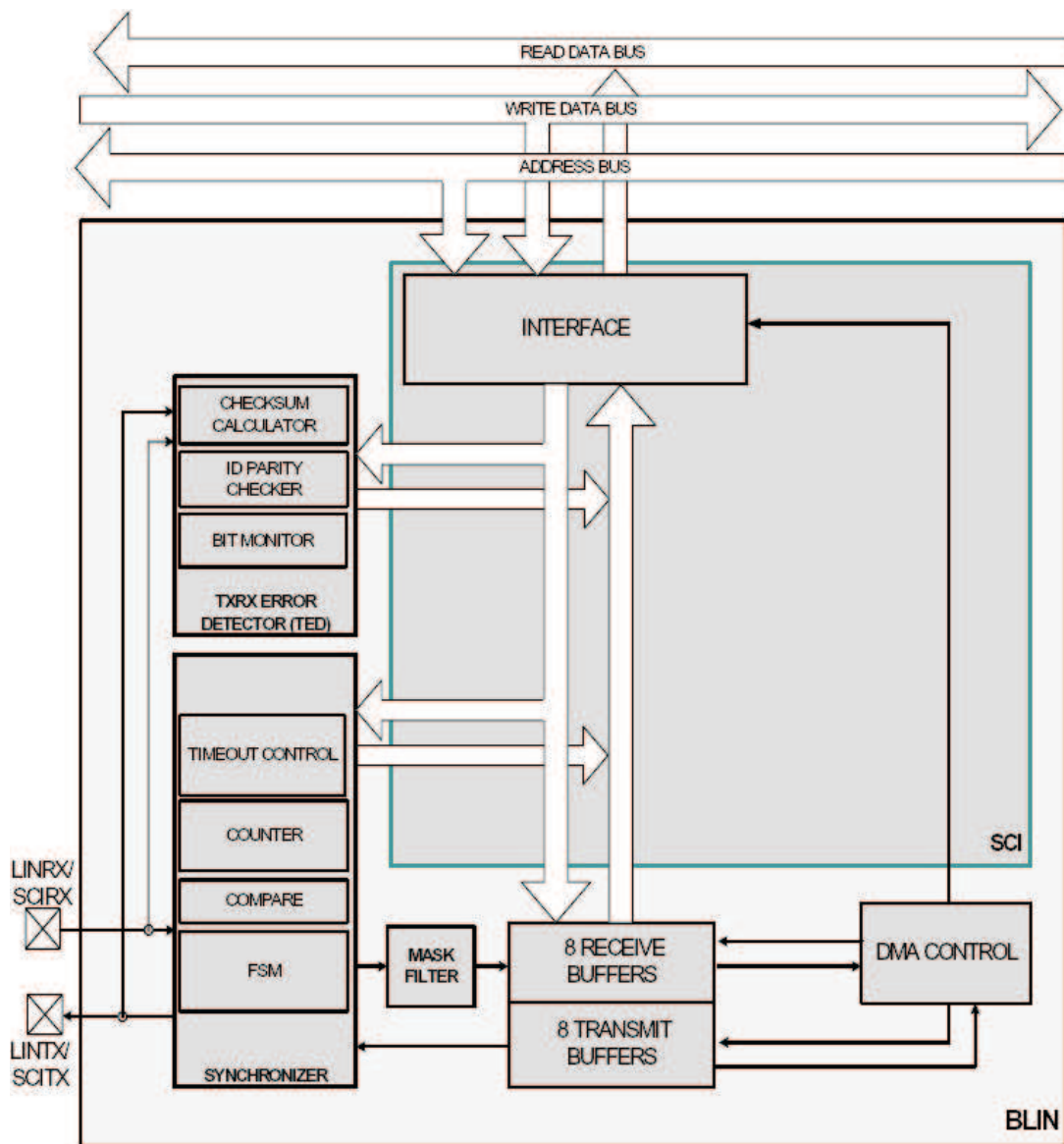


Figure 28-2. SCI/LIN Block Diagram



28.2 SCI Communication Formats

The SCI module can be configured to meet the requirements of many applications. Because communication formats vary depending on the specific application, many attributes of the SCI/LIN are user configurable. The list below describes these configuration options:

- SCI Frame format
- SCI Timing modes
- SCI Baud rate
- SCI Multiprocessor modes

28.2.1 SCI Frame Formats

The SCI uses a programmable frame format. All frames consist of the following:

- One start bit
- One to eight data bits
- Zero or one address bit
- Zero or one parity bit
- One or two stop bits

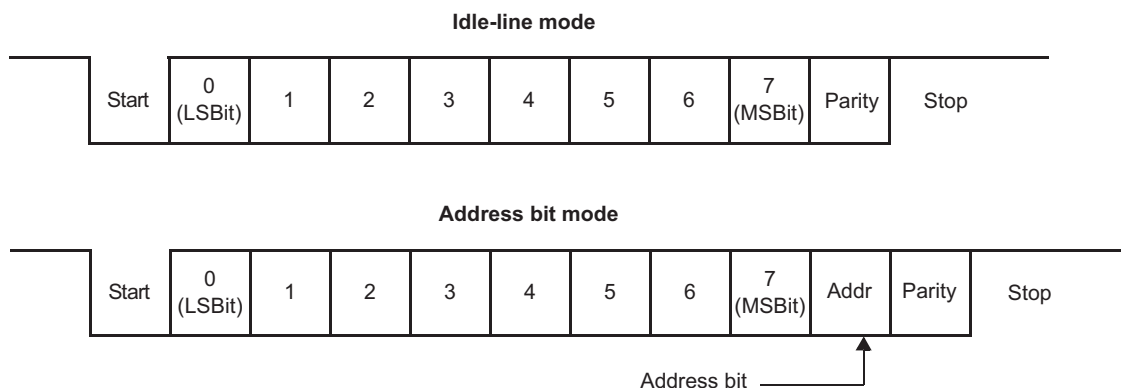
The frame format for both the transmitter and receiver is programmable through the bits in the SCIGCR1 register. Both receive and transmit data is in nonreturn to zero (NRZ) format, which means that the transmit and receive lines are at logic high when idle. Each frame transmission begins with a start bit, in which the transmitter pulls the SCI line low (logic low). Following the start bit, the frame data is sent and received least significant bit first (LSB).

An address bit is present in each frame if the SCI is configured to be in address-bit mode but is not present in any frame if the SCI is configured for idle-line mode. The format of frames with and without the address bit is illustrated in [Figure 28-3](#).

A parity bit is present in every frame when the PARITY ENA bit is set. The value of the parity bit depends on the number of one bits in the frame and whether odd or even parity has been selected via the PARITY ENA bit. Both examples in [Figure 28-3](#) have parity enabled.

All frames include one stop bit, which is always a high level. This high level at the end of each frame is used to indicate the end of a frame to ensure synchronization between communicating devices. Two stop bits are transmitted if the STOP bit in SCIGCR1 register is set. The examples shown in [Figure 28-3](#) use one stop bit per frame.

Figure 28-3. Typical SCI Data Frame Formats



28.2.2 SCI Timing Mode

The SCI can be configured to use asynchronous or isosynchronous timing using TIMING MODE bit in SCIGCR1 register.

28.2.2.1 Asynchronous Timing Mode

The asynchronous timing mode uses only the receive and transmit data lines to interface with devices using the standard universal asynchronous receiver-transmitter (UART) protocol.

In the asynchronous timing mode, each bit in a frame has a duration of 16 SCI baud clock periods. Each bit therefore consists of 16 samples (one for each clock period). When the SCI is using asynchronous mode, the baud rates of all communicating devices must match as closely as possible. Receive errors result from devices communicating at different baud rates.

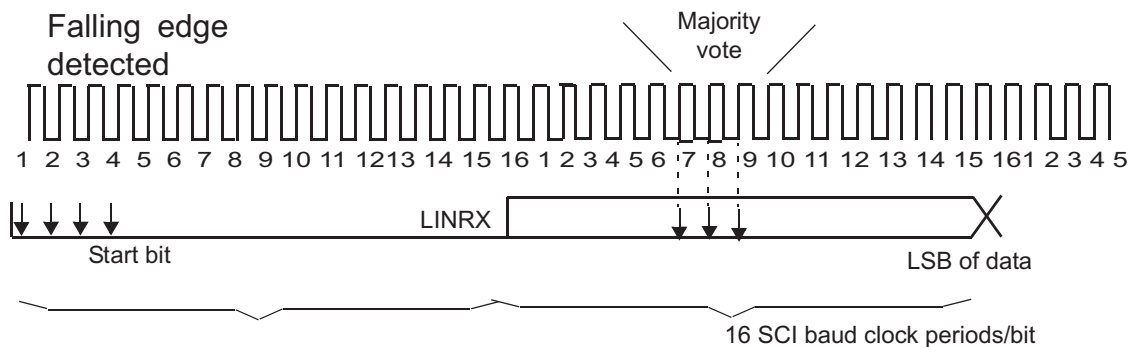
With the receiver in the asynchronous timing mode, the SCI detects a valid start bit if the first four samples after a falling edge on the LINRX pin are of logic level 0. As soon as a falling edge is detected on LINRX, the SCI assumes that a frame is being received and synchronizes itself to the bus.

To prevent interpreting noise as Start bit SCI expects LINRX line to be low for at least four contiguous SCI baud clock periods to detect a valid start bit. The bus is considered idle if this condition is not met. When a valid start bit is detected, the SCI determines the value of each bit by sampling the LINRX line value during the seventh, eighth, and ninth SCI baud clock periods. A majority vote of these three samples is used to determine the value stored in the SCI receiver shift register. By sampling in the middle of the bit, the SCI reduces errors caused by propagation delays and rise and fall times and data line noises.

Figure 28-4 illustrates how the receiver samples a start bit and a data bit in asynchronous timing mode.

The transmitter transmits each bit for a duration of 16 SCI baud clock periods. During the first clock period for a bit, the transmitter shifts the value of that bit onto the LINTX pin. The transmitter then holds the current bit value on LINTX for 16 SCI baud clock periods.

Figure 28-4. Asynchronous Communication Bit Timing



28.2.2.2 Isosynchronous Timing Mode

In isosynchronous timing mode, each bit in a frame has a duration of exactly 1 baud clock period and therefore consists of a single sample. With this timing configuration, the transmitter and receiver are required to make use of the SCICLK pin to synchronize communication with other SCI. **This mode is not supported on this device because SCICLK pin is not available.**

28.2.3 SCI Baud Rate

The SCI/LIN has an internally-generated serial clock determined by the peripheral VCLK and the prescalers P and M in this register. The SCI uses the 24-bit integer prescaler P value of the BRS register to select the required baud rates. The additional 4-bit fractional divider M refines the baud rate selection.

In asynchronous timing mode, the SCI generates a baud clock according to the following formula:

$$SCICLK \text{ Frequency} = \frac{VCLK \text{ Frequency}}{P + 1 + \frac{M}{16}}$$

$$\text{Asynchronous baud value} = \frac{SCICLK \text{ Frequency}}{16}$$

For P = 0,

$$\text{Asynchronous baud value} = \frac{VCLK \text{ Frequency}}{32} \quad (42)$$

28.2.3.1 Superfractional Divider, SCI Asynchronous Mode

The superfractional divider is available in SCI asynchronous mode (idle-line and address-bit mode). Building on the 4-bit fractional divider M (BRS[27:24]), the super fractional divider uses an additional 3-bit modulating value, illustrated in Table 28-2. The bits with a 1 in the table will have an additional VCLK period added to their T_{bit} . If the character length is more than 10, then the modulation table will be a rolled-over version of the original table (Table 28-1), as shown in Table 28-2.

The baud rate will vary over a data field to average according to the BRS[30:28] value by a “d” fraction of the peripheral internal clock: $0 < d < 1$. Refer Figure 28-5 for a simple Average “d” calculation based on “U” value (BRS[30:28]).

The instantaneous bit time is expressed in terms of T_{VCLK} as follows:

For all P other than 0, and all M and d (0 or 1),

$$T^{i}bit = \left[16 \left(P + 1 + \frac{M}{16} \right) + d \right] T_{VCLK} \quad (43)$$

For P = 0 $T_{bit} = 32T_{VCLK}$

The averaged bit time is expressed in terms of T_{VCLK} as follows:

For all P other than 0, and all M and d ($0 < d < 1$),

$$T^{a}bit = \left[16 \left(P + 1 + \frac{M}{16} \right) + d \right] T_{VCLK} \quad (44)$$

For P = 0 $T_{bit} = 32T_{VCLK}$

Table 28-1. Superfractional Bit Modulation for SCI Mode (Normal Configuration)⁽¹⁾

BRS[30:28]	Start Bit	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]	Stop Bit
0h	0	0	0	0	0	0	0	0	0	0
1h	1	0	0	0	0	0	0	0	1	0
2h	1	0	0	0	1	0	0	0	1	0
3h	1	0	1	0	1	0	0	0	1	0
4h	1	0	1	0	1	0	1	0	1	0
5h	1	1	1	0	1	0	1	0	1	1
6h	1	1	1	0	1	1	1	0	1	1
7h	1	1	1	1	1	1	1	0	1	1

⁽¹⁾ Normal configuration = Start + 8 Data Bits + Stop Bit

Table 28-2. Superfractional Bit Modulation for SCI Mode (Maximum Configuration)⁽¹⁾

BRS[30:28]	Start Bit	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]	Addr	Parity	Stop0	Stop1
0h	0	0	0	0	0	0	0	0	0	0	0	0	0
1h	1	0	0	0	0	0	0	0	1	0	0	0	0
2h	1	0	0	0	1	0	0	0	1	0	0	0	1
3h	1	0	1	0	1	0	0	0	1	0	1	0	1
4h	1	0	1	0	1	0	1	0	1	0	1	0	1
5h	1	1	1	0	1	0	1	0	1	1	1	0	1
6h	1	1	1	0	1	1	1	0	1	1	1	0	1
7h	1	1	1	1	1	1	1	0	1	1	1	1	1

⁽¹⁾ Maximum configuration = Start + 8 Data Bits + Addr Bit + Parity Bit + Stop Bit 0 + Stop Bit 1

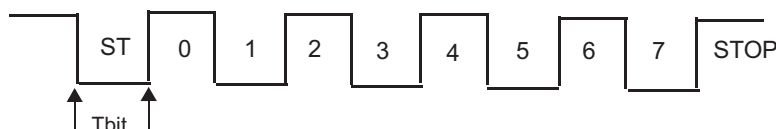
Table 28-3. SCI Mode (Minimum Configuration)⁽¹⁾

BRS[30:28]	Start Bit	D[0]	Stop
0h	0	0	0
1h	1	0	0
2h	1	0	0
3h	1	0	1
4h	1	0	1
5h	1	1	1
6h	1	1	1
7h	1	1	1

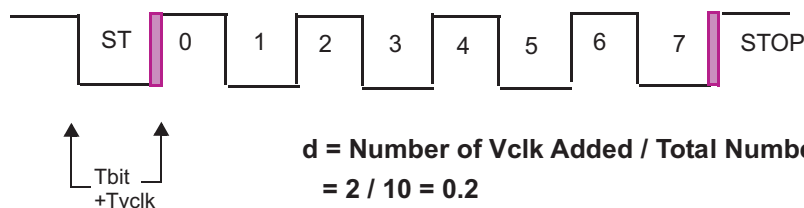
⁽¹⁾ Minimum configuration = Start + 1 Data Bits + Stop Bit

Figure 28-5. Superfractional Divider Example

Normal Data Frame with BRS[31:28] = 0



Normal Data Frame with BRS[31:28] = 1



28.2.4 SCI Multiprocessor Communication Modes

In some applications, the SCI may be connected to more than one serial communication device. In such a multiprocessor configuration, several frames of data may be sent to all connected devices or to an individual device. In the case of data sent to an individual device, the receiving devices must determine when they are being addressed. When a message is not intended for them, the devices can ignore the following data. When only two devices make up the SCI network, addressing is not needed, so multiprocessor communication schemes are not required.

SCI supports two multiprocessor communication modes which can be selected using COMM MODE bit:

- Idle-Line Mode
- Address Bit Mode

When the SCI is not used in a multiprocessor environment, software can consider all frames as data frames. In this case, the only distinction between the idle-line and address-bit modes is the presence of an extra bit (the address bit) in each frame sent with the address-bit protocol.

The SCI allows full-duplex communication where data can be sent and received via the transmit and receive pins simultaneously. However, the protocol used by the SCI assumes that only one device transmits data on the same bus line at any one time. No arbitration is done by the SCI.

28.2.4.1 Idle-Line Multiprocessor Modes

In idle-line multiprocessor mode, a frame that is preceded by an idle period (10 or more idle bits) is an address frame. A frame that is preceded by fewer than 10 idle bits is a data frame. [Figure 28-6](#) illustrates the format of several blocks and frames with idle-line mode.

There are two ways to transmit an address frame using idle-line mode:

Method 1: In software, deliberately leave an idle period between the transmission of the last data frame of the previous block and the address frame of the new block.

Method 2: Configure the SCI to automatically send an idle period between the last data frame of the previous block and the address frame of the new block.

Although Method 1 is only accomplished by a delay loop in software, Method 2 can be implemented by using the transmit buffer and the TXWAKE bit in the following manner:

Step 1: Write a 1 to the TXWAKE bit.

Step 2: Write a dummy data value to the SCITD register. This triggers the SCI to begin the idle period as soon as the transmitter shift register is empty.

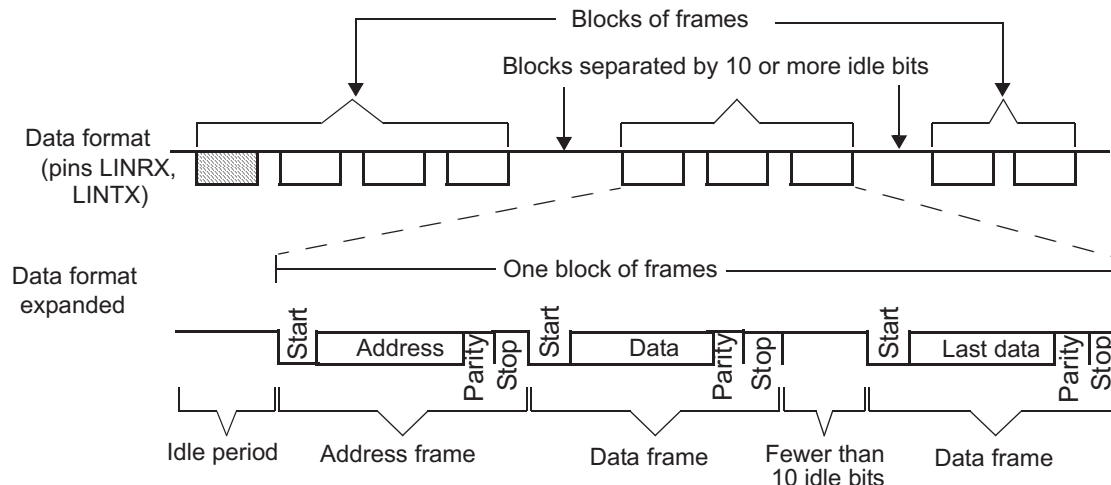
Step 3: Wait for the SCI to clear the TXWAKE flag.

Step 4: Write the address value to SCITD.

As indicated by Step 3, software should wait for the SCI to clear the TXWAKE bit. However, the SCI clears the TXWAKE bit at the same time it sets TXRDY (that is, transfers data from SCITD into SCITXSHF). Therefore, if the TX INT ENA bit is set, the transfer of data from SCITD to SCITXSHF causes an interrupt to be generated at the same time that the SCI clears the TXWAKE bit. If this interrupt method is used, software is not required to poll the TXWAKE bit waiting for the SCI to clear it.

When idle-line multiprocessor communications are used, software must ensure that the idle time exceeds 10 bit periods before addresses (using one of the methods mentioned above), and software must also ensure that data frames are written to the transmitter quickly enough to be sent without a delay of 10 bit periods between frames. Failure to comply with these conditions will result in data interpretation errors by other devices receiving the transmission.

Figure 28-6. Idle-Line Multiprocessor Communication Format



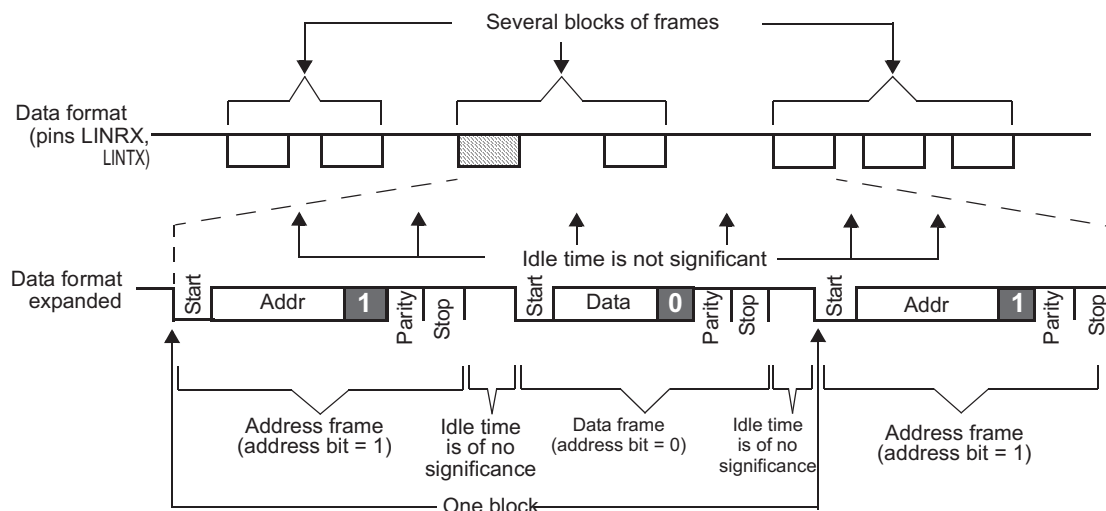
28.2.4.2 Address-Bit Multiprocessor Mode

In the address-bit protocol, each frame has an extra bit immediately following the data field called an address bit. A frame with the address bit set to 1 is an address frame; a frame with the address bit set to 0 is a data frame. The idle period timing is irrelevant in this mode. [Figure 28-7](#) illustrates the format of several blocks and frames with the address-bit mode.

When address-bit mode is used, the value of the TXWAKE bit is the value sent as the address bit. To send an address frame, software must set the TXWAKE bit. This bit is cleared as the contents of the SCITD are shifted from the TXWAKE register so that all frames sent are data except when the TXWAKE bit is written as a 1.

No dummy write to SCITD is required before an address frame is sent in address-bit mode. The first byte written to SCITD after the TXWAKE bit is written to 1 is transmitted with the address bit set when address-bit mode is used.

Figure 28-7. Address-Bit Multiprocessor Communication Format



28.2.5 SCI Multi-Buffered Mode

To reduce CPU load when Receiving or Transmitting data in interrupt mode or DMA mode, the SCI/LIN module has eight separate Receive and transmit buffers. Multi buffered mode is enabled by setting the MBUF MODE bit in the SCIGCR1 register.

The multi-buffer 3-bit counter counts the data bytes transferred from the SCIRXSHF register to the RDy receive buffers and TDy transmit buffers register to SCITXSHF register. The 3-bit compare register contains the number of data bytes expected to be received or transmitted. the LENGTH value in SCIFORMAT register indicates the expected length and is used to load the 3-bit compare register.

A receive interrupt (RX interrupt; see the SCIINTVECT0 and SCIINTVECT1 registers), and a receive ready RXRDY flag set in SCIFLR register, as well as a DMA request (RXDMA) could occur after receiving a response if there are no response receive errors for the frame (such as, there is, frame error, and overrun error).

A transmit interrupt (TX interrupt), and a transmit ready flag (TXRDY flag in SCIFLR register), and a DMA request (TXDMA) could occur after transmitting a response.

Figure 28-8 and Figure 28-9 shows the receive and transmit multi-buffer functional block diagram.

Figure 28-8. Receive Buffers

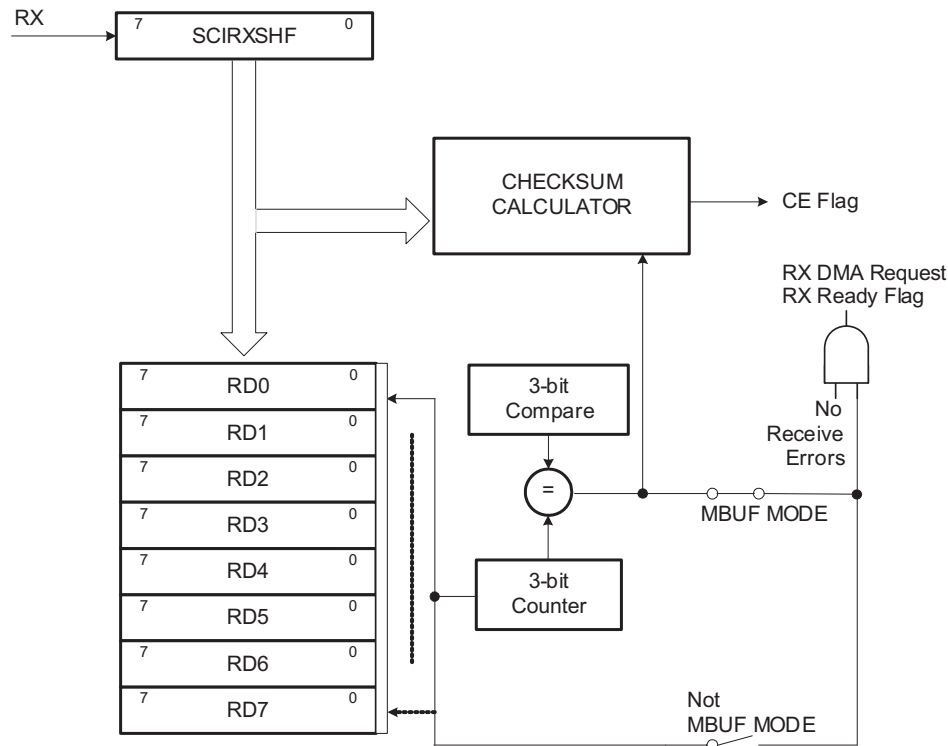
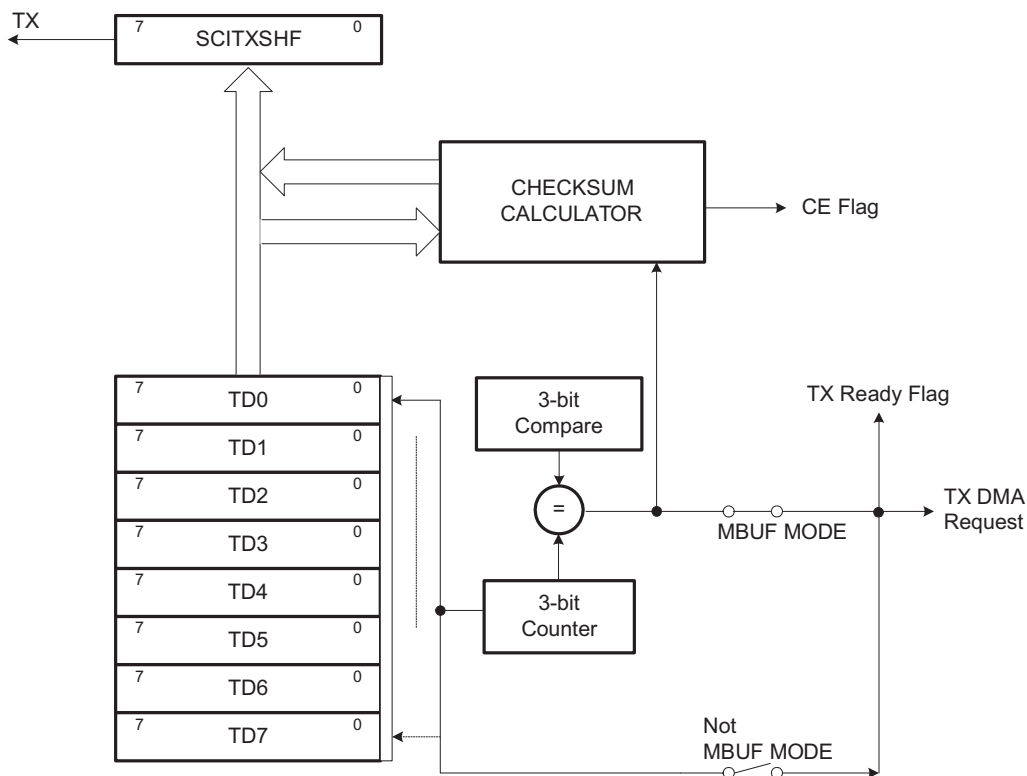


Figure 28-9. Transmit Buffers



28.3 SCI Interrupts

The SCI/LIN module has two interrupt lines, level 0 and level 1, to the vectored interrupt manager (VIM) module (see [Figure 28-10](#)). Two offset registers SCIINTVECT0 and SCIINTVECT1 determine which flag triggered the interrupt according to the respective priority encoders. Each interrupt condition has a bit to enable/disable the interrupt in the SCISSETINT and SCICLRINT registers respectively.

Each interrupt also has a bit that can be set as interrupt level 0 (INT0) or as interrupt level 1 (INT1). By default, interrupts are in interrupt level 0. SCISSETINTLVL sets a given interrupt to level 1. SCICLEARINTLVL resets a given interrupt level to the default level 0.

The interrupt vector registers SCIINTVECT0 and SCIINTVECT1 return the vector of the pending interrupt line INT0 or INT1. If more than one interrupt is pending, the interrupt vector register holds the highest priority interrupt.

Figure 28-10. General Interrupt Scheme

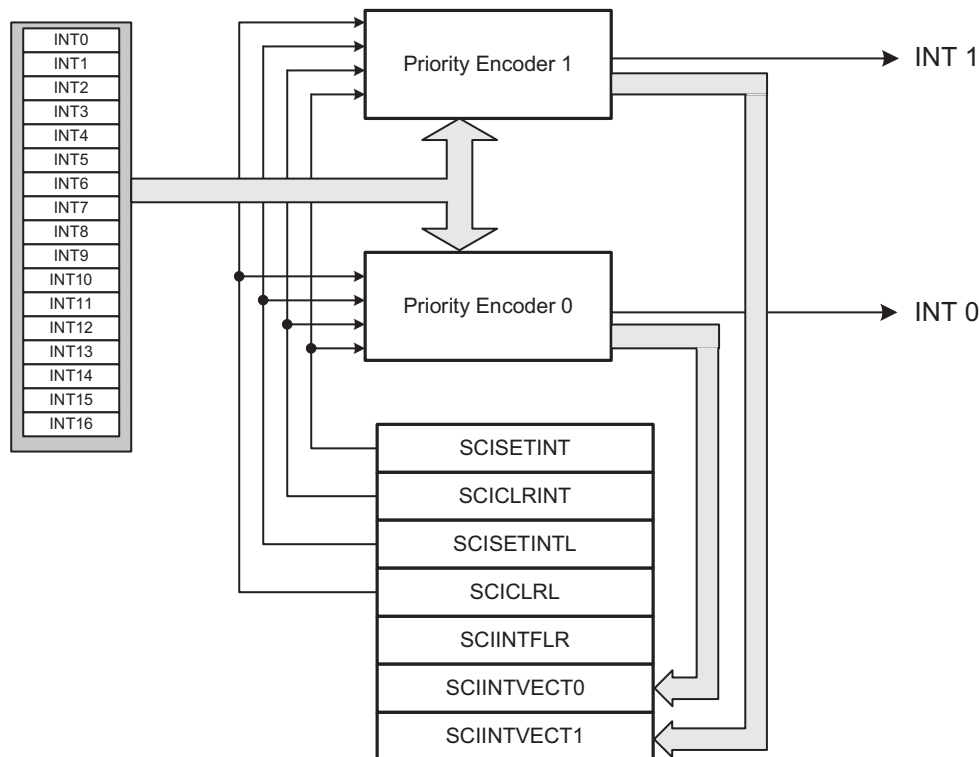
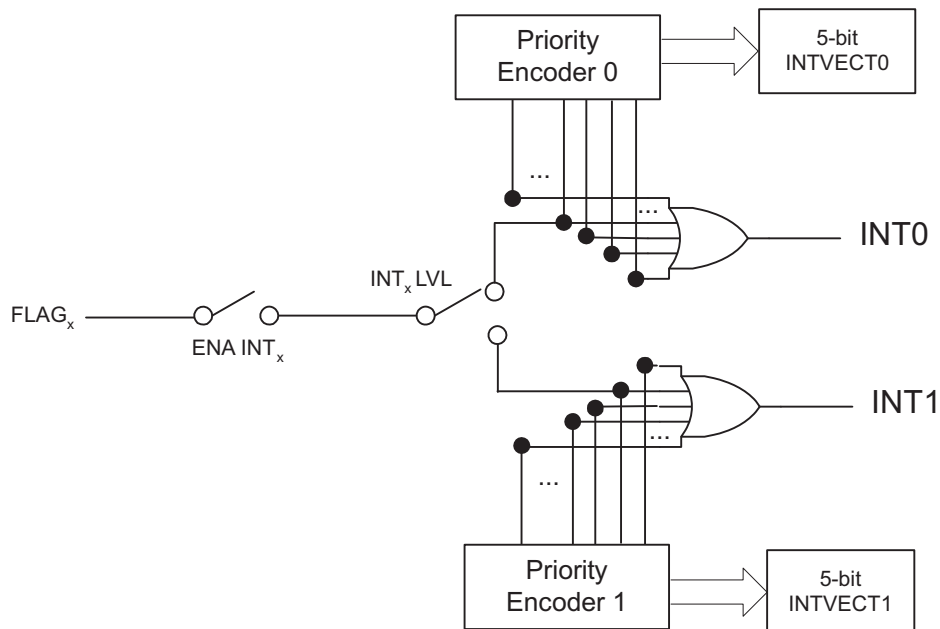


Figure 28-11. Interrupt Generation for Given Flags



28.3.1 Transmit Interrupt

To use transmit interrupt functionality, SET TX INT bit must be enabled and SET TX DMA bit must be cleared. The transmit ready (TXRDY) flag is set when the SCI transfers the contents of SCITD to the shift register, SCITXSHF. The TXRDY flag indicates that SCITD is ready to be loaded with more data. In addition, the SCI sets the TX EMPTY bit if both the SCITD and SCITXSHF registers are empty. If the SET TX INT bit is set, then a transmit interrupt is generated when the TXRDY flag goes high. Transmit Interrupt is not generated immediately after setting the SET TX INT bit unlike transmit DMA request. Transmit Interrupt is generated only after the first transfer from SCITD to SCITXSHF, that is first data has to be written to SCITD by the User before any interrupt gets generated. To transmit further data the user can write data to SCITD in the transmit Interrupt service routine.

Writing data to the SCITD register clears the TXRDY bit. When this data has been moved to the SCITXSHF register, the TXRDY bit is set again. The interrupt request can be suspended by setting the CLR TX INT bit; however, when the SET TX INT bit is again set to 1, the TXRDY interrupt is asserted again. The transmit interrupt request can be eliminated until the next series of values is written to SCITD, by disabling the transmitter via the TXENA bit, by a software reset SWnRST, or by a device hardware reset.

28.3.2 Receive Interrupt

The receive ready (RXRDY) flag is set when the SCI transfers newly received data from SCIRXSHF to SCIRD. The RXRDY flag therefore indicates that the SCI has new data to be read. Receive interrupts are enabled by the SET RX INT bit. If the SET RX INT is set when the SCI sets the RXRDY flag, then a receive interrupt is generated. The received data can be read in the Interrupt Service routine.

On a device with both SCI and a DMA controller, the bits SET RX DMA ALL and SET RX DMA must be cleared to select interrupt functionality.

28.3.3 WakeUp Interrupt

SCI sets the WAKEUP flag if bus activity on the RX line either prevents power-down mode from being entered, or RX line activity causes an exit from power-down mode. If enabled (SET WAKEUP INT), wakeup interrupt is triggered once WAKEUP flag is set.

28.3.4 Error Interrupts

The following error detection are supported with Interrupt by the SCI module:

- Parity errors (PE)
- Frame errors (FE)
- Break Detect errors (BRKDT)
- Overrun errors (OE)
- Bit errors (BE)

If all of these errors (PE, FE, BRKDT, OE, BE) are flagged, an interrupt for the flagged errors will be generated if enabled. A message is valid for both the transmitter and the receiver if there is no error detected until the end of the frame. Each of these flags is located in the receiver status (SCIFLR) register.

There are 16 interrupt sources in the SCI/LIN module, In SCI mode 8 interrupts are supported, as seen in [Table 28-4](#).

Table 28-4. SCI/LIN Interrupts

Offset ⁽¹⁾	Interrupt	Applicable to SCI	Applicable to LIN
0	No interrupt		
1	Wakeup	Yes	Yes
2	Inconsistent-synch-field error	No	Yes
3	Parity error	Yes	Yes
4	ID	No	Yes
5	Physical bus error	No	Yes
6	Frame error	Yes	Yes
7	Break detect	Yes	No
8	Checksum error	No	Yes
9	Overrun error	Yes	Yes
10	Bit error	Yes	Yes
11	Receive	Yes	Yes
12	Transmit	Yes	Yes
13	No-response error	No	Yes
14	Timeout after wakeup signal (150 ms)	No	Yes
15	Timeout after three wakeup signals (1.5 s)	No	Yes
16	Timeout (Bus Idle, 4s)	No	Yes

⁽¹⁾ Offset 1 is the highest priority. Offset 16 is the lowest priority.

28.4 SCI DMA Interface

DMA requests for receive (RXDMA request) and transmit (TXDMA request) are available for the SCI/LIN module. The DMA transfers depending on whether multi-buffer mode bit (MBUF MODE in the SCIGCR1 register) is enabled or is not enabled. See [Chapter 29](#) for additional information.

28.4.1 Receive DMA Requests

This DMA functionality is enabled/disabled by the CPU using the SET RX DMA/CLR RX DMA bits, respectively.

In Multi-Buffered SCI mode with DMA enabled, the receiver loads the RDy buffers for each received character. RX DMA request is triggered once the last character of the programmed number of characters (LENGTH) are received and copied to the corresponding RDy buffer successfully.

If the multi-buffer option is disabled, then DMA requests will be generated on a byte-per-byte basis.

In multiprocessor mode, the SCI can generate receiver interrupts for address frames and DMA requests for data frames or DMA requests for both. This is controlled by the SET RX DMA ALL bit.

In multiprocessor mode with the SLEEP bit set, no DMA is generated for received data frames. The software must clear the SLEEP bit before data frames can be received.

28.4.2 Transmit DMA Requests

DMA functionality is enabled/disabled by the CPU with SET TX DMA/CLR TX DMA bits, respectively.

In Multi-Buffered SCI mode, once the TXRDY bit is set or after a transmission of programmed number of characters (LENGTH) (up to eight data bytes stored in the transmit buffer(s) TDy in the LINTD0 and LINTD1 registers), a DMA request is generated in order to reload the transmit buffer for the next transmission. If the multi-buffer option is disabled, then DMA requests will be generated on a byte-per-byte basis.

28.5 SCI Configurations

Before the SCI sends or receives data, its registers should be properly configured. Upon power-up or a system-level reset, each bit in the SCI registers is set to a default state. The registers are writable only after the RESET bit in the SCIGCR0 register is set to 1. Of particular importance is the SWnRST bit in the SCIGCR1 register. The SWnRST is an active-low bit initialized to 0 and keeps the SCI in a reset state until it is programmed to 1. Therefore, all SCI configuration should be completed before a 1 is written to the SWnRST bit.

The following list details the configuration steps that software should perform prior to the transmission or reception of data. As long as the SWnRST bit is cleared to 0 the entire time that the SCI is being configured, the order in which the registers are programmed is not important.

- Enable SCI by setting the RESET bit to 1.
- Clear the SWnRST bit to 0 before SCI is configured.
- Select the desired frame format by programming the SCIGCR1 register.
- Set both the RX FUNC and TX FUNC bits in SCIPIO0 to 1 to configure the LINRX and LINTX pins for SCI functionality.
- Select the baud rate to be used for communication by programming the BRS register.
- Set the CLOCK bit in SCIGCR1 to 1 to select the internal clock.
- Set the CONT bit in SCIGCR1 to 1 to make SCI not halt for an emulation breakpoint until its current reception or transmission is complete (this bit is used only in an emulation environment).
- Set the LOOP BACK bit in SCIGCR1 to 1 to connect the transmitter to the receiver internally (this feature is used to perform a self-test).
- Set the RXENA bit in SCIGCR1 to 1, if data is to be received.
- Set the TXENA bit in SCIGCR1 to 1, if data is to be transmitted.
- Set the SWnRST bit to 1 after SCI is configured.
- Perform receiving or transmitting data (see [Section 28.5.1](#) or [Section 28.5.2](#)).

28.5.1 Receiving Data

The SCI receiver is enabled to receive messages if both the RX FUNC bit and the RXENA bit are set to 1. If the RX FUNC bit is not set, the LINRX pin functions as a general-purpose I/O pin rather than as an SCI function pin.

SCI module can receive data in one of the following modes:

- Single-Buffer (Normal) Mode
- Multi-Buffer Mode

After a valid idle period is detected, data is automatically received as it arrives on the LINRX pin.

28.5.1.1 Receiving Data in Single-Buffer Mode

Single-buffer mode is selected when the MBUF MODE bit in SCIGCR1 is cleared to 0. In this mode, SCI sets the RXRDY bit when it transfers newly received data from SCIRXSHF to SCIRD. The SCI clears the RXRDY bit after the new data in SCIRD has been read. Also, as data is transferred from SCIRXSHF to SCIRD, the SCI sets the FE, OE, or PE flags if any of these error conditions were detected in the received data. These error conditions are supported with configurable interrupt capability. The wake-up and break-detect status bits are also set if one of these errors occurs, but they do not necessarily occur at the same time that new data is being loaded into SCIRD.

You can receive data by:

1. Polling Receive Ready Flag
2. Receive Interrupt
3. DMA

In polling method, software can poll for the RXRDY bit and read the data from the SCIRD register once the RXRDY bit is set high. The CPU is unnecessarily overloaded by selecting the polling method. To avoid this, you can use either the interrupt or DMA method. To use the interrupt method, the SET RX INT bit is set. To use the DMA method, the SET RX DMA bit is set. Either an interrupt or a DMA request is generated the moment the RXRDY bit is set.

28.5.1.2 Receiving Data in Multi-Buffer Mode

Multi-buffer mode is selected when the MBUF MODE bit in SCIGCR1 is set to 1. In this mode, SCI sets the RXRDY bit after receiving the programmed number of data in the receive buffer, the complete frame. The error condition detection logic is similar to the single-buffer mode, except that it monitors for the complete frame. Like single-buffer mode, you can use the polling, interrupt, or DMA method to read the data. The SCI clears the RXRDY bit after the new data in SCIRD has been read.

28.5.2 Transmitting Data

The SCI transmitter is enabled if both the TX FUNC bit and the TXENA bit are set to 1. If the TX FUNC bit is not set, the LINTX pin functions as a general-purpose I/O pin rather than as an SCI function pin. Any value written to the SCITD before TXENA is set to 1 is not transmitted. Both of these control bits allow for the SCI transmitter to be held inactive independently of the receiver.

SCI module can transmit data in one of the following modes:

- Single-Buffer (Normal) Mode
- Multi-Buffered or Buffered SCI Mode

28.5.2.1 Transmitting Data in Single-Buffer Mode

Single-buffer mode is selected when the MBUF MODE bit in SCIGCR1 is cleared to 0. In this mode, SCI waits for data to be written to SCITD, transfers it to SCITXSHF, and transmits the data. The TXRDY and TX EMPTY bits indicate the status of the transmit buffers. That is, when the transmitter is ready for data to be written to SCITD, the TXRDY bit is set. Additionally, if both SCITD and SCITXSHF are empty, then the TX EMPTY bit is also set.

You can transmit data by:

1. Polling Transmit Ready Flag
2. Transmit Interrupt
3. DMA

In polling method, software can poll for the TXRDY bit to go high before writing the data to the SCITD register. The CPU is unnecessarily overloaded by selecting the polling method. To avoid this, you can use either the interrupt or DMA method. To use the interrupt method, the SET TX INT bit is set. To use the DMA method, the SET TX DMA bit is set. Either an interrupt or a DMA request is generated the moment the TXRDY bit is set. When the SCI has completed transmission of all pending frames, the SCITXSHF register and SCITD are empty, the TXRDY bit is set, and an interrupt/DMA request is generated, if enabled. Because all data has been transmitted, the interrupt/DMA request should be halted. This can either be done by disabling the transmit interrupt (CLR TX INT) / DMA request (CLR TX DMA bit) or by disabling the transmitter (clear TXENA bit).

NOTE: The TXRDY flag cannot be cleared by reading the corresponding interrupt offset in the SCIINTVECT0 or SCIINTVECT1 register.

28.5.2.2 Transmitting Data in Multi-Buffer Mode

Multi-buffer mode is selected when the MBUF MODE bit in SCIGCR1 is set to 1. Like single-buffer mode, you can use the polling, interrupt, or DMA method to write the data to be transmitted. The transmitted data has to be written to the SCITD registers. SCI waits for data to be written to the SCITD register and transfers the programmed number of bytes to SCITXSHF to transmit one by one automatically.

28.6 SCI Low-Power Mode

The SCI/LIN can be put in either local or global low-power mode. Global low-power mode is asserted by the system and is not controlled by the SCI/LIN. During global low-power mode, all clocks to the SCI/LIN are turned off so the module is completely inactive.

Local low-power mode is asserted by setting the POWERDOWN bit; setting this bit stops the clocks to the SCI/LIN internal logic and the module registers. Setting the POWERDOWN bit causes the SCI to enter local low-power mode and clearing the POWERDOWN bit causes SCI/LIN to exit from local low-power mode. All the registers are accessible during local power-down mode as any register access enables the clock to SCI for that particular access alone.

The wake-up interrupt is used to allow the SCI to exit low-power mode automatically when a low level is detected on the LINRX pin and also this clears the POWERDOWN bit. If wake-up interrupt is disabled, then the SCI/LIN immediately enters low-power mode whenever it is requested and also any activity on the LINRX pin does not cause the SCI to exit low-power mode.

NOTE: Enabling Local Low-Power Mode During Receive and Transmit

If the wake-up interrupt is enabled and low-power mode is requested while the receiver is receiving data, then the SCI immediately generates a wake-up interrupt to clear the powerdown bit and prevents the SCI from entering low-power mode and thus completes the current reception. Otherwise, if the wake-up interrupt is disabled, then the SCI completes the current reception and then enters the low-power mode.

28.6.1 Sleep Mode for Multiprocessor Communication

When the SCI receives data and transfers that data from SCIRXSHF to SCIRD, the RXRDY bit is set and if RX INT ENA is set, the SCI also generates an interrupt. The interrupt triggers the CPU to read the newly received frame before another one is received. In multiprocessor communication modes, this default behavior may be enhanced to provide selective indication of new data. When SCI receives an address frame that does not match its address, the device can ignore the data following this non-matching address until the next address frame by using sleep mode. Sleep mode can be used with both idle-line and address-bit multiprocessor modes.

If sleep mode is enabled by the SLEEP bit, then the SCI transfers data from SCIRXSHF to SCIRD only for address frames. Therefore, in sleep mode, all data frames are assembled in the SCIRXSHF register without being shifted into the SCIRD and without initiating a receive interrupt or DMA request. Upon reception of an address frame, the contents of the SCIRXSHF are moved into SCIRD, and the software must read SCIRD and determine if the SCI is being addressed by comparing the received address against the address previously set in the software and stored somewhere in memory (the SCI does not have hardware available for address comparison). If the SCI is being addressed, the software must clear the SLEEP bit so that the SCI will load SCIRD with the data of the data frames that follow the address frame.

When the SCI has been addressed and sleep mode has been disabled (in software) to allow the receipt of data, the SCI should check the RXWAKE bit (SCIFLR.12) to determine when the next address has been received. This bit is set to 1 if the current value in SCIRD is an address and set to 0 if SCIRD contains data. If the RXWAKE bit is set, then software should check the address in SCIRD against its own address. If it is still being addressed, then sleep mode should remain disabled. Otherwise, the SLEEP bit should be set again.

Following is a sequence of events typical of sleep mode operation:

- The SCI is configured and both sleep mode and receive actions are enabled.
- An address frame is received and a receive interrupt is generated.
- Software compares the received address frame against that set by software and determines that the SCI is not being addressed, so the value of the SLEEP bit is not changed.
- Several data frames are shifted into SCIRXSHF, but no data is moved to SCIRD and no receive interrupts are generated.
- A new address frame is received and a receive interrupt is generated.
- Software compares the received address frame against that set by software and determines that the SCI is being addressed and clears the SLEEP bit.
- Data shifted into SCIRXSHF is transferred to SCIRD, and a receive interrupt is generated after each data frame is received.
- In each interrupt routine, software checks RXWAKE to determine if the current frame is an address frame.
- Another address frame is received, RXWAKE is set, software determines that the SCI is not being addressed and sets the SLEEP bit back to 1. No receive interrupts are generated for the data frames following this address frame.

By ignoring data frames that are not intended for the device, fewer interrupts are generated. These interrupts would otherwise require CPU intervention to read data that is of no significance to this specific device. Using sleep mode can help free some CPU resources.

Except for the RXRDY flag, the SCI continues to update the receiver status flags (see [Table 28-12](#)) while sleep mode is active. In this way, if an error occurs on the receive line, an application can immediately respond to the error and take the appropriate corrective action.

Because the RXRDY bit is not updated for data frames when sleep mode is enabled, the SCI can enable sleep mode and use a polling algorithm if desired. In this case, when RXRDY is set, software knows that a new address has been received. If the SCI is not being addressed, then the software should not change the value of the SLEEP bit and should continue to poll RXRDY.

28.7 LIN Communication Formats

The SCI/LIN module can be used in LIN mode or SCI mode. The enhancements for baud generation, DMA controls and additional receive/transmit buffers necessary for LIN mode operation are also part of the enhanced buffered SCI module. LIN mode is selected by enabling LIN MODE bit in SCIGCR1 register.

NOTE: The SCI/LIN is built around the SCI platform and uses a similar sampling scheme: 16 samples for each bit with majority vote on samples 8, 9, and 10.

The SCI/LIN control registers are located at the SCI/LIN base address. For a detailed description of each register, see [Section 28.13](#).

28.7.1 LIN Standards

For compatibility with LIN2.0 standard the following additional features are implemented over LIN1.3:

- i. Support for LIN 2.0 checksum
- ii. Enhanced synchronizer FSM support for frame processing
- iii. Enhanced handling of extended frames
- iv. Enhanced baudrate generator
- v. Update wakeup/go to sleep

The LIN module covers the CPU performance-consuming features, defined in the *LIN Specification Package* Revision 1.3 and 2.0 by hardware.

The Master Mode of LIN module is compatible with LIN 2.1 standard.

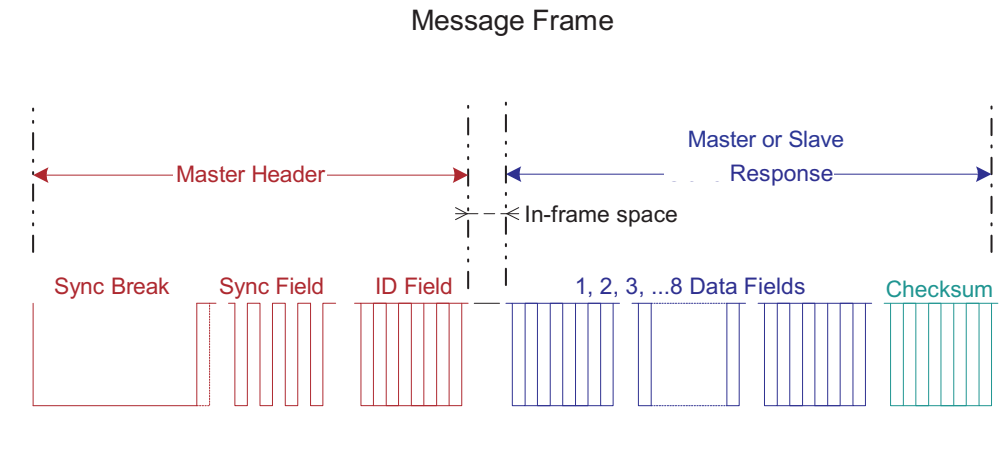
28.7.2 Message Frame

The LIN protocol defines a message frame format, illustrated in Figure 28-12. Each frame includes one master header, one response, one in-frame response space, and inter-byte spaces. In-frame-response and inter-byte spaces may be 0.

There is no arbitration in the definition of the LIN protocol; therefore, multiple slave nodes responding to a header might be detected as an error.

The LIN bus is a single channel wired-AND. The bus has a binary level: either dominant for a value of 0, or recessive for a value of 1.

Figure 28-12. LIN Protocol Message Frame Format: Master Header and Slave Response

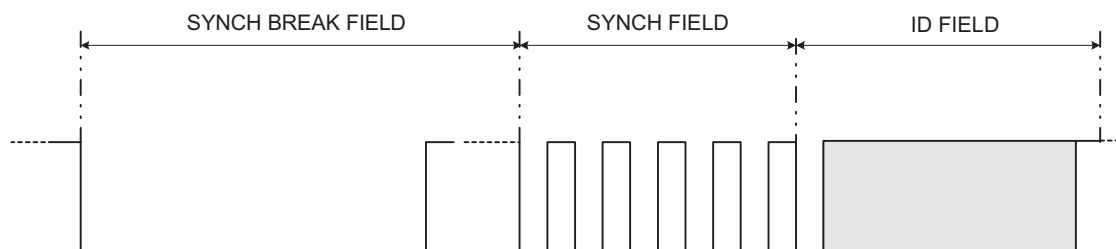


28.7.2.1 Message Header

The header of a message is initiated by a master (see Figure 28-13) and consists of a three field-sequence:

- The synch break field signaling the beginning of a message
- The synch field conveying bit rate information of the LIN bus
- The ID field denoting the content of a message

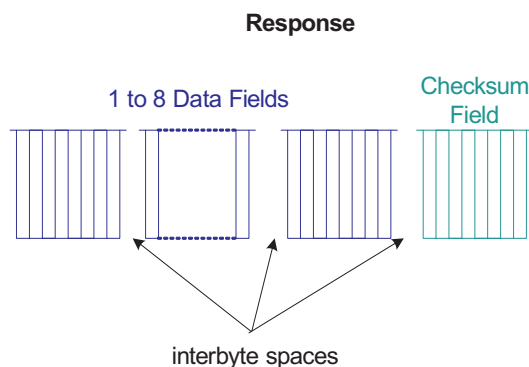
Figure 28-13. Header 3 Fields: Synch Break, Synch, and ID



28.7.2.2 Response

The format of the response is as illustrated in [Figure 28-14](#). There are two types of fields in a response: data and checksum. The data field consists of exactly one data byte, one start bit, and one stop bit, for a total of 10 bits. The LSB is transmitted first. The checksum field consists of one checksum byte, one start bit and one stop bit. The checksum byte is the inverted modulo-256 sum over all data bytes in the data fields of the response.

Figure 28-14. Response Format of LIN Message Frame



The format of the response is a stream of N data fields and one checksum field. Typically N is from 1 to 8, with the exception of the extended command frames ([Section 28.7.6](#)). The length N of the response is indicated either with the optional length control bits of the ID Field (this is used in standards earlier than LIN 1.x); see [Table 28-5](#), or by LENGTH value in SCIFORMAT[18:16] register; see [Table 28-6](#). The SCI/LIN module supports response lengths from 1 to 8 bytes in compliance with LIN 2.0.

Table 28-5. Response Length Info Using IDBYTE Field Bits [5:4] for LIN Standards Earlier than 1.3

ID5	ID4	Number of Data bytes
0	0	2
0	1	2
1	0	4
1	1	8

Table 28-6. Response Length with SCIFORMAT[18:16] Programming

SCIFORMAT[18:16]	No. of Bytes
000	1
001	2
010	3
011	4
100	5
101	6
110	7
111	8

28.7.3 Synchronizer

The synchronizer has three major functions in the messaging between master and slave nodes. It generates the master header data stream, it synchronizes to the LIN bus for responding, and it locally detects timeouts. A bit rate is programmed using the prescalers in the BRS register to match the indicated LIN_speed value in the LIN description file.

The LIN synchronizer will perform the following functions: master header signal generation, slave detection and synchronization to message header with optional baud rate adjustment, response transmission timing and timeout control.

The LIN synchronizer is capable of detecting an incoming break and initializing communication at all times.

28.7.4 Baud Rate

The transmission baud rate of any node is configured by the CPU at the beginning; this defines the bit time T_{bit} . The bit time is derived from the fields P and M in the baud rate selection register (BRS). There is an additional 3-bit fractional divider value, field U in the BRS register, which further fine tunes the data field baud rate.

The ranges for the prescaler values in the BRS are:

$$P = 0, 1, 2, 3, \dots, 2^{24} - 1$$

$$M = 0, 1, 2, \dots, 15$$

$$U = 0, 1, 2, 3, 4, 5, 6, 7$$

The P, M, and U values in the BRS are user programmable. The P and M dividers could be used for both SCI mode and LIN mode to select a baud rate. The U value is an additional 3-bit value determining that “**a** TVCLK” (with **a** = 0,1) is added to each T_{bit} as explained in [Section 28.7.4.2](#). If the ADAPT bit is set and the LIN slave is in adaptive baud rate mode, then all these divider values are automatically obtained during header reception when the synchronization field is measured.

The LIN protocol defines baud rate boundaries as follows:

$$1\text{kHz} \leq F_{LINCLK} \leq 20\text{kHz}$$

All transmitted bits are shifted in and out at T_{bit} periods.

28.7.4.1 Fractional Divider

The M field in the BRS modifies the integer prescaler P for fine tuning of the baud rate. The M value adds in increments of 1/16 of the P value.

The bit time, T_{bit} is expressed in terms of the VCLK period T_{VCLK} as follows:

For all P other than 0, and all M,

$$T_{bit} = 16 \left(P + 1 + \frac{M}{16} \right) T_{VCLK} \quad (45)$$

For P= 0 : $T_{bit} = 32T_{VCLK}$

Therefore, the LINCLK frequency is given by:

$$F_{LINCLK} = \frac{F_{VCLK}}{16(P + 1 + \frac{M}{16})} \quad \text{For all } P \text{ other than zero}$$

$$F_{LINCLK} = \frac{F_{VCLK}}{32} \quad \text{For } P = 0 \quad (46)$$

28.7.4.2 Superfractional Divider

The superfractional divider scheme applies to the following modes:

- LIN master mode (synch field + identifier field + response field + checksum field)
- LIN slave mode (response field + checksum field)

28.7.4.3 Superfractional Divider In LIN Mode

Building on the 4-bit fractional divider M (BRS[27:24], the super fractional divider uses an additional 3-bit modulating value, illustrated in Table 28-7. The sync field (0x55), the identifier field and the response field can all be seen as 8-bit data bytes flanked by a start bit and a stop bit. The bits with a 1 in the table will have an additional VCLK period added to their T_{bit} .

Table 28-7. Superfractional Bit Modulation for LIN Master Mode and Slave Mode⁽¹⁾

BRS[30:28]	Start Bit	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]	Stop Bit
0h	0	0	0	0	0	0	0	0	0	0
1h	1	0	0	0	0	0	0	0	1	0
2h	1	0	0	0	1	0	0	0	1	0
3h	1	0	1	0	1	0	0	0	1	0
4h	1	0	1	0	1	0	1	0	1	0
5h	1	1	1	0	1	0	1	0	1	1
6h	1	1	1	0	1	1	1	0	1	1
7h	1	1	1	1	1	1	1	0	1	1

⁽¹⁾

1. In LIN master mode, bit modulation applies to synch field + identifier field + response field
2. In LIN slave mode, bit modulation applies to identifier field + response field

The baud rate will vary over a LIN data field to average according to the BRS[30:28] value by a d fraction of the peripheral internal clock: $0 < d < 1$.

The instantaneous bit time is expressed in terms of T_{VCLK} as follows:

For all P other than 0, and all M and d (0 or 1),

$$T^{i}bit = \left[16 \left(P + 1 + \frac{M}{16} \right) + d \right] T_{VCLK} \quad (47)$$

For P = 0, $T_{bit} = 32T_{VCLK}$

The averaged bit time is expressed in terms of T_{VCLK} as follows:

For all P other than 0, and all M and d ($0 < d < 1$),

$$T^{a}bit = \left[16 \left(P + 1 + \frac{M}{16} \right) + d \right] T_{VCLK} \quad (48)$$

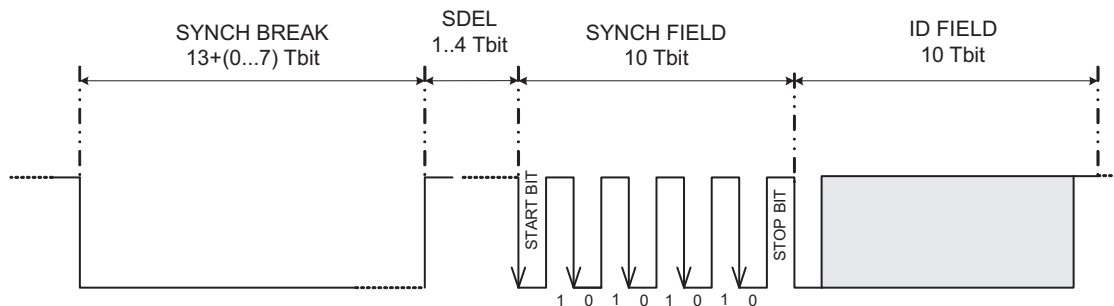
For P = 0, $T_{bit} = 32T_{VCLK}$

With the superfractional divider, a LIN baud rate of 20 kbps is achievable with an internal clock VCLK of 726 kHz. Furthermore, a rate of 400 kbps is achievable with an VCLK of 14.6 MHz.

28.7.5 Header Generation

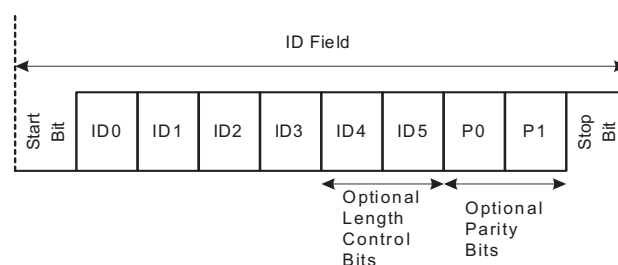
Automatic generation of the LIN protocol header data stream is supported without CPU interaction. The CPU or the DMA will trigger a message header generation and the LIN state machine will handle the generation itself. A master node initiates header generation on CPU or DMA writes to the IDBYTE in the LINID register. The header is always sent by the master to initiate a LIN communication and consists of three fields: break field, synchronization field, and identification field, as seen in Figure 28-15.

Figure 28-15. Message Header in Terms of T_{bit}



- The break field consists of two components:
 - The synchronization break (SYNCH BREAK) consists of a minimum of 13 (dominant) low bits to a maximum of 20 dominant bits. The synch break length may be extended from the minimum with the 3-bit SBREAK value in the LINCOMP register.
 - The synchronization break delimiter (SDEL) consists of a minimum of 1 (recessive) high bit to a maximum of 4 recessive bits. The delimiter marks the end of the synchronization break field. The synch break delimiter length depends on the 2-bit SDEL value in the LINCOMP register.
- The synchronization field (SYNCH FIELD) consists of one start bit, byte 0x55, and a stop bit. It is used to convey T_{bit} information and resynchronize LIN bus nodes.
- The identifier field's ID byte may use six bits as an identifier, with optional length control (see *Note: Optional Control Length Bits*), and two optional bits as parity of the identifier. The identifier parity is used and checked if the PARITY ENA bit is set. If length control bits are not used, then there can be a total of 64 identifiers plus parity. If neither length control or parity are used there can be up to 256 identifiers. See Figure 28-16 for an illustration of the ID field.

Figure 28-16. ID Field



NOTE: Optional Control Length Bits

The control length bits only apply to LIN standards prior to LIN 1.3. IDBYTE field conveys response length information if compliant to standards earlier than LIN1.3. The SCIFORMAT register stores the length of the response for later versions of the LIN protocol.

NOTE: If the BLIN module, configured as Slave in multi-buffer mode, is in the process of transmitting data while a new header comes in, the module might end up in responding with the data from the previous interrupted response (not the data corresponding to the new ID). To avoid this scenario the following procedure could be used:

1. Check for the Bit Error (BE) during the response transmission. If the BE flag is set, this indicates that a collision has happened on the LIN bus (here because of the new Synch Break).
2. In the Bit Error ISR, configure the TD0 and TD1 registers with the next set of data to be transmitted on a TX Match for the incoming ID. Before writing to TD0/TD1 make sure that there was not already an update because of a Bit Error; otherwise TD0/TD1 might be written twice for one ID.
3. Once the complete ID is received, based on the match, the newly configured data will be transmitted by the node.

28.7.5.1 Event Triggered Frame Handling Proposal

The LIN 2.0 protocol uses event-triggered frames that may occasionally cause collisions. Event-triggered frames have to be handled in software.

If no slave answers to an event triggered frame header, the master node will set the NRE flag, and a NRE interrupt will occur if enabled. If a collision occurs, a frame error and checksum error may arise before the NRE error. Those errors are flagged and the appropriate interrupts will occur, if enabled.

Frame errors and checksum errors depend on the behavior and synchronization of the responding slaves. If the slaves are totally synchronized and stop transmission once the collision occurred, it is possible that only the NRE error is flagged despite the occurrence of a collision. To detect if there has been a reception of one byte before the NRE error is flagged, the BUS BUSY flag can be used as an indicator.

The bus busy flag is set on the reception of the first bit of the header and remains set until the header reception is complete, and again is set on the reception of the first bit of the response. In the case of a collision the flag is cleared in the same cycle as the NRE flag is set.

Software could implement the following sequence:

- Once the reception of the header is done (poll for RXID flag), wait for the bus busy flag to get set or NRE flag to get set.
- If bus busy flag is not set before NRE flag, then it is a true no response case (no data has been transmitted onto the bus).
- If bus busy flag gets set, then wait for NRE flag to get set or for successful reception. If NRE flag is set, then in this case a collision has occurred on the bus.

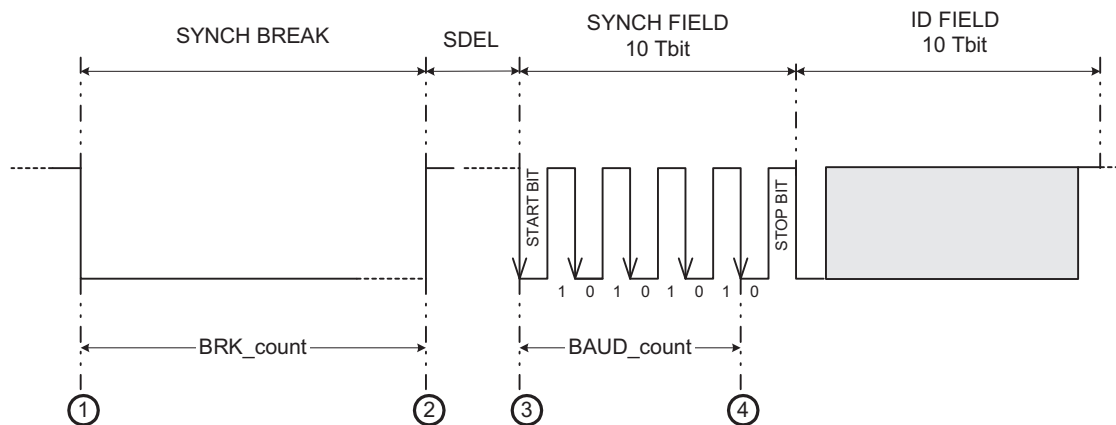
Even in the case of a collision, the received (corrupted) data is accessible in the RX buffers; registers LINRD0 and LINRD1.

28.7.5.2 Header Reception and Adaptive Baud Rate

A slave node baud rate can optionally be adjusted to the detected bit rate as an option to the LIN module. The adaptive baud rate option is enabled by setting the ADAPT bit. During header reception, a slave measures the baud rate during detection of the synch field. If ADAPT bit is set, then the measured baud rate is compared to the slave node's programmed baud rate and adjusted to the LIN bus baud rate if necessary.

The LIN synchronizer determines two measurements: BRK_count and BAUD_count (Figure 28-17). These values are always calculated during the Header reception for synch field validation (Figure 28-18).

Figure 28-17. Measurements for Synchronization



By measuring the values BRK_count and BAUD_count, a valid synch break sequence can be detected as described in Figure 28-18. The four numbered events in Figure 28-17 signal the start/stop of the synchronizer counter. The synchronizer counter uses VCLK as the time base.

The synchronizer counter is used to measure the synch break relative to the detecting node T_{bit} . For a slave node receiving the synch break, a threshold of $11 T_{bit}$ is used as required by the LIN protocol. For detection of the dominant data stream of the synch break, the synchronizer counter is started on a falling edge and stopped on a rising edge of the LINRX. On detection of the synch break delimiter, the synchronizer counter value is saved and then reset.

On detection of five consecutive falling edges, the BAUD_count is measured. Bit timing calculation and consistency to required accuracy is implemented following the recommendations of LIN revision 2.0. A slave node can calculate a single T_{bit} time by division of BAUD_count by 8. In addition, for consistency between the detected edges the following is evaluated:

$$BAUD_count + BAUD_count \gg 2 + BAUD_count \gg 3 \leq BRK_count$$

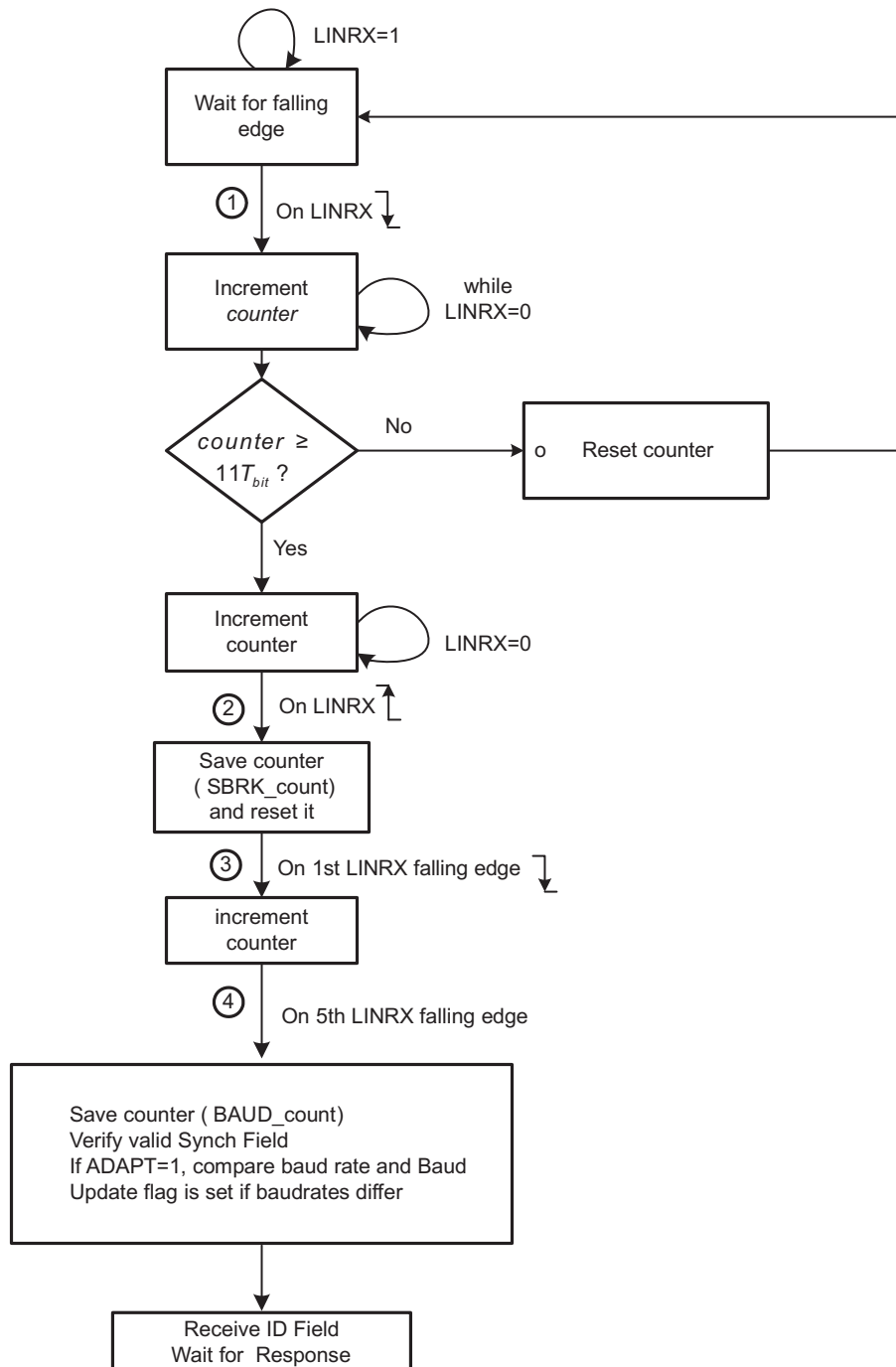
The BAUD_count value is shifted 3 times to the right and rounded using the first insignificant bit to obtain a T_{bit} unit. If the ADAPT bit is set, then the detected baud rate is compared to the programmed baud rate.

During the header reception processing as illustrated in Figure 28-18, if the measured BRK_count value is less than $11 T_{bit}$, the synch break is not valid according to the protocol for a fixed rate. If the ADAPT bit is set, then the MBRS register is used for measuring BRK_count and BAUD_count values and automatically adjusts to any allowed LIN bus rate (refer to LIN Specification Package 2.0).

NOTE: In adaptive mode, the MBRS divider should be set to allow a maximum baud rate that is not more than 10% above the expected operating baud rate in the LIN network. Otherwise, a 0x00 data byte could mistakenly be detected as a synch break.

The break-threshold relative to the slave node is $11 T_{bit}$. The break is $13 T_{bit}$ as specified in LIN version 1.3.

Figure 28-18. Synchronization Validation Process and Baud Rate Adjustment



If the synch field is not detected within the given tolerances, the inconsistent-synch-field-error (ISFE) flag will be set. An ISFE interrupt will be generated, if enabled by its respective bit in the SCISSETINT register. The ID byte should be received after the synch field validation was successful. Any time a valid break (larger than $11 T_{bit}$) is detected, the receiver's state machine should reset to reception of this new frame. This reset condition is only valid during response state, not if an additional synch break occurs during header reception.

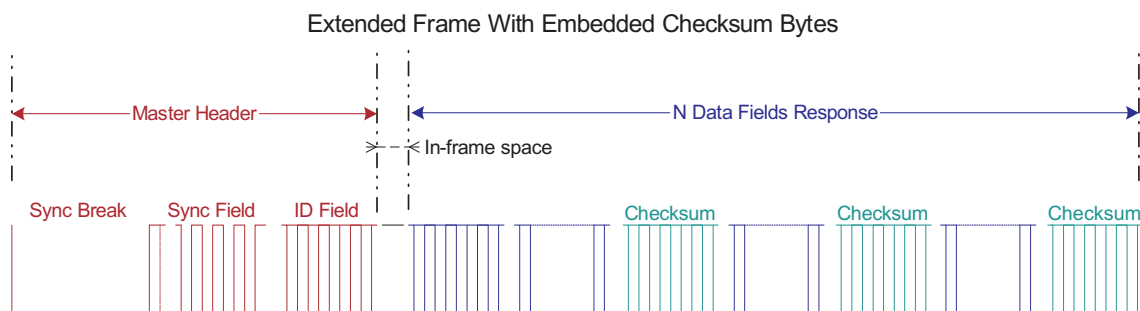
NOTE: When an inconsistent synch field (ISFE) error occurs, suggested action for the application is to reset the SWnRST bit and set the SWnRST bit to make sure that the internal state machines are back to their normal states

28.7.6 Extended Frames Handling

The LIN protocol 2.0 and prior includes two extended frames with identifiers 62 (user defined) and 63 (reserved extended). The response data length of the user-defined frame (ID 62, or 0x3E) is unlimited. The length for this identifier will be set at network configuration time to be shared with the LIN bus nodes.

Extended frame communication is triggered on reception of a header with identifier 0x3E; see [Figure 28-19](#). Once the extended frame communication is triggered, unlike normal frames, this communication needs to be stopped before issuing another header. To stop the extended frame communication the STOP EXT FRAME bit must be set.

Figure 28-19. Optional Embedded Checksum in Response for Extended Frames



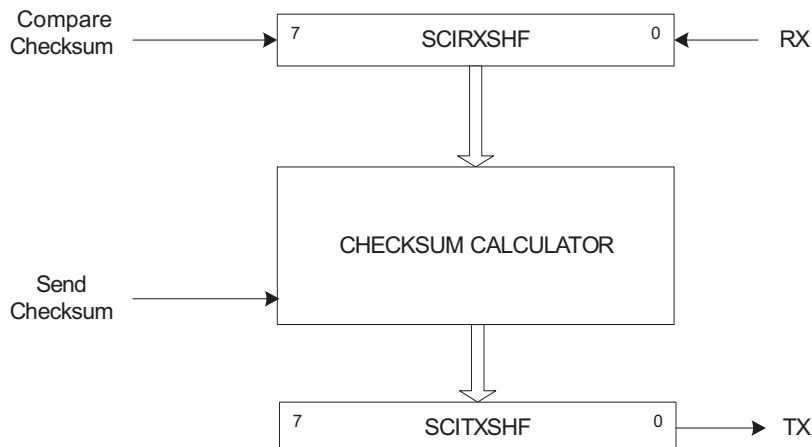
An ID interrupt will be generated (if enabled and there is a match) on reception of ID 62 (0x3E). This interrupt allows the CPU using a software counter to keep track of the bytes that are being sent out and decides when to calculate and insert a checksum byte (recommended at periodic rates). To handle this procedure, SC bit is used. A write to the send checksum bit SC will initiate an automatic send of the checksum byte. The last data field should always be a checksum in compliance with the LIN protocol.

The periodicity of the checksum insertion, defined at network configuration time, is used by the receiving node to evaluate the checksum of the ongoing message, and has the benefit of enhanced reliability.

For the sending node, the checksum is automatically embedded each time the send checksum bit SC is set. For the receiving node, the checksum is compared each time the compare checksum bit CC is set; see [Figure 28-20](#).

NOTE: The LIN 2.0 enhanced checksum does not apply to the reserved identifiers. The reserved identifiers always use the classic checksum.

Figure 28-20. Checksum Compare and Send for Extended Frames



28.7.7 Timeout Control

Any LIN node listening to the bus and expecting a response initiated from a master node could flag a no-response error timeout event. The LIN protocol defines four types of timeout events, which are all handled by the hardware of the LIN module. The four LIN protocol events are:

- No-response timeout error
- Bus idle detection
- Timeout after wakeup signal
- Timeout after three wakeup signals

28.7.7.1 No-Response Error (NRE)

The no-response error will occur when any node expecting a response waits for $T_{\text{FRAME_MAX}}$ time and the message frame is not fully completed within the maximum length allowed, $T_{\text{FRAME_MAX}}$. After this time a no-response error (NRE) is flagged in the NRE bit of the SCIFLR register. An interrupt is triggered if enabled.

As specified in the LIN 1.3 standard, the minimum time to transmit a frame is:

$$T_{\text{FRAME_MIN}} = T_{\text{HEADER_MIN}} + T_{\text{DATA_FIELD}} + T_{\text{CHECKSUM_FIELD}}$$

$$= 44 + 10N$$

where N = number of data fields.

And the maximum time frame is given by:

$$T_{\text{FRAME_MAX}} = T_{\text{FRAME_MIN}} * 1.4$$

$$= (44 + 10N) * 1.4$$

The timeout value $T_{\text{FRAME_MAX}}$ is derived from the N number of data fields value. The N value is either embedded in the header's ID field for messages or is part of the description file. In the latter case, the 3-bit CHAR value in SCIFORMAT register, will indicate the value for N .

NOTE: The length coding of the ID field does not apply to two extended frame identifiers, ID fields of 0x3E (62) and 0x3F (63). In these cases, the ID field can be followed by an arbitrary number of data byte fields. Also, the LIN 2.0 protocol specification mentions that ID field 0x3F (63) cannot be used. For these two cases, the NRE will not be handled by the LIN controller hardware.

Table 28-8. Timeout Values in T_{bit} Units

N	T_{DATA_FIELD}	T_{FRAME_MIN}	T_{FRAME_MAX}
1	10	54	76
2	20	64	90
3	30	74	104
4	40	84	118
5	50	94	132
6	60	104	146
7	70	114	160
8	80	124	174

28.7.7.2 Bus Idle Detection

The second type of timeout can occur when a node detects an inactive LIN bus: no transitions between recessive and dominant values are detected on the bus. This happens after a minimum of 4 s (this is 80,000 F_{LINCLK} cycles with the fastest bus rate of 20 kbps). If a node detects no activity in the bus as the TIMEOUT bit is set, then it can be assumed that the LIN bus is in sleep mode. Application software can use the Timeout flag to determine when the LIN bus is inactive and put the LIN into sleep mode by writing the POWERDOWN bit.

NOTE: After the timeout was flagged, a software RESET should be asserted before entering Low-Power Mode. This is required to reset the receiver in case that an incomplete frame was on the bus before the idle period.

28.7.7.3 Timeout after Wakeup Signal and Timeout after Three Wakeup Signals

The third and fourth types of timeout are related to the wakeup signal. A node initiating a wakeup should expect a header from the master within a defined amount of time: timeout after wakeup signal. See [Section 28.11.3](#) for more details.

28.7.8 TXRX Error Detector (TED)

The following sources of error are detected by the TXRX error detector logic (TED). The TED logic consists of a bit monitor, an ID parity checker, and a checksum error. The following errors are detected:

- Bit errors (BE)
- Physical bus errors (PBE)
- Identifier parity errors (PE)
- Checksum errors (CE)

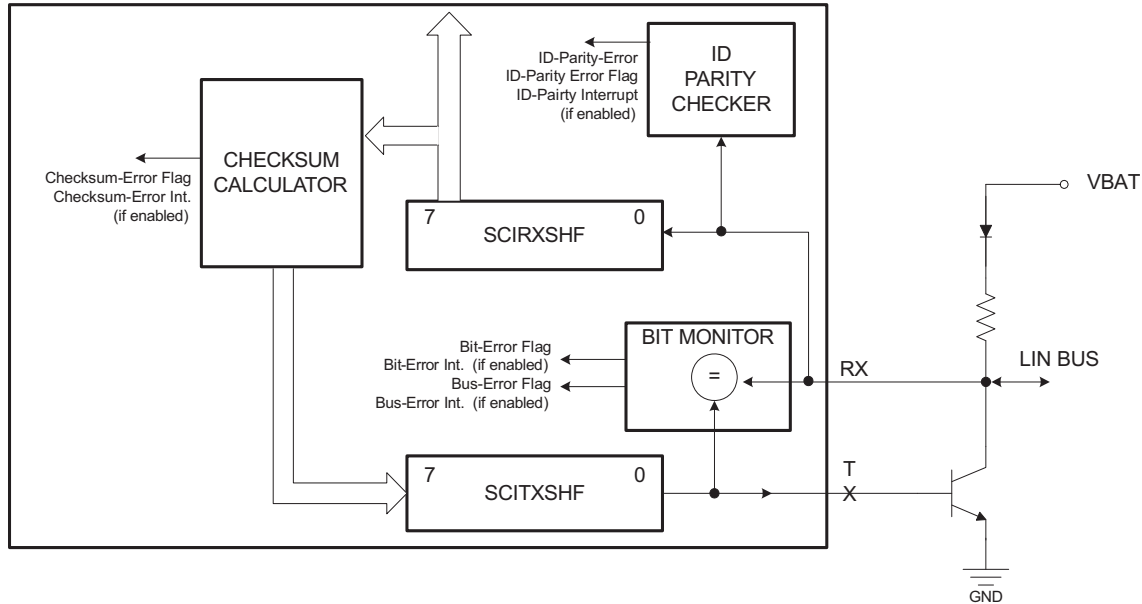
All of these errors (BE, PBE, PE, CE) are flagged. An interrupt for the flagged errors will be generated if enabled. A message is valid for both the transmitter and the receiver if there is no error detected until the end of the frame.

28.7.8.1 Bit Errors

A bit error (BE) is detected at the bit time when the bit value that is monitored is different from the bit value that is sent. A bit error is indicated by the BE flag in SCIFLR. After signaling a BE, the transmission is aborted no later than the next byte. The bit monitor ensures that the transmitted bit in LINTX is the correct value on the LIN bus by reading back on the LINRX pin as shown in Figure 28-21.

NOTE: If BE Occurs due to New Header reception during a Slave Response, NRE/TIMEOUT flag will not be set for the new Frame.

Figure 28-21. TXRX Error Detector



28.7.8.2 Physical Bus Errors

A Physical Bus Error (PBE) has to be detected by a master if no valid message can be generated on the bus (Bus shorted to GND or VBAT). The bit monitor detects a PBE during the header transmission, if no Synch Break can be generated (for example, because of a bus shortage to VBAT) or if no Synch break Delimiter can be generated (for example, because of a bus shortage to GND). Once the Synch Break Delimiter was validated, all other deviations between the monitored and the sent bit value are flagged as Bit Errors (BE) for this frame.

28.7.8.3 ID Parity Errors

If parity is enabled, an ID parity error (PE) is detected if any of the two parity bits of the sent ID byte are not equal to the calculated parity on the receiver node. The two parity bits are generated using the following mixed parity algorithm.

$$P0 = ID0 \oplus ID1 \oplus ID2 \oplus ID4 \text{ (even Parity)}$$

$$P1 = ID1 \oplus ID3 \oplus ID4 \oplus ID5 \text{ (odd Parity)}$$

(49)

If an ID-parity error is detected, the ID-parity error is flagged, and the received ID is not valid. See Section 28.7.9 for details.

28.7.8.4 Checksum Errors

A checksum error (CE) is detected and flagged at the receiving end if the calculated modulo-256 sum over all received data bytes (including the ID byte if it is the enhanced checksum type) plus the checksum byte does not result in 0xFF. The modulo-256 sum is calculated over each byte by adding with carry, where the carry bit of each addition is added to the LSB of its resulting sum.

For the transmitting node, the checksum byte sent at the end of a message is the inverted sum of all the data bytes (see [Figure 28-22](#)) for classic checksum implementation. The checksum byte is the inverted sum of the identifier byte and all the data bytes (see [Figure 28-23](#)) for the LIN 2.0 compliant enhanced checksum implementation. The classic checksum implementation should always be used for reserved identifiers 60 to 63; therefore, the CTYPE bit will be overridden in this case. For signal-carrying-frame identifiers (0 to 59) the type of checksum used depends on the CTYPE bit.

Figure 28-22. Classic Checksum Generation at Transmitting Node

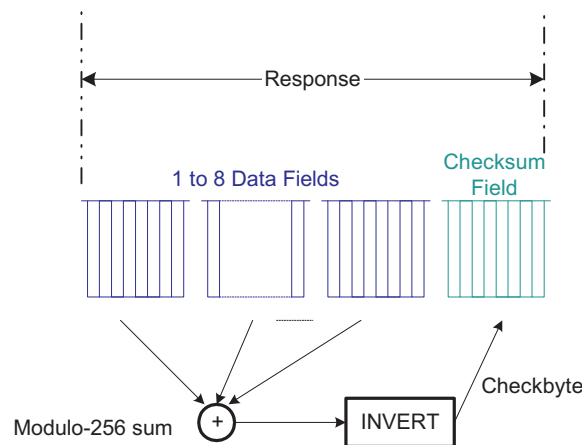
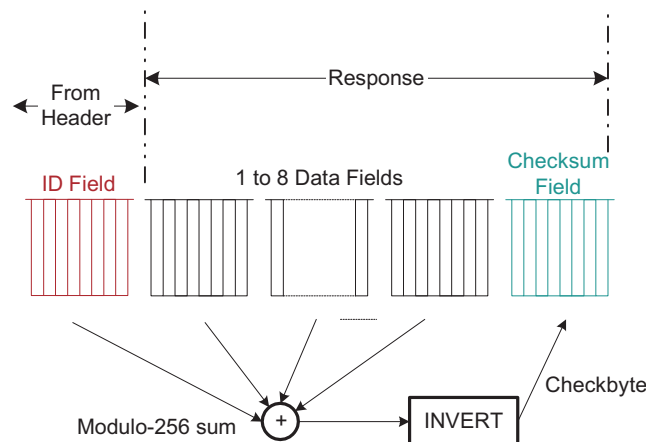


Figure 28-23. LIN 2.0-Compliant Checksum Generation at Transmitting Node



28.7.9 Message Filtering and Validation

Message filtering uses the entire identifier to determine which nodes will participate in a response, either receiving or transmitting a response. Therefore, two acceptance masks are used, as shown in [Figure 28-24](#). During header reception, all nodes filter the ID-Field (ID-Field is the part of the header explained in [Figure 28-16](#)) to determine whether they transmit a response or receive a response for the current message. There are two masks for message ID filtering: one to accept a response reception, the other to initiate a response transmission. See [Figure 28-24](#). All nodes compare the received ID to the identifier stored in the ID-SlaveTask BYTE of the LINID register and use the RX ID MASK and the TX ID MASK fields in the LINMASK register to filter the bits of the identifier that should not be compared.

If there is an RX match with no parity error and the RXENA bit is set, there will be an ID RX flag and an interrupt will be triggered if enabled. If there is a TX match with no parity error and the TXENA bit is set, there will be an ID TX flag and an interrupt will be triggered if enabled in the SCISSETINT register.

The masked bits become don't cares for the comparison. To build a mask for a set of identifiers, an XOR function could be used.

For example, to build a mask to accept IDs 0x26 and 0x25 using LINID[7:0] = 0x20; that is, compare 5 most significant bits (MSBs) and filter 3 least significant bits (LSBs), the acceptance mask could be:

$$(0x26 + 0x25) \oplus 0x20 = 0x07 \quad (50)$$

A mask of all zeros will compare all bits of the received identifier in the shift register with the ID-BYTE in LINID[7:0]. If HGEN CTRL is set to 1, a mask of 0xFF will always cause a match. A mask of all 1s will filter all bits of the received identifier, and thus there will be an ID match regardless of the content of the ID-SlaveTask BYTE field in the LINID register.

NOTE: When the HGEN CTRL bit = 0, the LIN nodes compare the received ID to the ID-BYTE field in the LINID register, and use the RX ID MASK and the TX ID MASK in the LINMASK register to filter the bits of the identifier that should not be compared.

If there is an RX match with no parity error and the RXENA bit is set, there will be an ID RX flag and an interrupt will be triggered if enabled. A mask of all 0s will compare all bits of the received identifier in the shift register with the ID-BYTE field in LINID[7:0]. A mask of all 1s will filter all bits of the received identifier and there will be no match.

During header reception, the received identifier is copied to the Received ID field LINID[23:16]. If there is no parity error and there is either a TX match or an RX match, then the corresponding TX or RX ID flag is set. If the ID interrupt is enabled, then an ID interrupt is generated.

After the ID interrupt is generated, the CPU may read the Received ID field LINID[23:16] and determine what response to load into the transmit buffers.

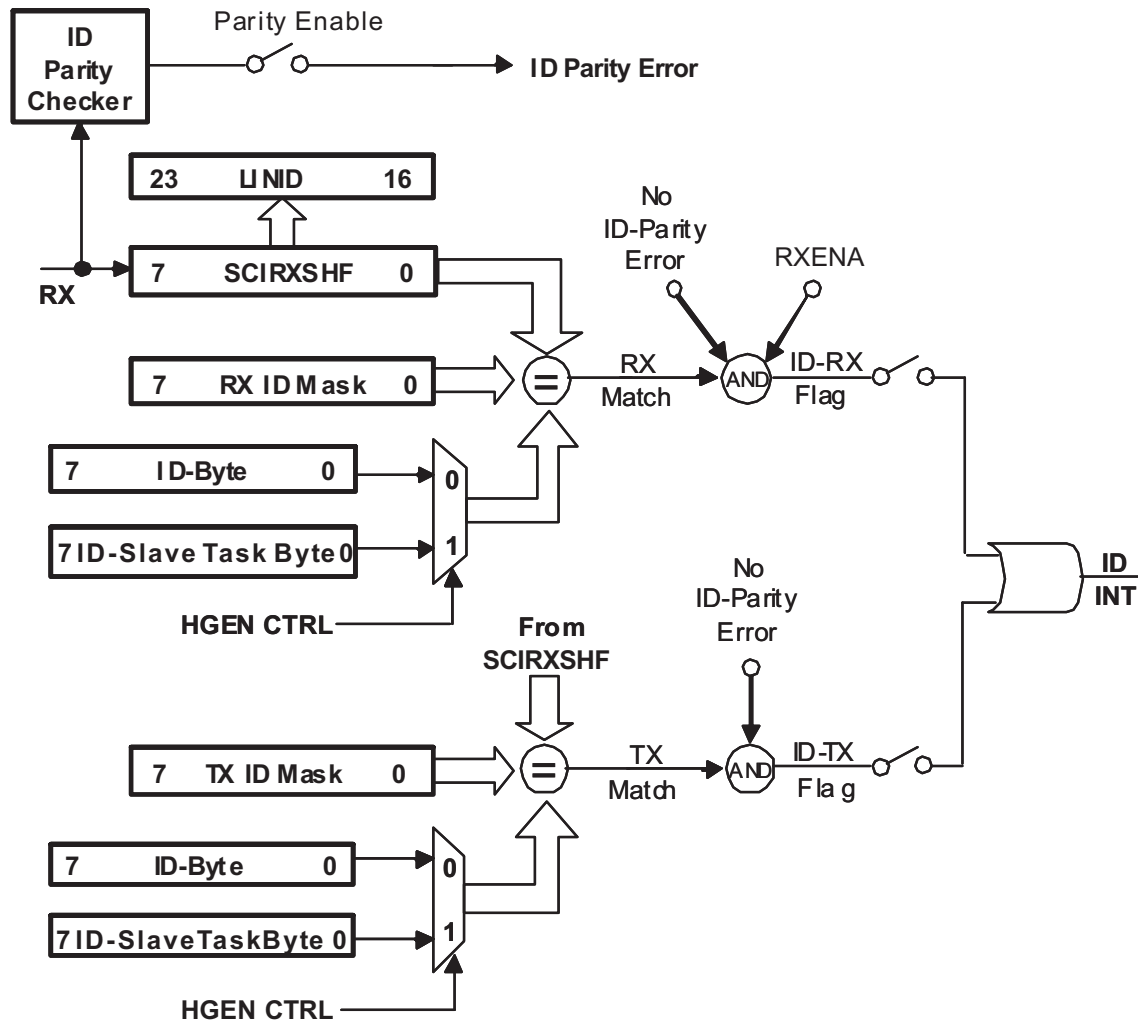
NOTE: When byte 0 is written to TD0 (LINTD0[31:24]), the response transmission is automatically generated.

In multi-buffer mode, the TXRDY flag will be set when all the response data bytes and checksum byte are copied to the shift register SCITXSHF. In non-multi-buffer mode, the TXRDY flag is set each time a byte is copied to the SCITXSHF register, and also for the last byte of the frame after the checksum byte is copied to the SCITXSHF register.

In multi-buffer mode, the TXEMPTY flag is set when both the transmit buffer(s) TDy and the SCITXSHF shift register are emptied and the checksum has been sent. In non multi-buffer mode, TXEMPTY is set each time TD0 and SCITXSHF are emptied, except for the last byte of the frame where the checksum byte must also be transmitted.

If parity is enabled, all slave receiving nodes will validate the identifier using all eight bits of the received ID byte. The SCI/LIN will flag a corrupted identifier if an ID-parity error is detected.

Figure 28-24. ID Reception, Filtering and Validation



28.7.10 Receive Buffers

To reduce CPU load when receiving a LIN N-byte (with N = 1–8) response in interrupt mode or DMA mode, the SCI/LIN module has eight receive buffers. These buffers can store an entire LIN response in the RDy receive buffers.

The checksum byte following the data bytes is validated by the internal checksum calculator. The checksum error (CE) flag indicates a checksum error and a CE interrupt will be generated if enabled in the SCISSETINT register.

The multi-buffer 3-bit counter counts the data bytes transferred from the SCIRXSHF register to the RDy receive buffers if multi-buffer mode is enabled, or to RD0 if multi-buffer mode is disabled. The 3-bit compare register contains the number of data bytes expected to be received. In cases where the ID BYTE field does not convey message length (see *Note: Optional Control Length Bits* in [Section 28.7.5](#)), the LENGTH value, indicates the expected length and is used to load the 3-bit compare register. Whether the length control field or the LENGTH value is used is selectable with the COMM MODE bit.

[Figure 28-8](#) illustrates the receive buffers.

A receive interrupt, and a receive ready RXRDY flag set as well as a DMA request (RXDMA) could occur after receiving a response if there are no response receive errors for the frame (such as, there is no checksum error, frame error, and overrun error). The checksum byte will be compared before acknowledging a reception. A DMA request can be generated for each received byte or for the entire response depending on whether the multi-buffer mode is enabled or not (MBUF MODE bit in the SCIGCR1 register).

NOTE: In multi-buffer mode, following are the scenarios associated with clearing the RXRDY flag bit:

1. The RXRDY flag cannot be cleared by reading the corresponding interrupt offset in the SCIINTVECT0/1 register.
 2. For LENGTH less than or equal to 4, Read to RD0 register will clear the RXRDY flag.
 3. For LENGTH greater than 4, Read to RD1 register will clear the RXRDY flag.
-

28.7.11 Transmit Buffers

To reduce the CPU load when transmitting a LIN N-byte (with N = 1–8) response in interrupt mode or DMA mode, the SCI/LIN module has eight transmit buffers, TD0–TD7 in LINTD0 and LINTD1. With these transmit buffers, an entire LIN response field can be preloaded in the TXy transmit buffers. Optionally, a DMA transfer could be done on a byte-per-byte basis when multi-buffer mode is not enabled (MBUF MODE bit in the SCIGCR1 register).

The multi-buffer 3-bit counter counts the data bytes transferred from the TDy transmit buffers register if multi-buffer mode is enabled, or from TD0 to SCITXSHF if multi-buffer mode is disabled. The 3-bit compare register contains the number of data bytes expected to be transmitted. If the ID field is not used to convey message length (see *Note: Optional Control Length Bits* in [Section 28.7.5](#)), the LENGTH value indicates the expected length and is used instead to load the 3-bit compare register. Whether the length control field or the LENGTH value is used is selectable with the COMM MODE bit.

A transmit interrupt (TX interrupt), and a transmit ready flag (TXRDY flag), and a DMA request (TXDMA) could occur after transmitting a response. A DMA request can be generated for each transmitted byte or for the entire response depending on whether multi-buffer mode is enabled or not (MBUF MODE bit in the SCIGCR1 register).

[Figure 28-9](#) illustrates the transmit buffers.

The checksum byte will be automatically generated by the checksum calculator and sent after the data-fields transmission is finished. The multi-buffer 3-bit counter counts the data bytes transferred from the TDy buffers into the SCITXSHF register.

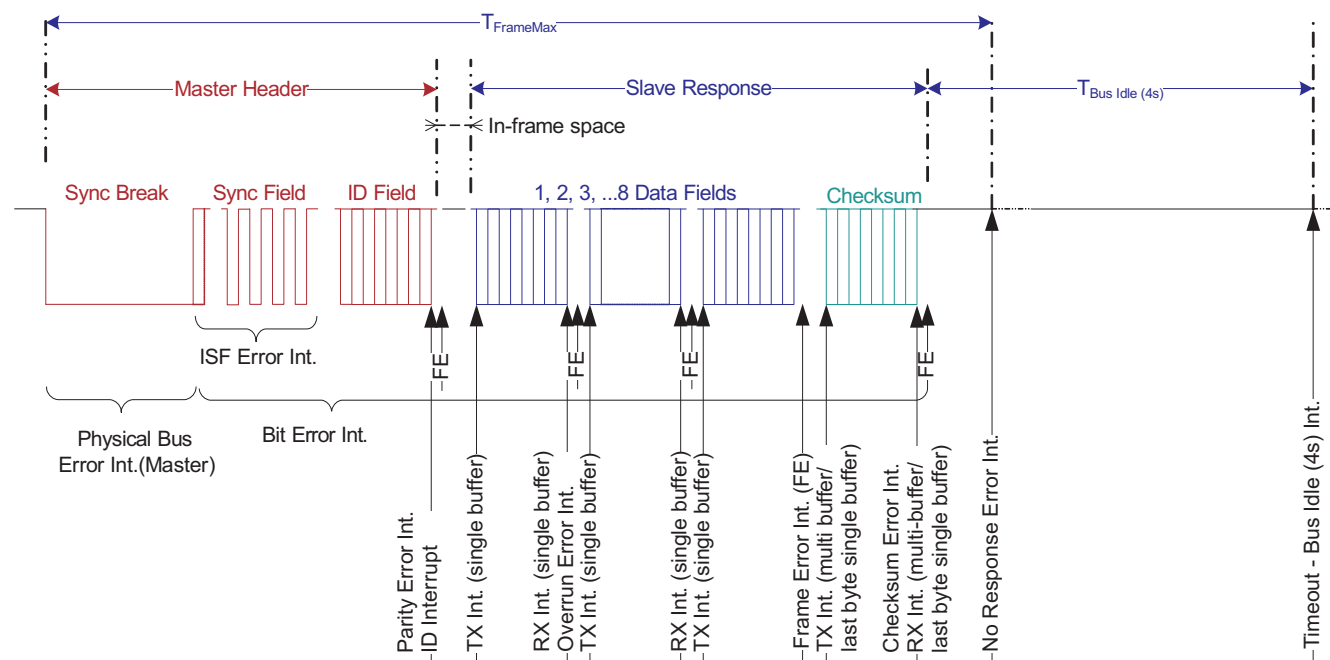
NOTE: The transmit interrupt request can be eliminated until the next series of data is written into the transmit buffers LINTD0 and LINTD1, by disabling the corresponding interrupt via the SCICLRINT register or by disabling the transmitter via the TXENA bit.

28.8 LIN Interrupts

LIN and SCI mode have a common Interrupt block as explained in [Section 28.3](#). There are 16 interrupt sources in the SCI/LIN module, with 8 of them being LIN mode only, as seen in [Table 28-4](#).

A LIN message frame indicating the timing and sequence of the LIN interrupts that could occur is shown in [Figure 28-25](#).

Figure 28-25. LIN Message Frame Showing LIN Interrupt Timing and Sequence



28.9 LIN DMA Interface

LIN DMA Interface uses the SCI DMA interface logic. DMA requests for receive (RXDMA request) and transmit (TXDMA request) are available for the SCI/LIN module. There are two modes for DMA transfers depending on whether multi-buffer mode is enabled or not via the multi-buffer enable control bit (MBUF MODE bit in the SCIGCR1 register).

28.9.1 LIN Receive DMA Requests

In LIN mode, when the multi-buffer option is enabled, if a received response (up to eight data bytes) is transferred to the receive buffers (RDy), then a DMA request is generated. If the multi-buffer option is disabled, then DMA requests will be generated on a byte-per-byte basis until all the expected response data fields are received. This DMA functionality is enabled/disabled by the user using SET RX DMA / CLR RX DMA bits, respectively.

28.9.2 LIN Transmit DMA Requests

In LIN mode with the multi-buffer option enabled, after a transmission (up to eight data bytes stored in the transmit buffer(s) TDy in the LINTD0 and LINTD1 registers), a DMA request is generated in order to reload the transmit buffer for the next transmission. If the multi-buffer option is disabled, then DMA requests will be generated on a byte-per-byte basis until all bytes are transferred. This DMA functionality is enabled/disabled by the user using SET TX DMA / CLR TX DMA bits, respectively.

28.10 LIN Configurations

The following list details the configuration steps that software should perform prior to the transmission or reception of data in LIN mode. As long as the SWnRST bit in the SCIGCR1 register is cleared to 0 the entire time that the LIN is being configured, the order in which the registers are programmed is not important.

- Enable LIN by setting the RESET bit in SCIGCR0 to 1.
- Clear the SWnRST bit to 0 before LIN is configured.
- Select the desired frame format by programming the SCIGCR1 register.
- Set both the RX FUNC and TX FUNC bits in SCIPIO0 to 1 to configure the LINRX and LINTX pins for LIN functionality.
- Set the LIN MODE bit in SCIGCR1 to 1 to enable LIN mode.
- Select Master or Slave mode by programming the CLOCK bit in SCIGCR1.
- Set the MBUF MODE bit in SCIGCR1 to 1 to select multi-buffer mode.
- Select the baud rate to be used for communication by programming the BRS register.
- Set the maximum baud rate to be used for communication by programming the MBRS register.
- Set the CONT bit in SCIGCR1 to 1 to make LIN not halt for an emulation breakpoint until its current reception or transmission is complete (this bit is used only in an emulation environment).
- Set the LOOP BACK bit in SCIGCR1 to 1 to connect the transmitter to the receiver internally (this feature is used to perform a self-test).
- Set the RXENA bit in SCIGCR1 to 1, if data is to be received.
- Set the TXENA bit in SCIGCR1 to 1, if data is to be transmitted.
- Select the RX ID MASK and the TX ID MASK fields in the LINMASK register.
- Set the SWnRST bit to 1 after LIN is configured.
- Perform receiving or transmitting data (see [Section 28.10.1](#) or [Section 28.10.2](#)).

28.10.1 Receiving Data

The LIN receiver is enabled to receive messages if both the RX FUNC bit and the RXENA bit are set to 1. If the RX FUNC bit is not set, the LINRX pin functions as a general-purpose I/O pin rather than as a LIN function pin.

The ID RX FLAG is set after a valid LIN ID is received with RX Match. An ID interrupt is generated, if enabled.

28.10.1.1 Receiving Data in Single-Buffer Mode

Single-buffer mode is selected when the MBUF MODE bit is cleared to 0. In this mode, LIN sets the RXRDY bit when it transfers newly received data from SCIRXSHF to RD0. The SCI clears the RXRDY bit after the new data in RD0 has been read. Also, as data is transferred from SCIRXSHF to RD0, the LIN sets the FE, OE, or PE flags if any of these error conditions were detected in the received data. These error conditions are supported with configurable interrupt capability.

You can receive data by:

1. Polling Receive Ready Flag
2. Receive Interrupt
3. DMA

In polling method, software can poll for the RXRDY bit and read the data from RD0 byte of the LINRD0 register once the RXRDY bit is set high. The CPU is unnecessarily overloaded by selecting the polling method. To avoid this, you can use either the interrupt or DMA method. To use the interrupt method, the SET RX INT bit is set. To use the DMA method, the SET RX DMA bit is set. Either an interrupt or a DMA request is generated the moment the RXRDY bit is set. If the checksum scheme is enabled by setting the Compare Checksum (CC) bit to 1, the checksum will be compared on the byte that is currently being received, which is expected to be the checksum byte. The CC bit will be cleared once the checksum is received. A CE will immediately be flagged if there is a checksum error.

28.10.1.2 Receiving Data in Multi-Buffer Mode

Multi-buffer mode is selected when the MBUF MODE bit is set to 1. In this mode, LIN sets the RXRDY bit after receiving the programmed number of data in the receive buffer and the checksum field, the complete frame. The error condition detection logic is similar to the single-buffer mode, except that it monitors for the complete frame. Like single-buffer mode, you can use the polling, interrupt, or DMA method to read the data. The received data has to be read from the LINRD0 and LINRD1 registers, based on the number of bytes. For a LENGTH less than or equal to 4, a read from the LINRD0 register clears the RXRDY flag. For a LENGTH greater than 4, a read from the LINRD1 register clears the RXRDY flag. If the checksum scheme is enabled by setting the Compare Checksum (CC) bit to 1 during the reception of the data, then the byte that is received after the reception of the programmed number of data bytes indicated by the LENGTH field is treated as a checksum byte. The CC bit will be cleared once the checksum is received and compared.

28.10.2 Transmitting Data

The LIN transmitter is enabled if both the TX FUNC bit and the TXENA bit are set to 1. If the TX FUNC bit is not set, the LINTX pin functions as a general-purpose I/O pin rather than as a LIN function pin. Any value written to the TD0 before the TXENA bit is set to 1 is not transmitted. Both of these control bits allow for the LIN transmitter to be held inactive independently of the receiver.

The ID TX flag is set after a valid LIN ID is received with TX Match. An ID interrupt is generated, if enabled.

28.10.2.1 Transmitting Data in Single-Buffer Mode

Single-buffer mode is selected when the MBUF MODE bit is cleared to 0. In this mode, LIN waits for data to be written to TD0, transfers it to SCITXSHF, and transmits the data. The TXRDY and TX EMPTY bits indicate the status of the transmit buffers. That is, when the transmitter is ready for data to be written to TD0, the TXRDY bit is set. Additionally, if both TD0 and SCITXSHF are empty, then the TX EMPTY bit is also set.

You can transmit data by:

1. Polling Transmit Ready Flag
2. Transmit Interrupt
3. DMA

In polling method, software can poll for the TXRDY bit to go high before writing the data to the TD0. The CPU is unnecessarily overloaded by selecting the polling method. To avoid this, you can use either the interrupt or DMA method. To use the interrupt method, the SET TX INT bit is set. To use the DMA method, the SET TX DMA bit is set. Either an interrupt or a DMA request is generated the moment the TXRDY bit is set. When the LIN has completed transmission of all pending frames, the SCITXSHF register and the TD0 are empty, the TXRDY bit is set, and an interrupt/DMA request is generated, if enabled. Because all data has been transmitted, the interrupt/DMA request should be halted. This can either be done by disabling the transmit interrupt (CLR TX INT) / DMA request (CLR TX DMA bit) or by disabling the transmitter (clear TXENA bit). If the checksum scheme is enabled by setting the Send Checksum (SC) bit to 1, the checksum byte will be sent after the current byte transmission. The SC bit will be cleared after the checksum byte has been transmitted.

NOTE: The TXRDY flag cannot be cleared by reading the corresponding interrupt offset in the SCIINTVECT0 or SCIINTVECT1 register.

28.10.2.2 Transmitting Data in Multi-Buffer Mode

Multi-buffer mode is selected when the MBUF MODE bit is set to 1. Like single-buffer mode, you can use the polling, interrupt, or DMA method to write the data to be transmitted. The transmitted data has to be written to the LINTD0 and LINTD1 registers, based on the number of bytes. LIN waits for data to be written to Byte 0 (TD0) of the LINTD0 register and transfers the programmed number of bytes to SCITXSHF to transmit one by one automatically. If the checksum scheme is enabled by setting the Send Checksum (SC) bit to 1, the checksum will be sent after transmission of the last byte of the programmed number of data bytes, indicated by the LENGTH field. The SC bit will be cleared after the checksum byte has been transmitted.

28.11 Low-Power Mode

The SCI/LIN module can be put in either local or global low-power mode. Global low-power mode is asserted by the system and is not controlled by the SCI/LIN module. During global low-power mode, all clocks to the SCI/LIN are turned off so the module is completely inactive. If global low-power mode is requested while the receiver is receiving data, then the SCI/BLIN completes the current reception and then enters the low-power mode, that is, module enters low-power mode only when Busy bit (SCIFLR.3) is cleared.

The BLIN module may enter low-power mode either when there was no activity on the LINRX pin for more than 4s (this can be either a constant recessive or dominant level) or when a Sleep Command frame was received. Once the Timeout flag (SCIFLR.4) was set or once a Sleep Command was received, the POWERDOWN bit (SCIGCR2.0) must be set by the application software to make the module enter local low-power mode. A wakeup signal will terminate the sleep mode of the LIN bus.

NOTE: Enabling Local Low-Power Mode During Receive and Transmit

If the wakeup interrupt is enabled and low-power mode is requested while the receiver is receiving data, then the SCI/LIN immediately generates a wake-up interrupt to clear the powerdown bit. Thus, the SCI/LIN is prevented from entering low-power mode and completes the current reception. Otherwise, if the wakeup interrupt is disabled, the SCI/LIN completes the current reception and then enters the low-power mode.

28.11.1 Entering Sleep Mode

In LIN protocol, a sleep command is used to broadcast the sleep mode to all nodes. The sleep command consists of a diagnostic master request frame with identifier 0x3C (60), with the first data field as 0x00. There should be no activity in the bus once all nodes receive the sleep command: the bus is in sleep mode.

Local low-power mode is asserted by setting the POWERDOWN bit; setting this bit stops the clocks to the SCI/LIN internal logic and registers. Clearing the POWERDOWN bit causes SCI/LIN to exit from local low-power mode. All the registers are accessible during local power-down mode. If a register is accessed in low-power mode, this access results in enabling the clock to the module for that particular access alone.

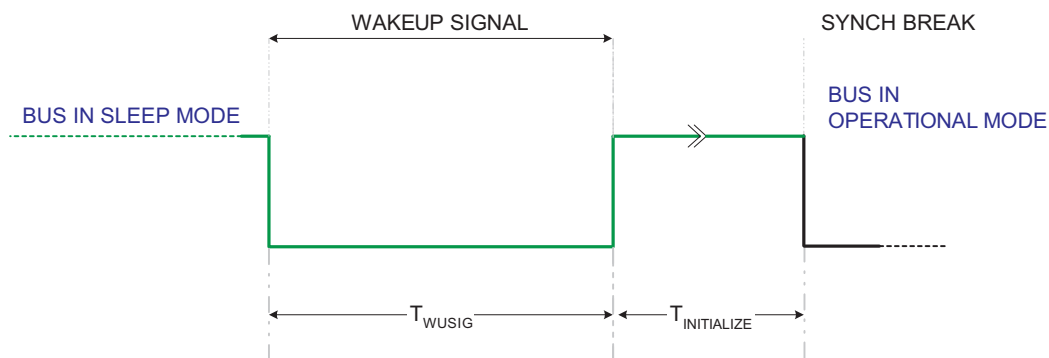
28.11.2 Wakeup

The wakeup interrupt is used to allow the SCI/LIN module to automatically exit low-power mode. A SCI/LIN wakeup is triggered when a low level is detected on the receive RX pin, and this clears the POWERDOWN bit.

NOTE: If the wakeup interrupt is disabled then the SCI/LIN enters low-power mode whenever it is requested to do so, but a low level on the receive RX pin does NOT cause the SCI/LIN to exit low-power mode.

In LIN mode, any node can terminate sleep mode by sending a wakeup signal; see [Figure 28-26](#). A slave node that detects the bus in sleep mode, and with a wakeup request pending, will send a wakeup signal. The wakeup signal is a dominant value on the LIN bus for T_{WUSIG} ; this is at least $5 T_{bits}$ for the LIN bus baud rates. The wakeup signal is generated by sending an 0xF0 byte containing 5 dominant T_{bits} and 5 recessive T_{bits} .

Figure 28-26. Wakeup Signal Generation



$$0.25\text{ms} \leq T_{WUSIG} \leq 5\text{ms}$$

(51)

Assuming a perfect bus with no noise or loading effects, a write of 0xF0 to TD0 will load the transmitter to meet the wakeup signal timing requirement for T_{WUSIG} . Then, setting the GENWU bit will transmit the preloaded value in TD0 for a wakeup signal transmission.

NOTE: The GENWU bit can be set/reset only when SWnRST is set to '1' and the node is in power down mode. The bit will be cleared on a valid synch break detection. A master sending a wakeup request, will exit power down mode upon reception of the wakeup pulse. The bit will be cleared on a SWnRST. This can be used to stop a master from sending further wakeup requests.

The TI TPIC1021 LIN transceiver, upon receiving a wakeup signal, will translate it to the microcontroller for wakeup with a dominant level on the RX pin, or a signal to the voltage regulator. While the POWERDOWN bit is set, if the LIN module detects a recessive-to-dominant edge (falling edge) on the RX pin, it will generate a wakeup interrupt if enabled in the SCISSETINT register.

According to LIN protocol 2.0, the TI TPIC1021 LIN transceiver detecting a dominant level on the bus longer than 150 ms will detect it as a wakeup request. The LIN controller's slave is ready to listen to the bus in less than 100 ms ($T_{INITIALIZE} < 100\text{ms}$) after a dominant-to-recessive edge (end-of-wakeup signal).

28.11.3 Wakeup Timeouts

The LIN protocol defines the following timeouts for a wakeup sequence. After a wakeup signal has been sent to the bus, all nodes wait for the master to send a header. If no synch field is detected before 150 ms (3,000 cycles at 20 kHz) after wakeup signal is transmitted, a new wakeup is sent by the same node that requested the first wakeup. This sequence is not repeated more than two times. After three attempts to wake up the LIN bus, wakeup signal generation is suspended for a 1.5 s (30,000 cycles at 20 kHz) period after three breaks.

NOTE: To achieve compatibility to LIN1.3 timeout conditions, the MBRS register must be set to assure that the LIN 2.0 (real-time-based) timings meet the LIN 1.3 bit time base. A node triggering the wakeup should set the MBRS register accordingly to meet the targeted time as 128 Tbits x programmed prescaler.

The LIN controller handles the wakeup expiration times defined by the LIN protocol with a hardware implementation.

28.12 Emulation Mode

In emulation mode, the CONT bit determines how the SCI/LIN operates when the program is suspended. The SCI/LIN counters are affected by this bit during debug mode. When set, the counters are not stopped and when cleared, the counters are stopped debug mode.

Any reads in emulation mode to a SCI/LIN register will not have any effect on the flags in the SCIFLR register.

NOTE: When emulation mode is entered during the Frame transmission or reception of the frame and CONT bit is not set, Communication is not expected to be successful. The suggested usage is to set CONT bit during emulation mode for successful communication.

28.13 SCI/LIN Control Registers

The SCI/LIN module registers are based on the SCI registers, with added functionality registers enabled by the LIN MODE bit in the SCIGCR1 register.

These registers are accessible in 8-, 16-, and 32-bit reads or writes. The SCI/LIN is controlled and accessed through the registers listed in [Table 28-9](#). Among the features that can be programmed are the LIN protocol mode, communication and timing modes, baud rate value, frame format, DMA requests, and interrupt configuration. The base address for the control registers is FFF7 E400h.

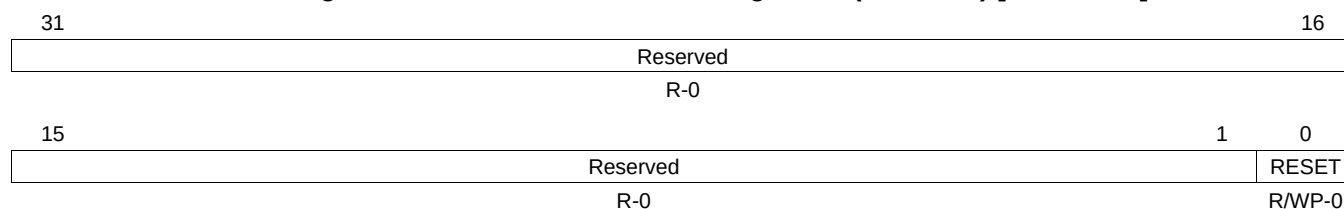
Table 28-9. SCI/LIN Control Registers

Offset	Acronym	Register Description	Section
00h	SCIGCR0	SCI Global Control Register 0	Section 28.13.1
04h	SCIGCR1	SCI Global Control Register 1	Section 28.13.2
08h	SCIGCR2	SCI Global Control Register 2	Section 28.13.3
0Ch	SCISSETINT	SCI Set Interrupt Register	Section 28.13.4
10h	SCICLEARINT	SCI Clear Interrupt Register	Section 28.13.5
14h	SCISSETINTLV	SCI Set Interrupt Level Register	Section 28.13.6
18h	SCICLEARINTLV	SCI Clear Interrupt Level Register	Section 28.13.7
1Ch	SCIFLR	SCI Flags Register	Section 28.13.8
20h	SCIINTVECT0	SCI Interrupt Vector Offset 0	Section 28.13.9
24h	SCIINTVECT1	SCI Interrupt Vector Offset 1	Section 28.13.10
28h	SCIFORMAT	SCI Format Control Register	Section 28.13.11
2Ch	BRS	Baud Rate Selection Register	Section 28.13.12
30h	SCIED	Receiver Emulation Data Buffer	Section 28.13.13.1
34h	SCIRD	Receiver Data Buffer	Section 28.13.13.2
38h	SCITD	Transmit Data Buffer	Section 28.13.13.3
3Ch	SCIPIO0	SCI Pin I/O Control Register 0	Section 28.13.14
40h	SCIPIO1	SCI Pin I/O Control Register 1	Section 28.13.15
44h	SCIPIO2	SCI Pin I/O Control Register 2	Section 28.13.16
48h	SCIPIO3	SCI Pin I/O Control Register 3	Section 28.13.17
4Ch	SCIPIO4	SCI Pin I/O Control Register 4	Section 28.13.18
50h	SCIPIO5	SCI Pin I/O Control Register 5	Section 28.13.19
54h	SCIPIO6	SCI Pin I/O Control Register 6	Section 28.13.20
58h	SCIPIO7	SCI Pin I/O Control Register 7	Section 28.13.21
5Ch	SCIPIO8	SCI Pin I/O Control Register 8	Section 28.13.22
60h	LINCOMPARE	LIN Compare Register	Section 28.13.23
64h	LINRD0	LIN Receive Buffer 0 Register	Section 28.13.24
68h	LINRD1	LIN Receive Buffer 1 Register	Section 28.13.25
6Ch	LINMASK	LIN Mask Register	Section 28.13.26
70h	LINID	LIN Identification Register	Section 28.13.27
74h	LINTD0	LIN Transmit Buffer 0	Section 28.13.28
78h	LINTD1	LIN Transmit Buffer 1	Section 28.13.29
7Ch	MBRS	Maximum Baud Rate Selection Register	Section 28.13.30
90h	IODFTCTRL	Input/Output Error Enable Register	Section 28.13.31

28.13.1 SCI Global Control Register 0 (SCIGCR0)

The SCIGCR0 register defines the module reset. [Figure 28-27](#) and [Table 28-10](#) illustrate this register.

Figure 28-27. SCI Global Control Register 0 (SCIGCR0) [offset = 00]



LEGEND: R/W = Read/Write; R = Read only; R/WP = Read/Write in privileged mode only; -n = value after reset

Table 28-10. SCI Global Control Register 0 (SCIGCR0) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	RESET	0 1	This bit resets the SCI/LIN module. This bit is effective in SCI and LIN mode. SCI/LIN module is in reset. SCI/LIN module is out of reset. Note: Read/Write in privileged mode only.

28.13.2 SCI Global Control Register 1 (SCIGCR1)

The SCIGCR1 register defines the frame format, protocol, and communication mode used by the SCI. [Figure 28-28](#) and [Table 28-11](#) illustrate this register.

Figure 28-28. SCI Global Control Register 1 (SCIGCR1) [offset = 04h]

31				26				25		24					
Reserved								TXENA		RXENA					
R-0								R/W-0		R/W-0					
23				18				17		16					
Reserved								CONT		LOOP BACK					
R-0								R/W-0		R/W-0					
15		14		13		12		11		10		9		8	
Reserved				STOP EXT FRAME		HGEN CTRL		CTYPE		MBUF MODE		ADAPT		SLEEP	
R-0				R/WL-0		R/WL-0		R/WL-0		R/W-0		R/WL-0		R/W-0	
7		6		5		4		3		2		1		0	
SWnRST		LIN MODE		CLOCK		STOP		PARITY		PARITY ENA		TIMING MODE		COMM MODE	
R/W-0		R/W-0		R/W-0		R/WC-0		R/WC-0		R/W-0		R/WC-0		R/W-0	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; WC = Write in SCI-compatible mode only; WL = Write in LIN mode only; -n = value after reset

Table 28-11. SCI Global Control Register 1 (SCIGCR1) Field Descriptions

Bit	Field	Value	Description
31-26	Reserved	0	Reads return 0. Writes have no effect.
25	TXENA	0	Transmit enable. This bit is effective in LIN and SCI modes. Data is transferred from SCITD, or the TDy (with y=0, 1,...7) buffers in LIN mode to the SCITXSHF shift out register only when the TXENA bit is set.
		0	Transfers from SCITD or TDy to SCITXSHF are disabled.
		1	Transfers from SCITD or TDy to SCITXSHF are enabled.
			Note: Data written to SCITD or the transmit multi-buffer before TXENA is set is not transmitted. If TXENA is cleared while transmission is ongoing, the data previously written to SCITD is sent (including the checksum byte in LIN mode).
24	RXENA	0	Receive enable. This bit is effective in LIN and SCI modes. RXENA allows or prevents the transfer of data from SCIRXSHF to SCIRD or the receive multi-buffers.
		0	The receiver will not transfer data from the shift buffer to the receive buffer or multi-buffers.
		1	The receiver will transfer data from the shift buffer to the receive buffer or multi-buffers.
			Note: Clearing RXENA stops received characters from being transferred into the receive buffer or multi-buffers, prevents the RX status flags (see Table 28-12) from being updated by receive data, and inhibits both receive and error interrupts. However, the shift register continues to assemble data regardless of the state of RXENA.
			Note: If RXENA is cleared before a frame is completely received, the data from the frame is not transferred into the receive buffer.
			Note: If RXENA is set before a frame is completely received, the data from the frame is transferred into the receive buffer. If RXENA is set while SCIRXSHF is in the process of assembling a frame, the status flags are not guaranteed to be accurate for that frame. To ensure that the status flags correctly reflect what was detected on the bus during a particular frame, RXENA should be set before the detection of that frame.
23-18	Reserved	0	Reads return 0. Writes have no effect.

Table 28-11. SCI Global Control Register 1 (SCIGCR1) Field Descriptions (continued)

Bit	Field	Value	Description
17	CONT	0 1	Continue on suspend. This bit is effective in LIN and SCI modes. This bit has an effect only when a program is being debugged with an emulator, and it determines how the SCI/LIN operates when the program is suspended. The SCI/LIN counters are affected by this bit: when the bit is set the counters are not stopped, when the bit is cleared the counters are stopped during debug mode. When debug mode is entered, the SCI/LIN state machine is frozen. Transmissions and LIN counters are halted and resume when debug mode is exited. When debug mode is entered, the SCI/LIN continues to operate until the current transmit and receive functions are complete.
16	LOOP BACK	0 1	Loopback bit. This bit is effective in LIN and SCI modes. The self-checking option for the SCI/LIN can be selected with this bit. If the LINITX and LINRX pins are configured with SCI/LIN functionality, then the LINTX pin is internally connected to the LINRX pin. Externally, during loop back operation, the LINTX pin outputs a high value and the LINRX pin is in a high-impedance state. If this bit value is changed while the SCI/LIN is transmitting or receiving data, errors may result. Loop back mode is disabled. Loop back mode is enabled.
15-14	Reserved	0	Reads return 0. Writes have no effect.
13	STOP EXT FRAME	0 1	Stop extended frame communication. This bit is effective in LIN mode only. This bit can be written only during extended frame communication. When the extended frame communication is stopped, this bit is cleared automatically. This bit has no effect. Extended frame communication will be stopped when current frame transmission/reception is completed.
12	HGEN CTRL	0 1	HGEN control. This bit is effective in LIN mode only. This bit controls the type of mask filtering comparison ID filtering using the ID-BYTE field in LIN Identification Register (LINID) occurs. Mask of FFh in LIN Mask Register (LINMASK) register will result in no match. ID filtering uses ID-SlaveTask BYTE (recommended). Mask of FFh in LIN Mask Register (LINMASK) register will result in ALWAYS match. Note: For software compatibility with future LIN modules, the HGEN CTRL bit must be set to 1, the RX ID MASK must be set to FFh, and the TX ID MASK must be set to FFh.
11	CTYPE	0 1	Checksum type. This bit is effective in LIN mode only. This bit controls the type of checksum to be used: classic or enhanced. Classic checksum is used. Enhanced checksum is used.
10	MBUF MODE	0 1	Multi-buffer mode. This bit is effective in LIN and SCI modes. This bit controls receive/transmit buffer usage, that is, whether the RX/TX multi-buffers are used or a single register, RD0/TD0, is used. Multi-buffer mode is disabled. Multi-buffer mode is enabled.
9	ADAPT	0 1	Adapt. This mode is effective in LIN mode only. This bit has an effect during the detection of the synch field. Two LIN protocol bit rate modes could be enabled with this bit according to the node capability file definition: automatic or select. The software and network configuration will decide which of these two modes are enabled. When this bit is cleared, the LIN 2.0 protocol fixed bit rate should be used. If the ADAPT bit is set, a SCI/LIN slave node detecting the baud rate will compare it to the prescalers in BRS register and update it if they are different. The BRS register will be updated with the new value. If this bit is not set there will be no adjustment to the BRS register. Automatic baud rate adjustment is disabled. Automatic baud rate adjustment is enabled.

Table 28-11. SCI Global Control Register 1 (SCIGCR1) Field Descriptions (continued)

Bit	Field	Value	Description
8	SLEEP	0 1	SCI sleep. This bit is effective in SCI mode only. In a multiprocessor configuration, this bit controls the receive sleep function. Clearing this bit brings the SCI/LIN out of sleep mode. Sleep mode is disabled. Sleep mode is enabled. Note: The receiver still operates when the SLEEP bit is set; however, RXRDY is updated and SCIRD is loaded with new data only when an address frame is detected. The remaining receiver status flags (see Table 28-12) are updated and an error interrupt is requested if the corresponding interrupt enable bit is set, regardless of the value of the SLEEP bit. In this way, if an error is detected on the receive data line while the SCI is asleep, software can promptly deal with the error condition. Note: The SLEEP bit is <i>not</i> automatically cleared when an address byte is detected. See Section 28.6.1 for more information on using the SLEEP bit for multiprocessor communication.
7	SWnRST	0 1	Software reset (active low). This bit is effective in LIN and SCI modes. The SCI/LIN is in its reset state; no data will be transmitted or received. Writing a 0 to this bit initializes the SCI/LIN state machines and operating flags as defined in Table 28-12 and Table 28-13. All affected logic is held in the reset state until a 1 is written to this bit. The SCI/LIN is in its ready state; transmission and reception can be done. After this bit is set to 1, the configuration of the module should not change. Note: The SCI/LIN should only be configured while SWnRST = 0.
6	LIN MODE	0 1	LIN mode. This bit is effective in LIN and SCI mode. This bit controls the module mode of operation. LIN mode is disabled; SCI mode is enabled. LIN mode is enabled; SCI mode is disabled.
5	CLOCK	0 1 0 1	SCI internal clock enable. The CLOCK bit determines the source of the module clock on the SCICLK pin. It also determines whether a LIN node is a slave or master. <i>SCI mode</i> The external SCICLK is the clock source. Note: If an external clock is selected, then the internal baud rate generator and baud rate registers are bypassed. The maximum frequency allowed for an externally sourced SCI clock is VCLK/16. The internal SCICLK is the clock source. <i>LIN mode</i> The node is in slave mode. The node is in master mode.
4	STOP	0 1	SCI number of stop bits per frame. This bit is effective in SCI mode only. One stop bit is used. Two stop bits are used. Note: The receiver checks for only one stop bit. However in idle-line mode, the receiver waits until the end of the second stop bit (if STOP = 1) to begin checking for an idle period.
3	PARITY	0 1	SCI parity odd/even selection. This bit is effective in SCI mode only. If the PARITY ENA bit is set, PARITY designates odd or even parity. Odd parity is used. Even parity is used. The parity bit is calculated based on the data bits in each frame and the address bit (in address-bit mode). The start and stop fields in the frame are not included in the parity calculation. For odd parity, the SCI transmits and expects to receive a value in the parity bit that makes odd the total number of bits in the frame with the value of 1. For even parity, the SCI transmits and expects to receive a value in the parity bit that makes even the total number of bits in the frame with the value of 1.

Table 28-11. SCI Global Control Register 1 (SCIGCR1) Field Descriptions (continued)

Bit	Field	Value	Description
2	PARITY ENA		Parity enable. This bit enables or disables the parity function.
			<i>SCI or buffered SCI mode:</i>
		0	Parity is disabled; no parity bit is generated during transmission or is expected during reception.
		1	Parity is enabled. A parity bit is generated during transmission and is expected during reception.
			<i>LIN mode:</i>
		0	ID field parity verification is disabled.
		1	ID field parity verification is enabled.
1	TIMING MODE		SCI timing mode bit. This bit is effective in SCI mode only. it selects the SCI timing mode.
		0	Synchronous timing is used.
		1	Asynchronous timing is used.
0	COMM MODE		SCI/LIN communication mode bit. In compatibility mode it selects the SCI communication mode. In LIN mode it selects length control option for ID-field bits ID4 and ID5.
			<i>SCI mode:</i>
		0	Idle-line mode is used.
		1	Address-bit mode is used.
			<i>LIN mode:</i>
		0	ID4 and ID5 are not used for length control.
		1	ID4 and ID5 are used for length control.

Table 28-12. SCI Receiver Status Flags

SCI Flag	Register	Bit	Value After Reset ⁽¹⁾
CE	SCIFLR	29	0
ISFE	SCIFLR	28	0
NRE	SCIFLR	27	0
FE	SCIFLR	26	0
OE	SCIFLR	25	0
PE	SCIFLR	24	0
RXWAKE	SCIFLR	12	0
RXRDY	SCIFLR	9	0
BUSY	SCIFLR	3	0
IDLE	SCIFLR	2	0
WAKE UP	SCIFLR	1	0
BRKDT	SCIFLR	0	0

⁽¹⁾ The flags are frozen with their reset value while SWnRST = 0.

Table 28-13. SCI Transmitter Status Flags

SCI Flag	Register	Bit	Value After Reset ⁽¹⁾
BE	SCIFLR	31	0
PBE	SCIFLR	30	0
TX WAKE	SCIFLR	10	0
TX EMPTY	SCIFLR	11	1
TXRDY	SCIFLR	8	1

⁽¹⁾ The flags are frozen with their reset value while SWnRST = 0.

28.13.3 SCI Global Control Register 2 (SCIGCR2)

The SCIGCR2 register is used to send or compare a checksum byte during extended frames, to generate a wakeup and for low-power mode control of the LIN module. [Figure 28-29](#) and [Table 28-14](#) illustrate this register.

Figure 28-29. SCI Global Control Register 2 (SCIGCR2) [offset = 08h]

31											18	17	16
Reserved											CC	SC	
R-0											R/WL-0	R/WL-0	
15	9			8	7				1	0			
Reserved					GEN WU	Reserved					POWERDOWN		
R-0					R/W-0		R-0			R/W-0			

LEGEND: R/W = Read/Write; R = Read only; WL = Write in LIN mode only; -n = value after reset

Table 28-14. SCI Global Control Register 2 (SCIGCR2) Field Descriptions

Bit	Field	Value	Description
31-18	Reserved	0	Reads return 0. Writes have no effect.
17	CC	0	Compare checksum. This bit is effective in LIN mode only. This bit is used by the receiver for extended frames to trigger a checksum compare. The user will initiate this transaction by writing a one to this bit. CC bit has to be set only after RX_RDY flag is set for the last received data. In non-multi-buffer mode, when the CC bit is set, the checksum will be compared on the byte that is expected to be the checksum byte. During multi-buffer mode, the following scenarios are associated with the CC bit: a) If the CC bit is set during the reception of the data, then the byte that is received after the reception of the programmed number of data bytes as indicated by SCIFORMAT[18:16] is treated as a checksum byte. b) If the CC bit is set during the idle period (that is, during the inter-frame space), then the immediate next byte will be treated as a checksum byte. c) CC bit will be auto cleared after the checkbyte has been received and compared. Checksum reception is not guaranteed if CC bit is write cleared by software during the checksum reception. See Section 28.7.6 for more details.
		0	No checksum compare will occur.
		1	Compare checksum on expected checksum byte.
16	SC	0	Send checksum byte. This bit is effective in LIN mode only. This bit is used by the transmitter with extended frames to send a checksum byte. In non-multi-buffer mode, the checksum byte will be sent after the current byte transmission. In multi-buffer mode, the checksum byte will be sent after the last byte count, indicated by the SCIFORMAT[18:16]). See Section 28.7.6 for more details. This byte will be cleared after the checksum byte has been transmitted. In non-multi-buffer mode, the checksum byte will be sent after the current byte transmission. During multi-buffer mode, the following scenarios are associated with the SC bit: a) The checkbyte will be sent after the last byte count, indicated by the SCIFORMAT[18:16]). b) Checksum will not be sent if SC is set before transmitting the very first byte(that is, during interframe space). c) SC bit will be auto cleared after the checkbyte has been transmitted. Checksum transmission is not guaranteed if SC bit is write cleared by software during the checksum transmission. See Section 28.7.6 for more details.
		0	No checksum byte will be sent.
		1	A checksum byte will be sent.
15-9	Reserved	0	Reads return 0. Writes have no effect.
8	GEN WU	0	Generate wakeup signal. This bit is effective in LIN mode only. This bit controls the generation of a wakeup signal, by transmitting the TDO buffer value. The LIN protocol specifies that this signal should be a dominant for T _{WUSIG} . This bit is cleared on reception of a valid synch break.
		0	No wakeup signal will be generated.
		1	The TDO buffer value will be transmitted for a wakeup signal. The bit will be cleared on a SWnRST.
7-1	Reserved	0	Reads return 0. Writes have no effect.

Table 28-14. SCI Global Control Register 2 (SCIGCR2) Field Descriptions (continued)

Bit	Field	Value	Description
0	POWERDOWN		Power down. This bit is effective in LIN or SCI mode. When this bit is set, the SCI/LIN module attempts to enter local low-power mode. If the POWERDOWN bit is set while the receiver is actively receiving data and the wakeup interrupt is disabled, then the SCI/LIN will delay entering low-power mode until the reception is completed. In LIN mode, the user may set the POWERDOWN bit after receiving a sleep command or on idle bus detection (more than 4 seconds). See Section 28.11 for more information on low-power mode.
		0	The SCI/LIN module is in normal operation.
		1	The SCI/LIN module enters local low-power mode.

28.13.4 SCI Set Interrupt Register (SCISSETINT)

Figure 28-30 and Table 28-15 illustrate this register. Refer Figure 28-30 for details on when different interrupt flags get set in a frame during LIN Mode.

Figure 28-30. SCI Set Interrupt Register (SCISSETINT) [offset = 0Ch]

31		30		29		28		27		26		25		24			
SET BE INT		SET PBE INT		SET CE INT		SET ISFE INT		SET NRE INT		SET FE INT		SET OE INT		SET PE INT			
R/WL-0		R/WL-0		R/WL-0		R/WL-0		R/WL-0		R/W-0		R/W-0		R/W-0			
23				Reserved				19				18		17		16	
SET RX DMA ALL								SET RX DMA		SET TX DMA							
R-0								R/WC-0		R/W-0		R/W-0					
15		14		13		12		10				9		8			
Reserved				SET ID INT		Reserved						SET RX INT		SET TX INT			
R-0				R/WL-0		R-0				R/W-0				R/W-0			
7		6		5		4		3		2		1		0			
SET TOA3WUS INT		SET TOAWUS INT		Reserved		SET TIMEOUT INT		Reserved				SET WAKEUP INT		SET BRKDT INT			
R/WL-0		R/WL-0		R-0		R/WL-0		R-0				R/W-0		R/WC-0			

LEGEND: R/W = Read/Write; R = Read only; WL = Write in LIN mode only; WC = Write in SCI-compatible mode only; -n = value after reset

Table 28-15. SCI Set Interrupt Register (SCISSETINT) Field Descriptions

Bit	Field	Value	Description
31	SET BE INT	0	Set bit error interrupt. This bit is effective in LIN mode only. Setting this bit enables the SCI/LIN module to generate an interrupt when there is a bit error. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
30	SET PBE INT	0	Set physical bus error interrupt. This bit is effective in LIN mode only. Setting this bit enables the SCI/LIN module to generate an interrupt when a physical bus error occurs. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
29	SET CE INT	0	Set checksum-error interrupt. This bit is effective in LIN mode only. Setting this bit enables the SCI/LIN module to generate an interrupt when there is a checksum error. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
28	SET ISFE INT	0	Set inconsistent-synch-field-error interrupt. This bit is effective in LIN mode only. Setting this bit enables the SCI/LIN module to generate an interrupt when there is an inconsistent synch field error. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
27	SET NRE INT	0	Set no-response-error interrupt. This bit is effective in LIN mode only. Setting this bit enables the SCI/LIN module to generate an interrupt when a no-response error occurs. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.

Table 28-15. SCI Set Interrupt Register (SCISSETINT) Field Descriptions (continued)

Bit	Field	Value	Description
26	SET FE INT	0	Set framing-error interrupt. This bit is effective in LIN or SCI-compatible mode. Setting this bit enables the SCI/LIN module to generate an interrupt when a framing error occurs. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
25	SET OE INT	0	Set overrun-error interrupt. This bit is effective in LIN or SCI-compatible mode. Setting this bit enables the SCI/LIN module to generate an interrupt when an overrun error occurs. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
24	SET PE INT	0	Set parity interrupt. This bit is effective in LIN or SCI-compatible mode. Setting this bit enables the SCI/LIN module to generate an interrupt when a parity error occurs. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
23-19	Reserved	0	Reads return 0. Writes have no effect.
18	SET RX DMA ALL	0	Set receive DMA all. This bit is effective in SCI-compatible mode only. This bit determines if a separate interrupt is generated for the address frames sent in multiprocessor communications. When this bit is 0, RX interrupt requests are generated for address frames and DMA requests are generated for data frames. When this bit is 1, RX DMA requests are generated for both address and data frames. <i>Read:</i> The DMA request is disabled for address frames (the receive interrupt request is enabled for address frames). <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read and write:</i> The DMA request is enabled for address and data frames.
17	SET RX DMA	0	Set receiver DMA. This bit is effective in LIN or SCI-compatible mode. To enable receiver DMA requests, this bit must be set. If it is cleared, interrupt requests are generated depending on the SET RX INT bit (SCISSETINT). <i>Read:</i> The DMA request is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read and write:</i> The DMA request is enabled for address and data frames.
16	SET TX DMA	0	Set transmit DMA. This bit is effective in LIN or SCI-compatible mode. To enable DMA requests for the transmitter, this bit must be set. If it is cleared, interrupt requests are generated depending on the SET TX INT bit (SCISSETINT). <i>Read:</i> Transmit DMA request is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read and write:</i> Transmit DMA request is enabled.
15-14	Reserved	0	Reads return 0. Writes have no effect.
13	SET ID INT	0	Set identification interrupt. This bit is effective in LIN mode only. This bit is set to enable an interrupt when a valid matching identifier is received. See Section 28.7.9 for more details. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
12-10	Reserved	0	Reads return 0. Writes have no effect.
9	SET RX INT	0	Receiver interrupt enable. This bit is effective in LIN or SCI-compatible mode. Setting this bit enables the SCI/LIN to generate a receive interrupt after a frame has been completely received and the data is being transferred from SCIRXSHF to SCIRD. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.

Table 28-15. SCI Set Interrupt Register (SCISSETINT) Field Descriptions (continued)

Bit	Field	Value	Description
8	SET TX INT	0	Set transmitter interrupt. This bit is effective in LIN or SCI-compatible mode. Setting this bit enables the SCI/LIN to generate a transmit interrupt as data is being transferred from SCITD to SCITXSHF and the TXRDY bit is being set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
7	SET TOA3WUS INT	0	Set timeout after three wakeup signals interrupt. This bit is effective in LIN mode only. Setting this bit enables the SCI/LIN to generate an interrupt when a timeout occurs after three wakeup signals have been sent. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
6	SET TOAWUS INT	0	Set timeout after wakeup signal interrupt. This bit is effective in LIN mode only. Setting this bit enables the SCI/LIN to generate an interrupt when a timeout occurs after one wakeup signal has been sent. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
5	Reserved	0	Reads return 0. Writes have no effect.
4	SET TIMEOUT INT	0	Set timeout interrupt. This bit is effective in LIN mode only. Setting this bit enables the SCI/LIN to generate an interrupt when there is no LIN bus activity (bus idle) for at least four seconds. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
3-2	Reserved	0	Reads return 0. Writes have no effect.
1	SET WAKEUP INT	0	Set wake-up interrupt. This bit is effective in LIN or SCI-compatible mode. Setting this bit enables the SCI/LIN to generate a wakeup interrupt and thereby exit low-power mode. If enabled, the wakeup interrupt is asserted when local low-power mode is requested while the receiver is busy or if a low level is detected on the LINRX pin during low-power mode. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.
0	SET BRKDT INT	0	Set break-detect interrupt. This bit is effective in SCI-compatible mode only. Setting this bit enables the SCI/LIN to generate an error interrupt if a break condition is detected on the LINRX pin. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt is enabled.

28.13.5 SCI Clear Interrupt Register (SCICLEARINT)

Figure 28-31 and Table 28-16 illustrate this register. SCICLEARINT register is used to clear the enabled interrupts without accessing SCISSETINT register.

Figure 28-31. SCI Clear Interrupt Register (SCICLEARINT) [offset = 10h]

31		30		29		28		27		26		25		24	
CLR BE INT		CLR PBE INT		CLR CE INT		CLR ISFE INT		CLR RE INT		CLR FE INT		CLR OE INT		CLR PE INT	
R/WL-0		R/WL-0		R/WL-0		R/WL-0		R/WL-0		R/W-0		R/W-0		R/W-0	
23								19		18		17		16	
Reserved										CLR RX DMA ALL		CLR RX DMA		CLR TX DMA	
R-0										R/WC-0		R/W-0		R/W-0	
15		14		13		12				10		9		8	
Reserved				CLR ID INT		Reserved						CLR RX INT		CLR TX INT	
R-0				R/WL-0				R-0				R/W-0		R/W-0	
7		6		5		4		3		2		1		0	
CLR TOA3WUS INT		CLR TOAWUS INT		Reserved		CLR TIMEOUT INT		Reserved				CLR WAKEUP INT		CLR BRKDT INT	
R/WL-0		R/WL-0		R-0		R/WL-0		R-0				R/W-0		R/WC-0	

LEGEND: R/W = Read/Write; R = Read only; WL = Write in LIN mode only; WC = Write in SCI-compatible mode only; -n = value after reset

Table 28-16. SCI Clear Interrupt Register (SCICLEARINT) Field Descriptions

Bit	Field	Value	Description
31	CLR BE INT	0	Clear bit error interrupt. This bit is effective in LIN mode only. This bit disables the bit error interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
30	CLR PBE INT	0	Clear physical bus error interrupt. This bit is effective in LIN mode only. This bit disables the physical-bus error interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
29	CLR CE INT	0	Clear checksum-error interrupt. This bit is effective in LIN mode only. This bit disables the checksum interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
28	CLR ISFE INT	0	Clear inconsistent-synch-field-error (ISFE) interrupt. This bit is effective in LIN mode only. This bit disables the ISFE interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.

Table 28-16. SCI Clear Interrupt Register (SCICLEARINT) Field Descriptions (continued)

Bit	Field	Value	Description
27	CLR NRE INT	0	Clear no-response-error interrupt. This bit is effective in LIN mode only. This bit disables the NRE interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
26	CLR FE INT	0	Clear framing-error interrupt. This bit is effective in LIN or SCI mode. This bit disables the framing-error interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
25	CLR OE INT	0	Clear overrun-error interrupt. This bit is effective in LIN or SCI mode. This bit disables the SCI/LIN overrun error interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
24	CLR PE INT	0	Clear parity interrupt. This bit is effective in LIN or SCI mode. This bit disables the parity error interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
23-19	Reserved	0	Reads return 0. Writes have no effect.
18	CLR RX DMA ALL	0	Clear receive DMA all. This bit is effective in SCI mode only. This bit clears the receive DMA request for address frames when set. Only receive data frames generate a DMA request. <i>Read:</i> Receive DMA request for address frames is disabled; Instead, RX interrupt requests are enabled for address frames. Receive DMA requests are still enabled for data frames. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> Receive DMA request for address and data frames is enabled. <i>Write:</i> Receive DMA request for address and data frames is disabled.
17	CLR RX DMA	0	Clear receive DMA request. This bit is effective in LIN or SCI mode. This bit disables the receive DMA request when set. <i>Read:</i> Receive DMA request is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> Receive DMA request is enabled. <i>Write:</i> Receive DMA request for is disabled.
16	CLR TX DMA	0	Clear transmit DMA request. This bit is effective in LIN or SCI mode. This bit disables the transmit DMA request when set. <i>Read:</i> Transmit DMA request is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> Transmit DMA request is enabled. <i>Write:</i> Transmit DMA request for is disabled.
15-14	Reserved	0	Reads return 0. Writes have no effect.

Table 28-16. SCI Clear Interrupt Register (SCICLEARINT) Field Descriptions (continued)

Bit	Field	Value	Description
13	CLR ID INT	0	Clear ID interrupt. This bit is effective in LIN mode only. This bit disables the ID interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
12-10	Reserved	0	Reads return 0. Writes have no effect.
9	CLR RX INT	0	Clear receiver interrupt. This bit is effective in LIN or SCI mode. This bit disables the receiver interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
8	CLR TX INT	0	Clear transmitter interrupt. This bit is effective in LIN or SCI mode. This bit disables the transmitter interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
7	CLR TOA3WUS INT	0	Clear timeout after three wakeup signals interrupt. This bit is effective in LIN mode only. This bit disables the timeout after three wakeup signals interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
6	CLR TOAWUS INT	0	Clear timeout after wakeup signal interrupt. This bit is effective in LIN mode only. This bit disables the timeout after one wakeup signal interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
5	Reserved	0	Reads return 0. Writes have no effect.
4	CLR TIMEOUT INT	0	Clear timeout interrupt. This bit is effective in LIN mode only. This bit disables the timeout (LIN bus idle) interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.
3-2	Reserved	0	Reads return 0. Writes have no effect.
1	CLR WAKEUP INT	0	Clear wake-up interrupt. This bit is effective in LIN or SCI-compatible mode. This bit disables the wakeup interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.

Table 28-16. SCI Clear Interrupt Register (SCICLEARINT) Field Descriptions (continued)

Bit	Field	Value	Description
0	CLR BRKDT INT	0	Clear break-detect interrupt. This bit is effective in SCI-compatible mode only. This bit disables the break-detect interrupt when set. <i>Read:</i> The interrupt is disabled. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt is enabled. <i>Write:</i> The interrupt is disabled.

28.13.6 SCI Set Interrupt Level Register (SCISSETINTLVL)

Figure 28-32 and Table 28-17 illustrate this register.

Figure 28-32. SCI Set Interrupt Level Register (SCISSETINTLVL) [offset = 14h]

31		30		29		28		27		26		25		24									
SET BE INT LVL		SET PBE INT LVL		SET CE INT LVL		SET ISFE INT LVL		SET NRE INT LVL		SET FE INT LVL		SET OE INT LVL		SET PE INT LVL									
R/WL-0		R/WL-0		R/WL-0		R/WL-0		R/WL-0		R/W-0		R/W-0		R/W-0									
23								19				18				17				16			
Reserved										SET RX DMA ALL INT LVL		Reserved											
R-0										R/WC-0						R-0							
15		14		13		12				10		9		8									
Reserved				SET ID INT LVL		Reserved						SET RX INT LVL		SET TX INT LVL									
R-0				R/WL-0		R-0						R/W-0		R/W-0									
7		6		5		4		3		2		1		0									
SET TOA3WUS INT LVL		SET TOAWUS INT LVL		Reserved		SET TIMEOUT INT LVL		Reserved				SET WAKEUP INT LVL		SET BRKDT INT LVL									
R/WL-0		R/WL-0		R-0		R/WL-0		R-0				R/W-0		R/WC-0									

LEGEND: R/W = Read/Write; R = Read only; WL = Write in LIN mode only; WC = Write in SCI-compatible mode only; -n = value after reset

Table 28-17. SCI Set Interrupt Level Register (SCISSETINTLVL) Field Descriptions

Bit	Field	Value	Description
31	SET BE INT LVL	0	Set bit error interrupt level. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
30	SET PBE INT LVL	0	Set physical bus error interrupt level. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
29	SET CE INT LVL	0	Set checksum-error interrupt level. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
28	SET ISFE INT LVL	0	Set inconsistent-synch-field-error interrupt level. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
27	SET NRE INT LVL	0	Set no-response-error interrupt level. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
26	SET FE INT LVL	0	Set framing-error interrupt level. This bit is effective in LIN or SCI-compatible mode. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.

Table 28-17. SCI Set Interrupt Level Register (SCISSETINTLVL) Field Descriptions (continued)

Bit	Field	Value	Description
25	SET OE INT LVL	0	Set overrun-error interrupt level. This bit is effective in LIN or SCI-compatible mode. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
24	SET PE INT LVL	0	Set parity error interrupt level. This bit is effective in LIN or SCI-compatible mode. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
23-19	Reserved	0	Reads return 0. Writes have no effect.
18	SET RX DMA ALL LVL	0	Set receive DMA all interrupt levels. This bit is effective in SCI mode only. <i>Read:</i> The receive interrupt request for address frames is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The receive interrupt request for address frames is mapped to the INT1 line.
17-14	Reserved	0	Reads return 0. Writes have no effect.
13	SET ID INT LVL	0	Set ID interrupt level. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
12-10	Reserved	0	Reads return 0. Writes have no effect.
9	SET RX INT LVL	0	Set receiver interrupt level. This bit is effective in LIN or SCI-compatible mode. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
8	SET TX INT LVL	0	Set transmitter interrupt level. This bit is effective in LIN or SCI-compatible mode. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
7	SET TOA3WUS INT LVL	0	Set timeout after three wakeup signals interrupt level. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
6	SET TOAWUS INT LVL	0	Set timeout after wakeup signal interrupt level. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
5	Reserved	0	Reads return 0. Writes have no effect.
4	SET TIMEOUT INT LVL	0	Set timeout interrupt level. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.
3-2	Reserved	0	Reads return 0. Writes have no effect.
1	SET WAKEUP INT LVL	0	Set wake-up interrupt level. This bit is effective in LIN or SCI-compatible mode. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.

Table 28-17. SCI Set Interrupt Level Register (SCISSETINTLVL) Field Descriptions (continued)

Bit	Field	Value	Description
0	SET BRKDT INT LVL	0	Set break-detect interrupt level. This bit is effective in SCI-compatible mode only. <i>Read:</i> The interrupt level is mapped to the INTO line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read or write:</i> The interrupt level is mapped to the INT1 line.

28.13.7 SCI Clear Interrupt Level Register (SCICLEARINTLVL)

Figure 28-33 and Table 28-18 illustrate this register.

Figure 28-33. SCI Clear Interrupt Level Register (SCICLEARINTLVL) [offset = 18h]

31		30		29		28		27		26		25		24	
CLR BE INT LVL		CLR PBE INT LVL		CLR CE INT LVL		CLR ISFE INT LVL		CLR NRE INT LVL		CLR FE INT LVL		CLR OE INT LVL		CLR PE INT LVL	
R/WL-0		R/WL-0		R/WL-0		R/WL-0		R/WL-0		R/W-0		R/W-0		R/W-0	
23								19		18		17		16	
Reserved										CLR RX DMA ALL INT LVL		Reserved			
R-0										R/WC-0		R-0			
15		14		13		12				10		9		8	
Reserved				CLR ID INT LVL		Reserved						CLR RX INT LVL		CLR TX INT LVL	
R-0				R/WL-0		R-0						R/W-0		R/W-0	
7		6		5		4		3		2		1		0	
CLR TOA3WUS INT LVL		CLR TOAWUS INT LVL		Reserved		CLR TIMEOUT INT LVL		Reserved				CLR WAKEUP INT LVL		CLR BRKDT INT LVL	
R/WL-0		R/WL-0		R-0		R/WL-0		R-0				R/W-0		R/WC-0	

LEGEND: R/W = Read/Write; R = Read only; WL = Write in LIN mode only; WC = Write in SCI-compatible mode only; -n = value after reset

Table 28-18. SCI Clear Interrupt Level Register (SCICLEARINTLVL) Field Descriptions

Bit	Field	Value	Description
31	CLR BE INT LVL	0	Clear bit error interrupt. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
30	CLR PBE INT LVL	0	Clear physical bus error interrupt. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
29	CLR CE INT LVL	0	Clear checksum-error interrupt. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
28	CLR ISFE INT LVL	0	Clear inconsistent-synch-field-error (ISFE) interrupt. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
27	CLR NRE INT LVL	0	Clear no-response-error interrupt. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.

Table 28-18. SCI Clear Interrupt Level Register (SCICLEARINTLVL) Field Descriptions (continued)

Bit	Field	Value	Description
26	CLR FE INT LVL	0	Clear framing-error interrupt. This bit is effective in LIN or SCI-compatible mode. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
25	CLR OE INT LVL	0	Clear overrun-error interrupt. This bit is effective in LIN or SCI-compatible mode. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
24	CLR PE INT LVL	0	Clear parity interrupt. This bit is effective in LIN or SCI-compatible mode. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
23-19	Reserved	0	Reads return 0. Writes have no effect.
18	CLR RX DMA ALL LVL	0	Clear receive DMA interrupt level. This bit is effective in SCI-compatible mode only. <i>Read:</i> The receive interrupt request for address frames is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The receive interrupt request for address frames is mapped to the INT1 line. <i>Write:</i> The receive interrupt request for address frames is mapped to the INT0 line.
17-14	Reserved	0	Reads return 0. Writes have no effect.
13	CLR ID INT LVL	0	Clear ID interrupt. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
12-10	Reserved	0	Reads return 0. Writes have no effect.
9	CLR RX INT LVL	0	Clear receiver interrupt. This bit is effective in LIN or SCI-compatible mode. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
8	CLR TX INT LVL	0	Clear transmitter interrupt. This bit is effective in LIN or SCI-compatible mode. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
7	CLR TOA3WUS INT LVL	0	Clear timeout after three wakeup signals interrupt. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.

Table 28-18. SCI Clear Interrupt Level Register (SCICLEARINTLVL) Field Descriptions (continued)

Bit	Field	Value	Description
6	CLR TOAWUS INT LVL	0	Clear timeout after wakeup signal interrupt. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
5	Reserved	0	Reads return 0. Writes have no effect.
4	CLR TIMEOUT INT LVL	0	Clear timeout interrupt. This bit is effective in LIN mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
3-2	Reserved	0	Reads return 0. Writes have no effect.
1	CLR WAKEUP INT LVL	0	Clear wake-up interrupt. This bit is effective in LIN or SCI-compatible mode. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.
0	CLR BRKDT INT LVL	0	Clear break-detect interrupt. This bit is effective in SCI-compatible mode only. <i>Read:</i> The interrupt level is mapped to the INT0 line. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> The interrupt level is mapped to the INT1 line. <i>Write:</i> The interrupt level is mapped to the INT0 line.

28.13.8 SCI Flags Register (SCIFLR)

Figure 28-34 and Table 28-19 illustrate this register.

Figure 28-34. SCI Flags Register (SCIFLR) [offset = 1Ch]

31		30		29		28		27		26		25		24	
BE		PBE		CE		ISFE		NRE		FE		OE		PE	
R/WL-0		R/WL-0		R/WL-0		R/WL-0		R/WL-0		R/W-0		R/W-0		R/W-0	
23														16	
Reserved															
R-0															
15		14		13		12		11		10		9		8	
Reserved		ID RX		ID TX		RX WAKE		TX EMPTY		TX WAKE		RX RDY		TX RDY	
R-0		R/WL-0		R/WL-0		R/WC-0		R/W-1		R/WC-0		R/W-0		R/W-1	
7		6		5		4		3		2		1		0	
TOA3WUS		TOAWUS		Reserved		TIMEOUT		BUSY		IDLE		WAKEUP		BRKDT	
R/WL-0		R/WL-0		R-0		R/WL-0		R/W-0		R-0		R/WL-0		R/WC-0	

LEGEND: R/W = Read/Write; R = Read only; WC = Write in SCI-compatible mode only; WL = Write in LIN mode only; -n = value after reset

Table 28-19. SCI Flags Register (SCIFLR) Field Descriptions

Bit	Field	Value	Description
31	BE		Bit error flag. This bit is effective in LIN mode only. This bit is set when a bit error has occurred. This is detected by the internal bit monitor. See Section 28.7.8 for more information. The bit error flag is cleared by any of the following:
		0	- Setting of the SWnRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - On reception of a new synch break - Reading the corresponding interrupt offset in SCIINTVECT0/1 Read: No error has been detected since this bit was last cleared. Write: Writing a 0 to this bit has no effect.
		1	Read: An error has been detected since this bit was last cleared. Write: The bit is cleared to 0.
30	PBE		Physical bus error flag. This bit is effective in LIN mode only. This bit is set when a physical bus error has been detected by the bit monitor in TED. See Section 28.7.8 for more information. The physical bus error flag is cleared by the following:
		0	- Setting of the SWnRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - On reception of a new synch break - Reading the corresponding interrupt offset in SCIINTVECT0/1 Note: The PBE will only be flagged, if no Synch Break can be generated (for example, because of a bus shortage to VBAT) or if no Synch Break Delimiter can be generated (for example, because of a bus shortage to GND). Read: No error has been detected since this bit was last cleared. Write: Writing a 0 to this bit has no effect.
		1	Read: An error has been detected since this bit was last cleared. Write: The bit is cleared to 0.

Table 28-19. SCI Flags Register (SCIFLR) Field Descriptions (continued)

Bit	Field	Value	Description
29	CE		Checksum error flag. This bit is effective in LIN mode only. This bit is set when a checksum error has been detected by a receiving node. This error is detected by the TED logic. See Section 28.7.8 for more information. The type of checksum to be used depends on the CTYPE bit in SCIGCR1. The checksum error flag is cleared by the following: - Setting of the SWnRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reception of a new synch break - Reading the corresponding interrupt offset in SCIINTVECT0/1
		0	<i>Read:</i> No error has been detected since this bit was last cleared. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> An error has been detected since this bit was last cleared. <i>Write:</i> The bit is cleared to 0.
28	ISFE		Inconsistent synch field error flag. This bit is effective in LIN mode only. This bit is set when an inconsistent synch field error has been detected by the synchronizer during header reception. See Section 28.7.5.2 for more information. The inconsistent synch field error flag is cleared by the following: - Setting of the SWnRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reception of a new synch break - Reading the corresponding interrupt offset in SCIINTVECT0/1
		0	<i>Read:</i> No inconsistent synch field error has been detected. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> An inconsistent synch field error has been detected. <i>Write:</i> The bit is cleared to 0.
27	NRE		No-response error flag. This bit is effective in LIN mode only. This bit is set when there is no response to a master's header completed within TFRAME_MAX. This timeout period is applied for message frames of known length (identifiers 0 to 61). This error is detected by the synchronizer. See Section 28.7.7 for more information. The no-response error flag is cleared by the following: - Setting of the SWnRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reception of a new synch break - Reading the corresponding interrupt offset in SCIINTVECT0/1
		0	<i>Read:</i> No no-response error has been detected since the last clear. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> A no-response error has been detected. <i>Write:</i> The bit is cleared to 0.

Table 28-19. SCI Flags Register (SCIFLR) Field Descriptions (continued)

Bit	Field	Value	Description
26	FE		Framing error flag. This bit is effective in LIN or SCI-compatible mode. This bit is set when an expected stop bit is not found. In SCI compatibility mode, only the first stop bit is checked. The missing stop bit indicates that synchronization with the start bit has been lost and that the character is incorrectly framed. Detection of a framing error causes the SCI/LIN to generate an error interrupt if the SET FE INT bit is set in the register SCISSETINT. The framing error flag is cleared by the following: - Setting of the SWnRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reading the corresponding interrupt offset in SCIINTVECT0/1 - Reception of a new character/frame, depending on whether the module is in SCI compatible or LIN mode In multi-buffer mode, the frame is defined in the SCIFORMAT register.
		0	<i>Read:</i> No framing error has been detected since the last clear. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> A framing error has been detected since the last clear. <i>Write:</i> The bit is cleared to 0.
25	OE		Overrun error flag. This bit is effective in LIN or SCI-compatible mode. This bit is set when the transfer of data from SCIRXSHF to SCIRD overwrites unread data already in SCIRD or the RDy buffers in LINRD0 and LINRD1. Detection of an overrun error causes the LIN to generate an error interrupt if the SET OE INT bit = 1. The OE flag is reset by the following: - Setting of the SWnRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reading the corresponding interrupt offset in SCIINTVECT0/1
		0	<i>Read:</i> No overrun error has been detected since the last clear. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> An overrun error has been detected. <i>Write:</i> The bit is cleared to 0.
24	PE		Parity error flag. This bit is effective in LIN or SCI-compatible mode. This bit is set when a parity error is detected in the received data. In SCI address-bit mode, the parity is calculated on the data and address bit fields of the received frame. In idle-line mode, only the data is used to calculate parity. An error is generated when a character is received with a mismatch between the number of 1s and its parity bit. If the parity function is disabled (SCIGCR[2] = 0), the PE flag is disabled and read as 0. Detection of a parity error causes the LIN to generate an error interrupt if the SET PE INT bit = 1. The PE bit is reset by the following: - Setting of the SWnRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reception of a new character or frame, depending on whether the module is in SCI compatible or LIN mode, respectively. - Reading the corresponding interrupt offset in SCIINTVECT0/1
		0	<i>Read:</i> No parity error has been detected since the last clear. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> A parity error has been detected. <i>Write:</i> The bit is cleared to 0.
23-15	Reserved	0	Reads return 0. Writes have no effect.

Table 28-19. SCI Flags Register (SCIFLR) Field Descriptions (continued)

Bit	Field	Value	Description
14	ID RX FLAG		Identifier on receive flag. This bit is effective in LIN mode only. This flag is set once an identifier is received with an receive match and no ID-parity error. See Section 28.7.9 for more details. This flag indicates that a new valid identifier has been received on an RX match. This bit is cleared by the following: - Setting of the SWnRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reading the LINID register - Reception of a new synch break - Reading the corresponding interrupt offset in SCIINTVECT0/1
		0	<i>Read:</i> No valid ID has been received since the last clear. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> A valid ID RX has been received in LINID[23:16] on an RX match. <i>Write:</i> The bit is cleared to 0.
13	ID TX FLAG		Identifier on transmit flag. This bit is effective in LIN mode only. This flag is set when an identifier is received with a transmit match and no ID-parity error. See Section 28.7.9 for more details. This flag indicates that a new valid identifier has been received on a TX match. This bit is cleared by the following: - Setting the SWnRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reading the LINID register - Receiving a new synch break - Reading the corresponding interrupt offset in SCIINTVECT0/1
		0	<i>Read:</i> No valid ID has been received since the last clear. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> A valid ID TX has been received in LINID[23:16] on an TX match. <i>Write:</i> The bit is cleared to 0.
12	RXWAKE		Receiver wakeup detect flag. This bit is effective in SCI-compatible mode only. The SCI sets this bit to indicate that the data currently in SCIRD is an address. RXWAKE is cleared by the following: - Setting of the SWnRST bit - Setting of the RESET bit - A system reset - Upon receipt of a data frame.
		0	The data in SCIRD is not an address.
		1	The data in SCIRD is an address.
11	TX EMPTY		Transmitter empty flag. This flag indicates the transmitter's buffer register(s) (SCITD/TDy) and shift register (SCITXSHF) are empty. In multi-buffer mode, this flag indicates the TDx registers and shift register (SCITXSHF) are empty. In non-multi-buffer mode, this flag indicates the LINTD0 byte and the shift register (SCITXSHF) are empty. Note: The RESET bit, an active SWnRST (SCIGCR1[7]) or a system reset sets this bit. This bit does not cause an interrupt request. <i>SCI mode or LIN non-multi-buffer mode:</i>
		0	Transmitter buffer or shift register (or both) are loaded with data.
		1	Transmitter buffer and shift registers are both empty.
			<i>In LIN mode using multi-buffer mode:</i>
		0	Multi-buffer or shift register (or all) are loaded with data.
		1	Multi-buffer and shift registers are all empty.

Table 28-19. SCI Flags Register (SCIFLR) Field Descriptions (continued)

Bit	Field	Value	Description
10	TXWAKE		Transmitter wakeup method select. This bit is effective in SCI mode only. The TXWAKE bit controls whether the data in SCITD should be sent as an address or data frame using multiprocessor communication format. This bit is set to 1 or 0 by software before a byte is written to SCITD and is cleared by the SCI when data is transferred from SCITD to SCITXSHF or by a system reset. Note: TXWAKE is not cleared by the SWnRST bit. <i>Address-bit mode:</i> 0 Frame to be transmitted will be data (address bit = 0). 1 Frame to be transmitted will be an address (address bit = 1). <i>Idle-line mode:</i> 0 The frame to be transmitted will be data. 1 The following frame to be transmitted will be an address (writing a 1 to this bit followed by writing dummy data to the SCITD will result in a idle period of 11 bit periods before the next frame is transmitted).
9	RXRDY		Receiver ready flag. In SCI-compatible mode, the receiver sets this bit to indicate that the SCIRD contains new data and is ready to be read by the CPU. In <i>LIN mode</i> , RXRDY is set once a valid frame is received in multi-buffer mode, a valid frame being a message frame received with no errors. In <i>non-multi-buffer mode</i> , RXRDY is set for each received byte and will be set for the last byte of the frame if there are no errors. The SCI/LIN generates a receive interrupt when RXRDY flag bit is set if the SET RX INT bit is set (SCISSETINT[9]); RXRDY is cleared by the following: - Setting of the SWnRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reading the SCIRD register in compatibility mode - Reading the last data byte RDy of the response in LIN mode Note: The RXRDY flag cannot be cleared by reading the corresponding interrupt offset in the SCIINTVECT0/1 register. 0 <i>Read:</i> No new data is in SCIRD. <i>Write:</i> Writing a 0 to this bit has no effect. 1 <i>Read:</i> New data is ready to be read from SCIRD. <i>Write:</i> The bit is cleared to 0.
8	TXRDY		Transmitter buffer register ready flag. When set, this bit indicates that the transmit buffer(s) register(s) (SCITD in compatibility mode and LINTD0/LINTD1 in multi-buffer mode) are ready to get another character from a CPU write. <i>In SCI</i> , writing data to SCITD automatically clears this bit. <i>In LIN mode</i> , this bit is cleared once byte 0 (TD0) is written to LINTD0. This bit is set after the data of the TX buffer is shifted into the SCITXSHF register. This event can trigger a transmit interrupt after data is copied to the TX shift register SCITXSHF, if the SET TX INT bit is set. Note: 1) TXRDY is also set to 1 by setting of the RESET bit, enabling SWnRST, or by a system reset. 2) The TXRDY flag cannot be cleared by reading the corresponding interrupt offset in the SCIINTVECT0/1 register. 3) The transmit interrupt request can be eliminated until the next series of data written into the transmit buffers LINTD0 and LINTD1, by disabling the corresponding interrupt via the SCICLEARINT register or by disabling the transmitter via the TXENA bit. <i>SCI mode:</i> 0 SCITD is full. 1 SCITD is ready to receive the next character. <i>LIN mode:</i> 0 Multi-buffers are full. 1 Multi-buffers are ready to receive the next character. For more information on transmit interrupt handling, see the SCI document for compatibility mode and Section 28.7.9 for LIN mode.

Table 28-19. SCI Flags Register (SCIFLR) Field Descriptions (continued)

Bit	Field	Value	Description
7	TOA3WUS		Timeout after three wakeup signals flag. This bit is effective in LIN mode only. This flag is set if there is no synch break received after three wakeup signals and a period of 1.5 seconds has passed. Such expiration time is used before issuing another round of wakeup signals. This bit is cleared by the following: - Setting of the SWnRST bit - Setting of the RESET bit - A system reset - Writing a 1 to this bit - Reading the corresponding interrupt offset in SCIINTVECT0/1 See Section 28.11.3 for more information.
		0	<i>Read:</i> No timeout occurred after three wakeup signals. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> Timeout occurred after three wakeup signals and 1.5 seconds time. <i>Write:</i> The bit is cleared to 0.
6	TOAWUS		Timeout after wakeup signal flag. This bit is effective in LIN mode only. This bit is set if there is no synch break received after a wakeup signal has been sent. A minimum of 150 ms expiration time is used before issuing another wakeup signal. This bit is cleared by the following: - Setting the SWnRST bit - Setting of the RESET bit - A system reset occurring - Writing a 1 to this bit - Reading the corresponding interrupt offset in SCIINTVECT0/1 See Section 28.11.3 for more information.
		0	<i>Read:</i> No timeout occurred after one wakeup signal (150 ms). <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> Timeout occurred after one wakeup signal. <i>Write:</i> The bit is cleared to 0.
5	Reserved	0	Reads return 0. Writes have no effect.
4	TIMEOUT		LIN bus idle timeout flag. This bit is effective in LIN mode only. This bit is set earliest after at least four seconds of bus inactivity. Bus inactivity is defined as no transactions between recessive and dominant (and vice versa). This bit is cleared by the following: - Setting the SWnRST bit - Setting of the RESET bit - A system reset occurring - Writing a 1 to this bit - Reading the corresponding interrupt offset in SCIINTVECT0/1 See Section 28.7.7 for more information.
		0	<i>Read:</i> No bus idle has been detected since this bit was last cleared. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> A LIN bus idle has been detected. <i>Write:</i> The bit is cleared to 0.
3	BUSY		Bus busy flag. This bit is effective in LIN mode and SCI-compatible mode. This bit indicates whether the receiver is in the process of receiving a frame. As soon as the receiver detects the beginning of a start bit, the BUSY bit is set to 1. When the reception of a frame is complete, the SCI/LIN clears the BUSY bit. If SET WAKEUP INT is set and power down is requested while this bit is set, the SCI/LIN automatically prevents low-power mode from being entered and generates wakeup interrupt. The BUSY bit is controlled directly by the SCI/LIN receiver, but this bit can also be cleared by the following: - Setting the SWnRST bit - Setting of the RESET bit - A system reset occurring
		0	The receiver is not currently receiving a frame.
		1	The receiver is currently receiving a frame.

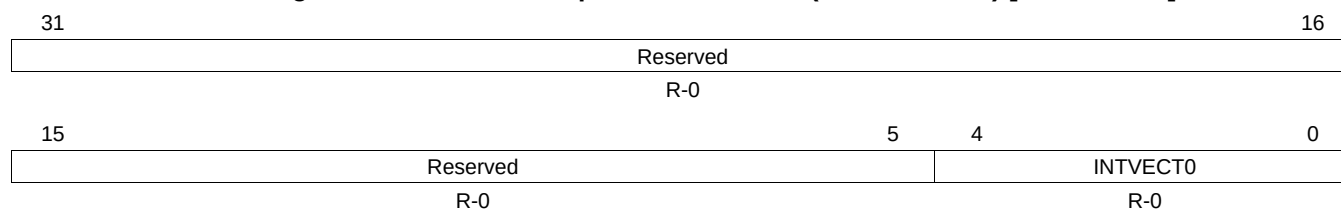
Table 28-19. SCI Flags Register (SCIFLR) Field Descriptions (continued)

Bit	Field	Value	Description
2	IDLE	0	SCI receiver in idle state. This bit is effective in SCI-compatible mode only. While this bit is set, the SCI looks for an idle period to resynchronize itself with the bit stream. The receiver does not receive any data while the bit is set. The bus must be idle for 11 bit periods to clear this bit. The SCI enters the idle state if one of the following events occurs: • A system reset • An SCI software reset • A power down • The RX pin is configured as a general I/O pin The idle period has been detected; the SCI is ready to receive.
		1	The idle period has not been detected; the SCI will not receive any data.
1	WAKEUP	0	Wakeup flag. This bit is effective in LIN mode only. This bit is set by the SCI/LIN when receiver or transmitter activity has taken the module out of power-down mode. An interrupt is generated if the SET WAKEUP INT bit (SCISSETINT[2]) is set. It is cleared by the following: • Setting of the SWnRST bit • Setting of the RESET bit • A system reset • Writing a 1 to this bit • Reading the corresponding interrupt offset in SCIINTVECT0/1 For compatibility mode, see the SCI document for more information on low-power mode. <i>Read:</i> The module will not wake up from power-down mode. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> Wake up from power-down mode. <i>Write:</i> The bit is cleared to 0.
0	BRKDT	0	SCI break-detect flag. This bit is effective in SCI-compatible mode only. This bit is set when the SCI detects a break condition on the LINRX pin. A break condition occurs when the LINRX pin remains continuously low for at least 10 bits after a missing first stop bit, that is, after a framing error. Detection of a break condition causes the SCI to generate an error interrupt if the SET BRKDT INT bit is set. The BRKDT bit is reset by the following: • Setting of the SWnRST bit • Setting of the RESET bit • A system reset • Writing a 1 to this bit • Reading the corresponding interrupt offset in SCIINTVECT0/1 <i>Read:</i> No break condition has been detected since the last clear. <i>Write:</i> Writing a 0 to this bit has no effect.
		1	<i>Read:</i> A break condition has been detected. <i>Write:</i> The bit is cleared to 0.

28.13.9 SCI Interrupt Vector Offset 0 (SCIINTVECT0)

Figure 28-35 and Table 28-20 illustrate this register.

Figure 28-35. SCI Interrupt Vector Offset 0 (SCIINTVECT0) [offset = 20h]



LEGEND: R = Read only; -n = value after reset

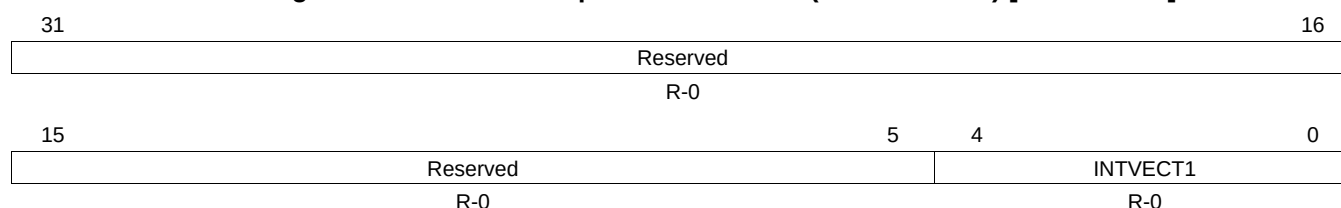
Table 28-20. SCI Interrupt Vector Offset 0 (SCIINTVECT0) Field Descriptions

Bit	Field	Value	Description
31-5	Reserved	0	Reads return 0. Writes have no effect.
4-0	INVECT0	0-1Fh	Interrupt vector offset for INT0. This register indicates the offset for interrupt line INT0. A read to this register updates its value to the next highest priority pending interrupt in SCIFLR and clears the flag in SCIFLR corresponding to the offset that was read. See Table 28-4 for a list of the interrupts. Note: The flags for the receive (SCIFLR[9]) and the transmit (SCIFLR[8]) interrupt cannot be cleared by reading the corresponding offset vector in this register (see detailed description in SCIFLR register).

28.13.10 SCI Interrupt Vector Offset 1 (SCIINTVECT1)

Figure 28-36 and Table 28-21 illustrate this register.

Figure 28-36. SCI Interrupt Vector Offset 1 (SCIINTVECT1) [offset = 24h]



LEGEND: R = Read only; -n = value after reset

Table 28-21. SCI Interrupt Vector Offset 1 (SCIINTVECT1) Field Descriptions

Bit	Field	Value	Description
31-5	Reserved	0	Reads return 0. Writes have no effect.
4-0	INVECT1	0-1Fh	Interrupt vector offset for INT1. This register indicates the offset for interrupt line INT1. A read to this register updates its value to the next highest priority pending interrupt in SCIFLR and clears the flag in SCIFLR corresponding to the offset that was read. See Table 28-4 for list of interrupts. Note: The flags for the receive (SCIFLR[9]) and the transmit (SCIFLR[8]) interrupt cannot be cleared by reading the corresponding offset vector in this register (see detailed description in SCIFLR register).

28.13.11 SCI Format Control Register (SCIFORMAT)

Figure 28-37 and Table 28-22 illustrate this register.

Figure 28-37. SCI Format Control Register (SCIFORMAT) [offset = 28h]

31		19	18	16
Reserved				LENGTH
R-0				R/W-0
15		3	2	0
Reserved				CHAR
R-0				R/WC-0

LEGEND: R/W = Read/Write; R = Read only; WC = Write in SCI-compatible mode only; -n = value after reset

Table 28-22. SCI Format Control Register (SCIFORMAT) Field Descriptions

Bit	Field	Value	Description
31-19	Reserved	0	Reads return 0. Writes have no effect.
18-16	LENGTH	0 1h 2h 3h 4h 5h 6h 7h	Frame length control bits. In <i>LIN mode</i>, these bits indicate the number of bytes in the response field from 1 to 8 bytes. In <i>buffered SCI mode</i>, these bits indicate the number of characters, with the number of bits per character specified in CHAR (SCIFORMAT[2:0]). When these bits are used to indicate LIN response length (SCIGCR1[0] = 1), then when there is an ID RX match, this value should be updated with the expected length of the response. In buffered SCI mode, these bits indicate the transmitter/receiver format for the number of characters: 1 to 8. There can be up to eight characters with eight bits each. The response field has 1 byte/character. The response field has 2 bytes/characters. The response field has 3 bytes/characters. The response field has 4 bytes/characters. The response field has 5 bytes/characters. The response field has 6 bytes/characters. The response field has 7 bytes/characters. The response field has 8 bytes/characters.
15-3	Reserved	0	Reads return 0. Writes have no effect.
2-0	CHAR	0 1h 2h 3h 4h 5h 6h 7h	Character length control bits. These bits are effective in non-buffered SCI and buffered SCI modes only. These bits set the SCI character length from 1 to 8 bits. In non-buffered SCI and buffered SCI mode, when data of fewer than eight bits in length is received, it is left justified in SCIRD/RDy and padded with trailing zeros. Data read from the SCIRD should be shifted by software to make the received data right justified. Data written to the SCITD should be right justified but does not need to be padded with leading zeros. The character is 1 bit long. The character is 2 bits long. The character is 3 bits long. The character is 4 bits long. The character is 5 bits long. The character is 6 bits long. The character is 7 bits long. The character is 8 bits long.

28.13.12 Baud Rate Selection Register (BRS)

This section describes the baud rate selection register. [Figure 28-38](#) and [Table 28-23](#) illustrate this register.

Figure 28-38. Baud Rate Selection Register (BRS) [offset = 2Ch]

31	30	28	27	24	23	16
Rsvd	U	M				PRESCALER P
R-0	R/W-0	R/W-0				R/W-0
15						0
PRESCALER P						
R/W-0						

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 28-23. Baud Rate Selection Register (BRS) Field Descriptions

Bit	Field	Value	Description
31	Reserved	0	Reads return 0. Writes have no effect.
30-28	U	0-2h	SCI/LIN super fractional divider selection. These bits are effective in LIN or SCI asynchronous mode. These bits are an additional fractional part for the baud rate specification. These bits allow a super-fine tuning of the fractional baud rate with seven more intermediate values for each of the M fractional divider values. See Section 28.7.4.1 for more details.
27-24	M	0-3h	SCI/LIN 4-bit fractional divider selection. These bits are effective in LIN or SCI asynchronous mode. These bits are used to select a baud rate for the SCI/LIN module, and they are a fractional part for the baud rate specification. The M divider allows fine-tuning of the baud rate over the P prescaler with 15 additional intermediate values for each of the P integer values. See Section 28.7.4.1 for more details.
23-0	PRESCALER P	0-FF FFFFh	These bits are used to select a baud rate for the SCI/LIN module. These bits are effective in LIN mode and SCI compatibility. The SCI/LIN has an internally generated serial clock determined by the VCLK and the prescalers P and M in this register. The LIN uses the 24-bit integer prescaler P value of this register to select one of over 16,700,000. The additional 4-bit fractional divider M refines the baud rate selection PRESCALER[27:24]. NOTE: In LIN mode, ONLY the asynchronous mode and baud rate values are used. The baud rate can be calculated using the following formulas: $\text{Asynchronous baud value} = \left(\frac{\text{VCLK Frequency}}{16 \left(P + 1 + \frac{M}{16} \right)} \right) \quad (52)$ $\text{Isosynchronous baud value} = \left(\frac{\text{VCLK Frequency}}{P + 1} \right) \quad (53)$ For P = 0, $\text{Asynchronous baud value} = \left(\frac{\text{VCLK Frequency}}{32} \right) \quad (54)$ $\text{Isosynchronous baud value} = \left(\frac{\text{VCLK Frequency}}{2} \right) \quad (55)$ Table 28-24 contains comparative baud values for different P values, with VCLK = 50 MHz, for asynchronous mode.

Table 28-24. Comparative Baud Values for Different P Values, Asynchronous Mode⁽¹⁾⁽²⁾

24-Bit Register Value		Baud Selected		Percent Error
Decimal	Hex	Ideal	Actual	
26	00001A	115200	115740	0.47
53	000035	57600	57870	0.47
80	000050	38400	38580	0.47
162	0000A2	19200	19172	-0.15
299	00012B	10400	10417	0.16
325	000145	9600	9586	-0.15
399	00018F	7812.5	7812.5	0.00
650	00028A	4800	4800	0.00
15624	003BA0	200	200	0.00
624999	098967	5	5	0.00

⁽¹⁾ VCLK = 50 MHz

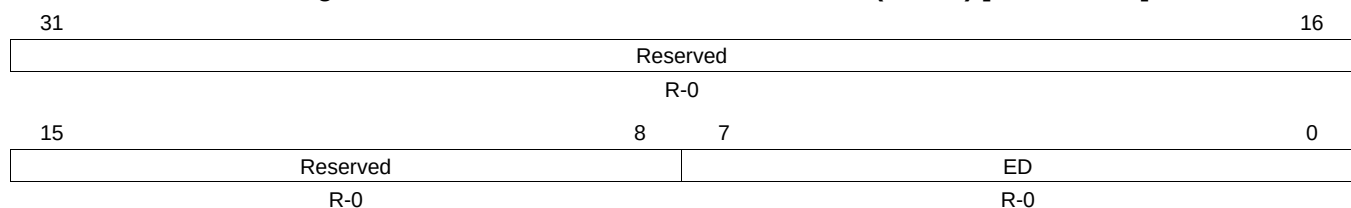
⁽²⁾ Values are in decimal except for column 2.

28.13.13 SCI Data Buffers (SCIED, SCIRD, SCITD)

The SCI has three addressable registers in which transmit and receive data is stored. These three registers are available in SCI mode only.

28.13.13.1 Receiver Emulation Data Buffer (SCIED)

The SCIED register is addressed at a location different from SCIRD, but is physically the same register. [Figure 28-39](#) and [Table 28-25](#) illustrate this register.

Figure 28-39. Receiver Emulation Data Buffer (SCIED) [offset = 30h]


LEGEND: R = Read only; -n = value after reset

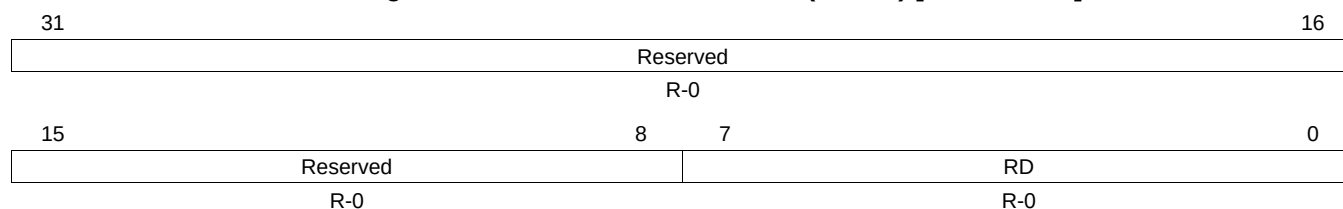
Table 28-25. Receiver Emulation Data Buffer (SCIED) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	ED	0-FFh	Emulator data. This bit is effective in SCI-compatible mode only. Reading SCIED[7:0] does not clear the RXRDY flag, unlike reading SCIRD. This register should be used only by an emulator that must continually read the data buffer without affecting the RXRDY flag.

28.13.13.2 Receiver Data Buffer (SCIRD)

This register provides a location for the receiver data. [Figure 28-40](#) and [Table 28-26](#) illustrate this register.

Figure 28-40. Receiver Data Buffer (SCIRD) [offset = 34h]



LEGEND: R = Read only; -n = value after reset

Table 28-26. Receiver Data Buffer (SCIRD) Field Descriptions

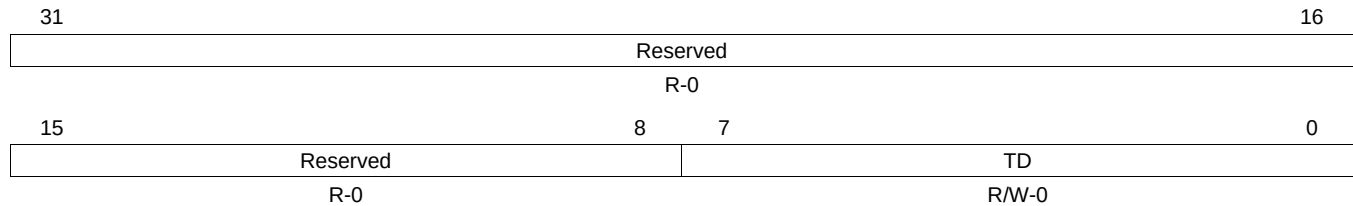
Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	RD	0-FFh	Receiver data. This bit is effective in SCI-compatible mode only. When a frame has been completely received, the data in the frame is transferred from the receiver shift register SCIRXSHF to this register. As this transfer occurs, the RXRDY flag is set and a receive interrupt is generated if SET RX INT is set. Note: When the data is read from SCIRD, the RXRDY flag is automatically cleared.

NOTE: When the SCI receives data that is fewer than eight bits in length, it loads the data into this register in a left justified format padded with trailing zeros. Therefore, your software should perform a logical shift on the data by the correct number of positions to make it right justified.

28.13.13.3 Transmit Data Buffer Register (SCITD)

Data to be transmitted is written to the SCITD register. [Figure 28-41](#) and [Table 28-27](#) illustrate this register.

Figure 28-41. Transmit Data Buffer Register (SCITD) [offset = 38h]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 28-27. Transmit Data Buffer Register (SCITD) Field Descriptions

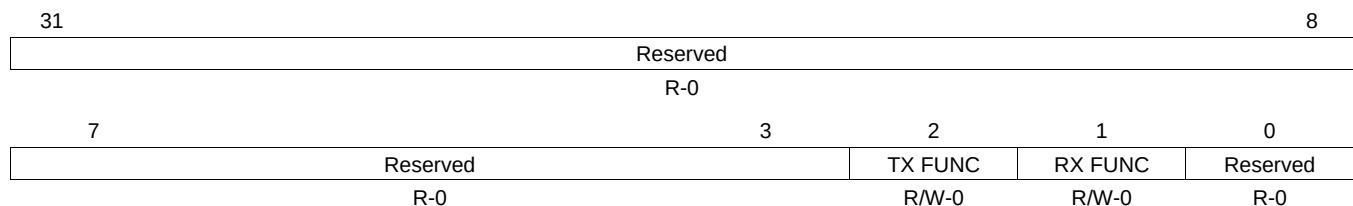
Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	TD	0-FFh	Transmit data. This bit is effective in SCI-compatible mode only. Data to be transmitted is written to the SCITD register. The transfer of data from this register to the transmit shift register SCITXSHF sets the TXRDY flag, which indicates that SCITD is ready to be loaded with another byte of data. Note: If SET TX INT is set, this data transfer also causes an interrupt.

NOTE: Data written to the SCITD register that is fewer than eight bits long must be right justified, but it does not need to be padded with leading zeros.

28.13.14 SCI Pin I/O Control Register 0 (SCIPIO0)

[Figure 28-42](#) and [Table 28-28](#) illustrate this register.

Figure 28-42. SCI Pin I/O Control Register 0 (SCIPIO0) [offset = 3Ch]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 28-28. SCI Pin I/O Control Register 0 (SCIPIO0) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX FUNC	0 1	Transfer function. This bit is effective in LIN or SCI mode. This bit defines the function of pin LINTX. LINTX is a general-purpose digital I/O pin. LINTX is the SCI/LIN transmit pin.
1	RX FUNC	0 1	Receive function. This bit is effective in LIN or SCI mode. This bit defines the function of pin LINRX. LINRX is a general-purpose digital I/O pin. LINRX is the SCI/LIN receive pin.
0	Reserved	0	Reads return 0. Writes have no effect.

28.13.15 SCI Pin I/O Control Register 1 (SCIPIO1)

Figure 28-43 and Table 28-29 illustrate this register.

Figure 28-43. SCI Pin I/O Control Register 1 (SCIPIO1) [offset = 40h]

31													8
Reserved													
R-0													
7							3	2	1	0			
Reserved							TX DIR		RX DIR		Reserved		
R-0							R/W-0		R/W-0		R-0		

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 28-29. SCI Pin I/O Control Register 1 (SCIPIO1) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX DIR	0 1	Transmit pin direction. This bit is effective in LIN or SCI mode. This bit determines the data direction on the LINTX pin if it is configured with general-purpose I/O functionality (TX FUNC = 0). See Table 28-30 for the LINTX pin control with this bit and others. LINTX is a general-purpose input pin. LINTX is a general-purpose output pin.
1	RX DIR	0 1	Receive pin direction. This bit is effective in LIN or SCI mode. This bit determines the data direction on the LINRX pin if it is configured with general-purpose I/O functionality (RX FUNC = 0). See Table 28-31 for the LINRX pin control with this bit and others. LINRX is a general-purpose input pin. LINRX is a general-purpose output pin.
0	Reserved	0	Reads return 0. Writes have no effect.

Table 28-30. LINTX Pin Control

Function	TX IN ⁽¹⁾	TX OUT	TX FUNC	TX DIR
LINTX	X	X	1	X
General purpose input	X	X	0	0
General purpose output, high	X	1	0	1
General purpose output, low	X	0	0	1

⁽¹⁾ TX IN is a read-only bit. Its value always reflects the level of the SCITX pin.

Table 28-31. LINRX Pin Control

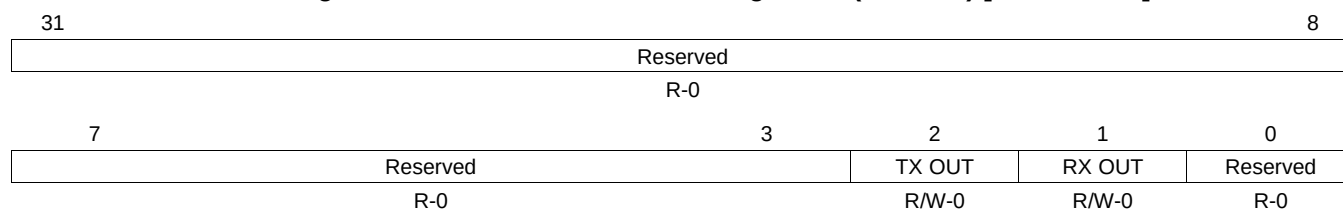
Function	RX IN ⁽¹⁾	RX OUT	RX FUNC	RX DIR
LINRX	X	X	1	X
General purpose input	X	X	0	0
General purpose output, high	X	1	0	1
General purpose output, low	X	0	0	1

⁽¹⁾ RX IN is a read-only bit. Its value always reflects the level of the SCIRX pin.

28.13.17 SCI Pin I/O Control Register 3 (SCIPIO3)

Figure 28-45 and Table 28-33 illustrate this register.

Figure 28-45. SCI Pin I/O Control Register 3 (SCIPIO3) [offset = 48h]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 28-33. SCI Pin I/O Control Register 3 (SCIPIO3) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX OUT	0 1	Transmit pin out. This bit is effective in LIN or SCI mode. This bit specifies the logic to be output on pin LINTX if the following conditions are met: - TX FUNC = 0 (LINTX pin is a general-purpose I/O.) - TX DIR = 1 (LINTX pin is a general-purpose output.) See Table 28-30 for an explanation of this bit's effect in combination with other bits. The output on the LINTX is at logic low (0). The output on the LINTX pin is at logic high (1). (Output voltage is V_{OH} or higher if TXPDR = 0 and output is in high impedance state if TXPDR = 1.)
1	RX OUT	0 1	Receive pin out. This bit is effective in LIN or SCI mode. This bit specifies the logic to be output on pin LINRX if the following conditions are met: - RX FUNC = 0 (LINRX pin is a general-purpose I/O.) - RX DIR = 1 (LINRX pin is a general-purpose output.) See Table 28-31 for an explanation of this bit's effect in combination with the other bits The output on the LINRX pin is at logic low (0). The output on the LINRX pin is at logic high (1). (Output voltage is V_{OH} or higher if RXPDR = 0, and output is in high impedance state if RXPDR = 1.)
0	Reserved	0	Reads return 0. Writes have no effect.

28.13.18 SCI Pin I/O Control Register 4 (SCIPIO4)

Figure 28-46 and Table 28-34 illustrate this register.

Figure 28-46. SCI Pin I/O Control Register 4 (SCIPIO4) [offset = 4Ch]

31																													8		
Reserved																															
R-0																															
7																							3	2	1	0					
Reserved																							TX SET			RX SET			Reserved		
R-0																							R/W-0			R/W-0			R-0		

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

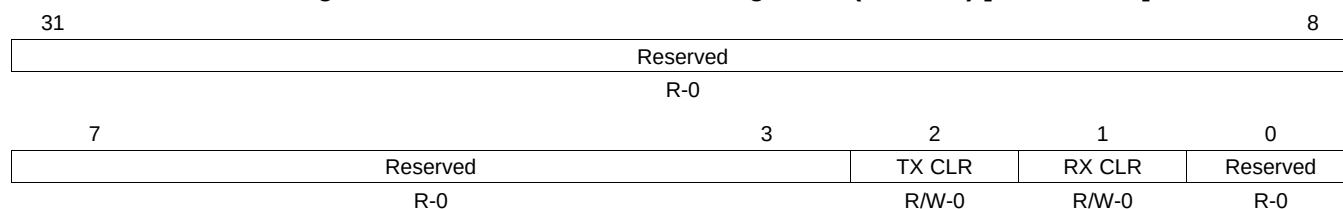
Table 28-34. SCI Pin I/O Control Register 4 (SCIPIO4) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX SET	0	Transmit pin set. This bit is effective in LIN or SCI mode. This bit sets the logic to be output on pin LINTX if the following conditions are met: - TX FUNC = 0 (LINTX pin is a general-purpose I/O.) - TX DIR = 1 (LINTX pin is a general-purpose output.) See Table 28-30 for an explanation of this bit's effect in combination with other bits. Read: The output on LINTX is at logic low (0). Write: Writing a 0 to this bit has no effect.
		1	Read and write: The output on LINTX is at logic high (1).
1	RX SET	0	Receive pin set. This bit is effective in LIN or SCI mode. This bit sets the logic to be output on pin LINRX if the following conditions are met: - RX FUNC = 0 (LINRX pin is a general-purpose I/O.) - RX DIR = 1 (LINRX pin is a general-purpose output.) See Table 28-31 for an explanation of this bit's effect in combination with the other bits. Read: The output on LINRX is at logic low (0). Write: Writing a 0 to this bit has no effect.
		1	Read and write: The output on LINRX is at logic high (1).
0	Reserved	0	Reads return 0. Writes have no effect.

28.13.19 SCI Pin I/O Control Register 5 (SCIPIO5)

Figure 28-47 and Table 28-35 illustrate this register.

Figure 28-47. SCI Pin I/O Control Register 5 (SCIPIO5) [offset = 50h]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 28-35. SCI Pin I/O Control Register 5 (SCIPIO5) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX CLR	0	Transmit pin clear. This bit is effective in LIN or SCI mode. This bit clears the logic to be output on pin LINTX if the following conditions are met: - TX FUNC = 0 (LINTX pin is a general-purpose I/O.) - TX DIR = 1 (LINTX pin is a general-purpose output.) Read: The output on LINTX is at logic low (0). Write: Writing a 0 to this bit has no effect.
		1	Read: The output on LINTX is at logic high (1). Write: The output on LINTX is at logic low (0).
1	RX CLR	0	Receive pin clear. This bit is effective in LIN or SCI mode. This bit clears the logic to be output on pin LINRX if the following conditions are met: - RX FUNC = 0 (LINRX pin is a general-purpose I/O.) - RX DIR = 1 (LINRX pin is a general-purpose output.) Read: The output on LINRX is at logic low (0). Write: Writing a 0 to this bit has no effect.
		1	Read: The output on LINRX is at logic high (1). Write: The output on LINRX is at logic low (0).
0	Reserved	0	Reads return 0. Writes have no effect.

28.13.20 SCI Pin I/O Control Register 6 (SCIPIO6)

Figure 28-48 and Table 28-36 illustrate this register.

Figure 28-48. SCI Pin I/O Control Register 6 (SCIPIO6) [offset = 54h]

31													8
Reserved													
R-0													
</													

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 28-36. SCI Pin I/O Control Register 6 (SCIPIO6) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX PDR	0 1	Transmit pin open drain enable. This bit is effective in LIN or SCI mode. This bit enables open-drain capability in the output pin LINTX, if the following conditions are met: - TX FUNC = 0 (LINTX pin is a general-purpose I/O.) - TX DIR = 1 (LINTX pin is a general-purpose output.) 0 Open-drain functionality is disabled; the output voltage is V_{OL} or lower if TXOUT = 0, and is V_{OH} or higher if TXOUT = 1. 1 Open-drain functionality is enabled; the output voltage is V_{OL} or lower if TXOUT = 0, and is high-impedance if TXOUT = 1.
1	RX PDR	0 1	Receive pin open drain enable. This bit is effective in LIN or SCI mode. This bit enables open-drain capability in the output pin LINRX, if the following conditions are met: - RX FUNC = 0 (LINRX pin is a general-purpose I/O.) - RX DIR = 1 (LINRX pin is a general-purpose output.) 0 Open-drain functionality is disabled; the output voltage is V_{OL} or lower if RXOUT = 0, and is V_{OH} or higher if RXOUT = 1. 1 Open-drain functionality is enabled; the output voltage is V_{OL} or lower if RXOUT = 0, and is high-impedance if RXOUT = 1.
0	Reserved	0	Reads return 0. Writes have no effect.

28.13.21 SCI Pin I/O Control Register 7 (SCIPIO7)

Figure 28-49 and Table 28-37 illustrate this register.

Figure 28-49. SCI Pin I/O Control Register 7 (SCIPIO7) [offset = 58h]

31																													8
Reserved																													
R-0																													
7															3	2	1	0											
Reserved														TX PD				RX PD				Reserved							
R-0														R/W-n				R/W-n				R-n							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset, Refer to the Terminal Functions in the device datasheet for default pin settings.

Table 28-37. SCI Pin I/O Control Register 7 (SCIPIO7) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX PD	0 1	Transmit pin pull control disable. This bit is effective in LIN or SCI mode. This bit disables pull control capability on the input pin LINTX. The pull control on the LINTX pin is enabled. The pull control on the LINTX pin is disabled.
1	RX PD	0 1	Receive pin pull control disable. This bit is effective in LIN or SCI mode. This bit disables pull control capability on the input pin LINRX. Pull control on the LINRX pin is enabled. Pull control on the LINRX pin is disabled.
0	Reserved		Writes have no effect.

28.13.22 SCI Pin I/O Control Register 8 (SCIPIO8)

Figure 28-50 and Table 28-38 illustrate this register.

Figure 28-50. SCI Pin I/O Control Register 8 (SCIPIO8) [offset = 5Ch]

31																8
Reserved																
R-0																
7											3	2	1	0		
Reserved										TX PSL		RX PSL		Reserved		
R-0										R/W-n		R/W-n		R-n		

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset, Refer to the Terminal Functions in the device datasheet for default pin settings.

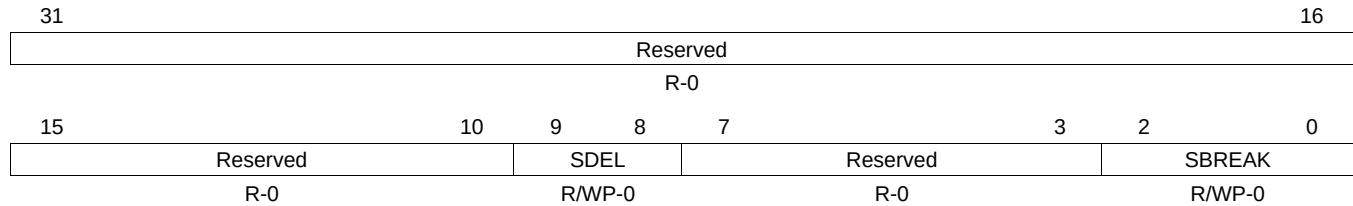
Table 28-38. SCI Pin I/O Control Register 8 (SCIPIO8) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Reads return 0. Writes have no effect.
2	TX PSL	0 1	Transmit pin pull select. This bit is effective in LIN or SCI mode. This bit selects pull type in the input pin LINTX. The LINTX pin is a pull down. The LINTX pin is a pull up.
1	RX PSL	0 1	Receive pin pull select. This bit is effective in LIN or SCI mode. This bit selects pull type in the input pin LINRX. The LINRX pin is a pull down. The LINRX pin is a pull up.
0	Reserved		Writes have no effect.

28.13.23 LIN Compare Register (LINCOMPARE)

Figure 28-51 and Table 28-39 illustrate this register.

Figure 28-51. LIN Compare Register (LINCOMPARE) [offset = 60h]



LEGEND: R/W = Read/Write; R = Read only; R/WP = Read/Write in privileged mode only; --n = value after reset

Table 28-39. LIN Compare Register (LINCOMPARE) Field Descriptions

Bit	Field	Value	Description
31-10	Reserved	0	Reads return 0. Writes have no effect.
9-8	SDEL	0 1h 2h 3h	2-bit synch delimiter compare. These bits are effective in LIN mode only. These bits are used to configure the number of T_{bit} for the synch delimiter in the synch field. The default value is 0. The formula to program the value (in T_{bits}) for the synchronization delimiter is: $T_{SDEL} = (SDEL + 1) T_{bit}$ The synch delimiter has 1 T_{bit}. The synch delimiter has 2 T_{bit}. The synch delimiter has 3 T_{bit}. The synch delimiter has 4 T_{bit}.
7-3	Reserved	0	Reads return 0. Writes have no effect.
2-0	SBREAK	0 1h 2h 3h 4h 5h 6h 7h	Synch break extend. These bits are effective in LIN mode only. These bits are used to configure the number of T_{bit} for the synch break to extend the minimum 13 T_{bit} break field to a maximum of 20 T_{bit} long. Note: The default value is 0, which adds nothing to the automatically generated SYNCH BREAK. The formula to program the value (in T_{bits}) for the SYNCH BREAK is: $T_{SYNBRK} = 13T_{bit} + (SBREAK \times T_{bit})$ The synch break has no additional T_{bit}. The synch break has 1 additional T_{bit}. The synch break has 2 additional T_{bit}. The synch break has 3 additional T_{bit}. The synch break has 4 additional T_{bit}. The synch break has 5 additional T_{bit}. The synch break has 6 additional T_{bit}. The synch break has 7 additional T_{bit}.

28.13.24 LIN Receive Buffer 0 Register (LINRD0)

Figure 28-52 and Table 28-40 illustrate this register.

Figure 28-52. LIN Receive Buffer 0 Register (LINRD0) [offset = 64h]

31	24	23	16
RD0		RD1	
R-0		R-0	
15	8	7	0
RD2		RD3	
R-0		R-0	

LEGEND: R = Read only; -n = value after reset

Table 28-40. LIN Receive Buffer 0 Register (LINRD0) Field Descriptions

Bit	Field	Value	Description
31-24	RD0	0-FFh	Receive buffer 0. Byte 0 of the response data byte. Each response data-byte that is received in the SCIRXSHFT register is transferred to the corresponding RDy bit field according to the number of bytes received. A read of this byte clears the RXDY byte. Note: RD<x-1> is equivalent to data byte <x> of the LIN frame.
23-16	RD1	0-FFh	Receive buffer 1. Byte 1 of the response data byte. Each response data-byte that is received in the SCIRXSHFT register is transferred to the corresponding RDy register according to the number of bytes received.
15-8	RD2	0-FFh	Receive buffer 2. Byte 2 of the response data byte. Each response data-byte that is received in the SCIRXSHFT register is transferred to the corresponding RDy register according to the number of bytes received.
7-0	RD3	0-FFh	Receive buffer 3. Byte 3 of the response data byte. Each response data-byte that is received in the SCIRXSHFT register is transferred to the corresponding RDy register according to the number of bytes received.

28.13.25 LIN Receive Buffer 1 Register (LINRD1)

Figure 28-53 and Table 28-41 illustrate this register.

Figure 28-53. LIN Receive Buffer 1 Register (RD1) [offset = 68h]

31	24	23	16
RD4		RD5	
R-0		R-0	
15	8	7	0
RD6		RD7	
R-0		R-0	

LEGEND: R = Read only; -n = value after reset

Table 28-41. LIN Receive Buffer 1 Register (RD1) Field Descriptions

Bit	Field	Value	Description
31-24	RD4	0-FFh	Receive buffer 4. Byte 4 of the response data byte. Each response data-byte that is received in the SCIRXSHFT register is transferred to the corresponding register according to the number of bytes received. Note: RD<x-1> is equivalent to data byte <x> of the LIN frame.
23-16	RD5	0-FFh	Receive buffer 5. Byte 5 of the response data byte. Each response data-byte that is received in the SCIRXSHFT register is transferred to the corresponding register according to the number of bytes received.
15-8	RD6	0-FFh	Receive buffer 6. Byte 6 of the response data byte. Each response data-byte that is received in the SCIRXSHFT register is transferred to the corresponding register according to the number of bytes received.
7-0	RD7	0-FFh	Receive buffer 7. Byte 7 of the response data byte. Each response data-byte that is received in the SCIRXSHFT register is transferred to the corresponding register according to the number of bytes received.

28.13.26 LIN Mask Register (LINMASK)

Figure 28-54 and Table 28-42 illustrate this register.

Figure 28-54. LIN Mask Register (LINMASK) [offset = 6Ch]

31	24	23	16
Reserved		RX ID MASK	
R-0		R/WL-0	
15	8	7	0
Reserved		TX ID MASK	
R-0		R/WL-0	

LEGEND: R/W = Read/Write; R = Read only; WL = Write in LIN mode only; -n = value after reset

Table 28-42. LIN Mask Register (LINMASK) Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-16	RX ID MASK	0-FFh	Receive ID mask. These bits are effective in LIN mode only. This 8-bit mask is used for filtering an incoming ID message and comparing it to the ID-byte. A compare match of the received ID with the RX ID MASK will set the ID RX flag and trigger an ID interrupt if enabled (SET ID INT in SCISSETINT). A 0 bit in the mask indicates that bit is compared to the ID-byte. A 1 bit in the mask indicates that bit is filtered and therefore is not used in the compare.
15-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	TX ID MASK	0-FFh	Transmit ID mask. These bits are effective in LIN mode only. This 8-bit mask is used for filtering an incoming ID message and comparing it to the ID-byte. A compare match of the received ID with the TX ID MASK will set the ID TX flag and trigger an ID interrupt if enabled (SET ID INT in SCISSETINT). A 0 bit in the mask indicates that bit is compared to the ID-byte. A 1 bit in the mask indicates that bit is filtered and therefore is not used for the compare.

28.13.27 LIN Identification Register (LINID)

Figure 28-55 and Table 28-43 illustrate this register.

Figure 28-55. LIN Identification Register (LINID) [offset = 70h]

31		24	23		16
Reserved				RECEIVED ID	
R-0				R-0	
15		8	7		0
ID-SLAVETASK BYTE				ID BYTE	
R/WL-0				R/WL-0	

LEGEND: R/W = Read/Write; R = Read only; WL = Write in LIN mode only; -n = value after reset

Table 28-43. LIN Identification Register (LINID) Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-16	RECEIVED ID	0-FFh	Received identification. These bits are effective in LIN mode only. This byte contains the current message identifier. During header reception the received ID is copied from the SCIRXSHF register to this byte if there is no ID-parity error and there has been an RX/TX match.
15-8	ID-SLAVETASK BYTE	0-FFh	ID-SlaveTask Byte. These bits are effective in LIN mode only. This field contains the identifier to which the received ID of an incoming header will be compared to decide whether a receive response, a transmit response, or no action needs to be performed by the LIN node when a header with that particular ID is received.
7-0	ID BYTE	0-FFh	ID byte. This field is effective in LIN mode only. This byte is the LIN mode message ID. On a master node, a write to this register by the CPU initiates a header transmission. For a slave task, this byte is used for message filtering when HGEN CTRL = 0.

NOTE: For software compatibility with future LIN modules, the HGEN CTRL bit must be set to 1, the RX ID MASK field must be set to FFh, and the TX ID MASK field must be set to FFh.

28.13.28 LIN Transmit Buffer 0 Register (LINTD0)

Figure 28-56 and Table 28-44 illustrate this register.

Figure 28-56. LIN Transmit Buffer 0 Register (LINTD0) [offset = 74h]

31	24	23	16
TD0		TD1	
R/W-0		R/W-0	
15	8	7	0
TD2		TD3	
R/W-0		R/W-0	

LEGEND: R/W = Read/Write; -n = value after reset

Table 28-44. LIN Transmit Buffer 0 Register (LINTD0) Field Descriptions

Bit	Field	Value	Description
31-24	TD0	0-FFh	8-Bit transmit buffer 0. Byte 0 to be transmitted is written into this register and then copied to SCITXSHF for transmission. Once byte 0 is written in TD0 buffer, transmission will be initiated. Note: TD<x-1> is equivalent to data byte <x> of the LIN frame.
23-16	TD1	0-FFh	8-Bit transmit buffer 1. Byte 1 to be transmitted is written into this register and then copied to SCITXSHF for transmission.
15-8	TD2	0-FFh	8-Bit transmit buffer 2. Byte 2 to be transmitted is written into this register and then copied to SCITXSHF for transmission.
7-0	TD3	0-FFh	8-Bit transmit buffer 3. Byte 3 to be transmitted is written into this register and then copied to SCITXSHF for transmission.

28.13.29 LIN Transmit Buffer 1 Register (LINTD1)

Figure 28-57 and Table 28-45 illustrate this register.

Figure 28-57. LIN Transmit Buffer 1 Register (LINTD1) [offset = 78h]

31	24	23	16
TD4		TD5	
R/W-0		R/W-0	
15	8	7	0
TD6		TD7	
R/W-0		R/W-0	

LEGEND: R/W = Read/Write; -n = value after reset

Table 28-45. LIN Transmit Buffer 1 Register (LINTD1) Field Descriptions

Bit	Field	Value	Description
31-24	TD4	0-FFh	8-Bit transmit buffer 4. Byte 4 to be transmitted is written into this register and then copied to SCITXSHF for transmission. Note: TD<x-1> is equivalent to data byte <x> of the LIN frame.
23-16	TD5	0-FFh	8-Bit transmit buffer 5. Byte 5 to be transmitted is written into this register and then copied to SCITXSHF for transmission.
15-8	TD6	0-FFh	8-Bit transmit buffer 6. Byte 6 to be transmitted is written into this register and then copied to SCITXSHF for transmission.
7-0	TD7	0-FFh	8-Bit transmit buffer 7. Byte 7 to be transmitted is written into this register and then copied to SCITXSHF for transmission.

28.13.30 Maximum Baud Rate Selection Register (MBRS)

Figure 28-58 and Table 28-46 illustrate this register.

Figure 28-58. Maximum Baud Rate Selection Register (MBRS) [offset = 7Ch]

31													16
Reserved													
R-0													
15	13	12											0
Reserved		MBR											
R-0		R/WL-0DAC _h											

LEGEND: R/W = Read/Write; R = Read only; WL = Write in LIN mode only; -n = value after reset

Table 28-46. Maximum Baud Rate Selection Register (MBRS) Field Descriptions

Bit	Field	Value	Description
31-13	Reserved	0	Reads return 0. Writes have no effect.
12-0	MBR	0-1FFF _h	Maximum baud rate prescaler. This bit is effective in LIN mode only. This 13-bit prescaler is used during the synchronization phase (see Section 28.7.5.2) of a slave module if the ADAPT bit is set. In this way, a SCI/LIN slave using an automatic or select bit rate modes detects any LIN bus legal rate automatically. The MBR value should be programmed to allow a maximum baud rate that is not more than 10% above the expected operating baud rate in the LIN network. Otherwise, a 0x00 data byte could mistakenly be detected as a sync break. The default value for a 70Mhz VCLK is 0xDAC. This MBR prescaler is used by the wake-up and idle time counters for a constant expiration time relative to a 20 kHz rate. $MBR = \frac{0.9 \times VCLK}{maxbaudrate}$

(56)

28.13.31 Input/Output Error Enable (IODFTCTRL) Register

All the bits in the IODFTCTRL register are used in IODFT (I/O design for test) mode only. [Figure 28-59](#) and [Table 28-47](#) illustrate this register. After the basic SCI/LIN module configuration, enable the required Error mode to be created followed by IODFT Key enable.

- NOTE:**
- 1) All the bits are used in IODFT mode only.
 - 2) Each IODFT are expected to be checked individually.
 - 3) ISFE Error will not be Flagged during IODFT mode.

Figure 28-59. Input/Output Error Enable Register (IODFTCTRL) [offset = 90h]

31	30	29	28	27	26	25	24
BEN	PBEN	CEN	ISFE	Reserved	FEN	PEN	BRKDT ENA
R/WL-0	R/WL-0	R/WL-0	R/WL-0	R-0	R/W-0	R/WC-0	R/WC-0
23		21	20	19	18		16
Reserved			PIN SAMPLE MASK			TX SHIFT	
R-0			R/W-0			R/W-0	
15			12	11			8
Reserved				IODFTENA			
R-0				R/WP-5h			
7					2	1	0
Reserved						LPB ENA	RXP ENA
R-0						R/WP-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WL = Write in LIN mode only; WC = Write in SCI-compatible mode only; WP = Write in privilege mode only; -n = value after reset

Table 28-47. Input/Output Error Enable Register (IODFTCTRL) Field Descriptions

Bit	Field	Value	Description
31	BEN	0 1	Bit error enable. This bit is effective in LIN mode only. This bit is used to create a bit error. No bit error is created. The bit received is ORed with 1 and passed to the bit monitor circuitry.
30	PBEN	0 1	Physical bus error enable. This bit is effective in LIN mode only. This bit is used to create a physical bus error. No error is created. The bit received during synch break field transmission is ORed with 1 and passed to the bit monitor circuitry.
29	CEN	0 1	Checksum error enable. This bit is effective in LIN mode only. This bit is used to create a checksum error. No error is created. The polarity of the CTYPE (checksum type) in the receive checksum calculator is changed so that a checksum error is occurred.
28	ISFE	0 1	Inconsistent synch field (ISF) error enable. This bit is effective in LIN mode only. This bit is used to create an ISF error. No error is created. The bit widths in the synch field are varied so that the ISF check fails and the error flag is set.
27	Reserved	0	Reads return 0. Writes have no effect.
26	FEN	0 1	Frame error enable. This bit is used to create a frame error. No error is created. The stop bit received is ANDed with 0 and passed to the stop bit check circuitry.

Table 28-47. Input/Output Error Enable Register (IODFTCTRL) Field Descriptions (continued)

Bit	Field	Value	Description
25	PEN	0 1	Parity error enable. This bit is effective in SCI-compatible mode only. This bit is used to create a parity error. No parity error occurs. The parity bit received is toggled so that a parity error occurs.
24	BRKDT ENA	0 1	Break detect error enable. This bit is effective in SCI-compatible mode only. This bit is used to create a BRKDT error. No error is created. The stop bit of the frame is ANDed with 0 and passed to the RSM so that a frame error occurs. Then the RX pin is forced to continuous low for 10 T _{bit} so that a BRKDT error occurs.
32-21	Reserved	0	Reads return 0. Writes have no effect.
20-19	PIN SAMPLE MASK	0 1h 2h 3h	Pin sample mask. These bits define the sample number at which the TX pin value that is being transmitted will be inverted to verify the receive pin samples majority detection circuitry. Note: In IODFT mode testing for pin_sample mask must be done with prescaler P programmed greater than 2 (P > 2). No mask is used. Invert the TX Pin value at TBIT_CENTER. Invert the TX Pin value at TBIT_CENTER + SCLK. Invert the TX Pin value at TBIT_CENTER + 2 SCLK.
18-16	TX SHIFT	0 1h 2h 3h 4h 5h 6h 7h	Transmit shift. These bits define the amount by which the value on TX pin is delayed so that the value on the RX pin is asynchronous. This feature is not applicable to the start bit. No delay occurs. The value is delayed by 1 SCLK. The value is delayed by 2 SCLK. The value is delayed by 3 SCLK. The value is delayed by 4 SCLK. The value is delayed by 5 SCLK. The value is delayed by 6 SCLK. The value is delayed by 7 SCLK.
15-12	Reserved	0	Reads return 0. Writes have no effect.
11-8	IODFTENA	Ah All Others	IODFT enable key. Write access permitted in Privilege mode only. IODFT is enabled. IODFT is disabled.
7-2	Reserved	0	Reads return 0. Writes have no effect.
1	LPB ENA	0 1	Module loopback enable. Write access permitted in Privilege mode only. Note: In analog loopback mode the complete communication path through the I/Os can be tested, whereas in digital loopback mode the I/O buffers are excluded from this path. Digital loopback is enabled. Analog loopback is enabled in module I/O DFT mode when IODFTENA = 1010.
0	RXP ENA	0 1	Module analog loopback through receive pin enable. Write access permitted in Privilege mode only. This bit defines whether the I/O buffers for the transmit or the receive pin are included in the communication path (in analog loopback mode). Analog loopback through the transmit pin is enabled. Analog loopback through the receive pin is enabled.

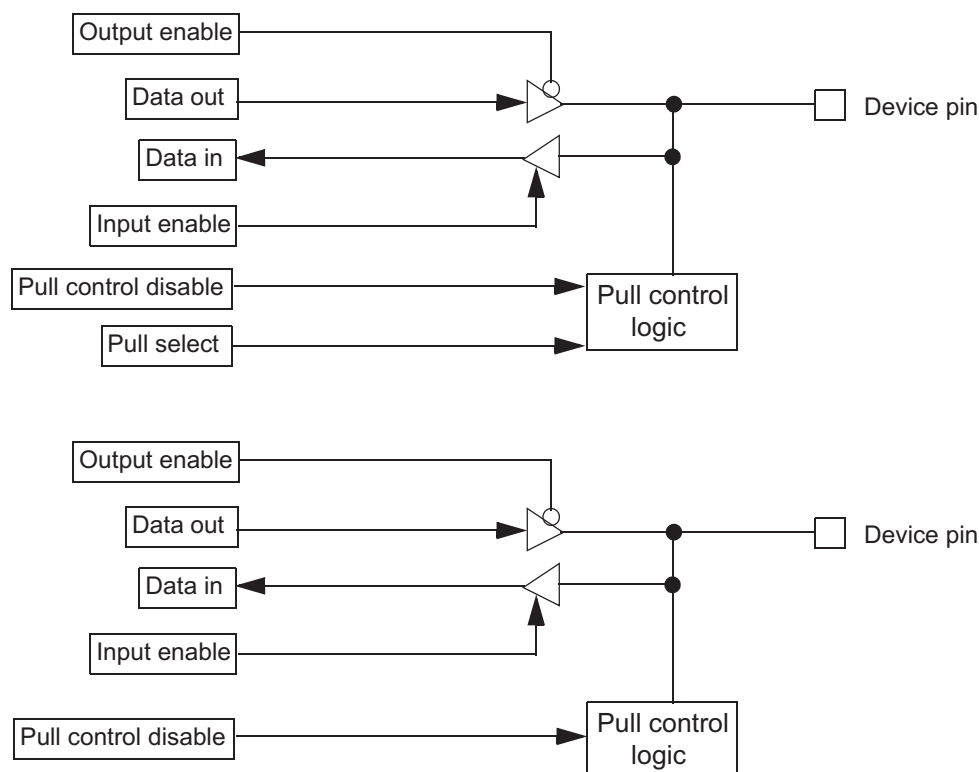
28.14 GPIO Functionality

The following sections apply to all device pins that can be configured as functional or general-purpose I/O pins.

28.14.1 GPIO Functionality

Figure 28-60 illustrates the GPIO functionality.

Figure 28-60. GPIO Functionality



28.14.2 Under Reset

The following apply if a device is under reset:

- Pull control. The reset pull control on the pins is enabled.
- Input buffer. The input buffer is enabled.
- Output buffer. The output buffer is disabled.

28.14.3 Out of Reset

The following apply if the device is out of reset:

- Pull control. The pull control is enabled by clearing the PD (pull control disable) bit in the SCIPIO7 register (Section 28.13.21). In this case, if the PSL (pull select) bit in the SCIPIO8 register (Section 28.13.22) is set, the pin will have a pull-up. If the PSL bit is cleared, the pin will have a pull-down. If the PD bit is set in the control register, there is no pull-up or pull-down on the pin.
- Input buffer. The input buffer is permanently enabled in this device.

NOTE: The pull-disable logic depends on the pin direction. It is independent of whether the device is in I/O or functional mode. If the pin is configured as output or transmit, then the pulls are disabled automatically.

- Output buffer. A pin can be driven as an output pin if the TX DIR bit is set in the pin direction control register (SCIPIO1; Section 28.13.15) AND the open-drain feature is not enabled in the SCIPIO6 register (Section 28.13.20).

28.14.4 Open-Drain Feature Enabled on a Pin

The following apply if the open-drain feature is enabled on a pin:

- The output buffer is enabled if a low signal is being driven on to the pin.
- The output buffer is disabled (the direction control signal DIR is internally forced low) if a high signal is being driven on to the pin.

NOTE: The open-drain feature is available only in I/O mode (SCIPIO0; Section 28.13.14).

28.14.5 Summary

The behavior of the input buffer, output buffer, and the pull control is summarized in Table 28-48.

Table 28-48. Input Buffer, Output Buffer, and Pull Control Behavior as GPIO Pins⁽¹⁾

Device under Reset?	Pin Direction (DIR) ⁽²⁾	Pull Disable (PULDIS) ⁽³⁾	Pull Select (PULSEL) ⁽⁴⁾	Pull Control	Output Buffer	Input Buffer
Yes	X	X	X	Enabled	Disabled	Enabled
No	0	0	0	Pull down	Disabled	Enabled
No	0	0	1	Pull up	Disabled	Enabled
No	0	1	0	Disabled	Disabled	Enabled
No	0	1	1	Disabled	Disabled	Enabled
No	1	X	X	Disabled	Enabled	Enabled

⁽¹⁾ X = Don't care

⁽²⁾ DIR = 0 for input, = 1 for output

⁽³⁾ PULDIS = 0 for enabling pull control, = 1 for disabling pull control

⁽⁴⁾ PULSEL = 0 for pull-down functionality, = 1 for pull-up functionality